

8

COLORED PATTERNS AND TILINGS

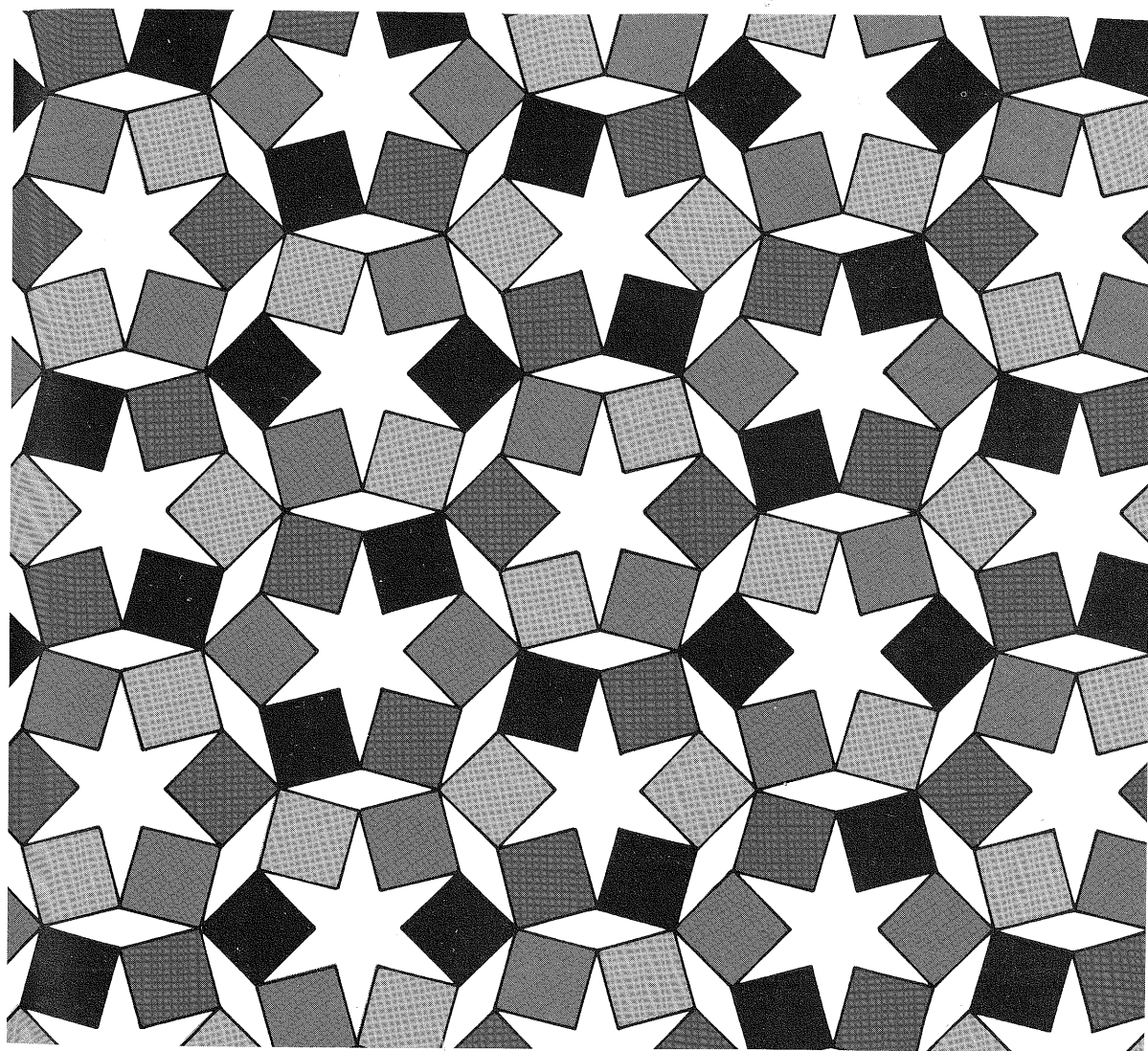


Figure 8.0.1

An attractive and ingenious pattern of squares from Islamic art (Bourgoin [1879, Plate 13]) perfectly colored in four colors. This pattern is of type PP48A and it can be perfectly k -colored if k has one of the values $3n$, $6n$, n^2 , $2n^2$, $3n^2$ or $6n^2$ for a positive integer n (see Senechal [1979a]).

Although the coloring of patterns and tilings is almost universal in the decorative arts, there exists only a small amount of mathematical literature devoted to the subject. Moreover, much of this is unsatisfactory because the authors tend to adopt *ad hoc* definitions and systems of classification; as a result, comparisons are often difficult to make and the scope of the achievements is not very clear.

In this chapter we shall treat the topic in a much more systematic manner—a procedure made possible by our basic approach to classifying geometric objects by symmetry, as discussed in Chapter 7. The application of these principles to colored patterns and tilings is straightforward, and the resulting classifications seem to be precisely what some earlier authors were seeking. We begin by defining the necessary concepts, which lead us to the coarsest classification—by color groups. This is refined, in the following sections, in various ways.

The subject abounds with open problems. Many of these concern the enumeration of the number of types of colorings of patterns or tilings with a given number of colors, when classified according to one of the methods. A great deal of work in this area remains to be done.

8.1 COLOR SYMMETRIES

Let $\mathcal{P}$ be a discrete pattern and suppose that to each copy of the motif in $\mathcal{P}$ we assign one of a given finite set of colors. Then we obtain what is called a *colored pattern*, which we denote by $\mathcal{P}^\wedge$. More formally, we may denote the colors by integers $1, 2, \dots, k$ (for some

k) and then define $\mathcal{P}^\wedge$ to be a pair $(\mathcal{P}, \chi)$ where $\mathcal{P}$ is the (uncolored) pattern that *underlies* $\mathcal{P}^\wedge$ and χ is a function which maps each copy M_i of the motif M of $\mathcal{P}$ onto an element of the set $\{1, 2, \dots, k\}$. Thus each copy M_i is assigned the “color” $\chi(M_i)$; the pair $M_i^\wedge = (M_i, \chi(M_i))$ is called a *colored copy* of the motif of $\mathcal{P}$ or of $\mathcal{P}^\wedge$. All the copies of the motif of $\mathcal{P}^\wedge$ assigned to a given color j (that is to say, all those M_i such that $\chi(M_i) = j$) are said to form a *color class* and it will be convenient to refer to this as the color class j . If k is minimal in the sense that each of the colors $1, 2, \dots, k$ is assigned to at least one copy of the motif of $\mathcal{P}$, then $\mathcal{P}^\wedge$ is called a *k-colored pattern*.*

All these definitions apply, with obvious modifications, to colored tilings, and we shall use symbols like $\mathcal{T}^\wedge$, or $\chi(T_i)$ and terms like “colored tile” or “k-colored tiling” without further explanation. The terminology extends that already introduced in Sections 1.3 and 2.9.

Examples of colored tilings have already been given in Figures 2.9.2 and 5.2.13. In the latter the tilings are 3-colored, and the three colors are indicated by white, black and stippling (instead of by integers 1, 2 and 3). A similar representation is employed in many of the diagrams of this chapter; various kinds of shading and hatching are used for the colors. In general, color 1 is indicated by white, and colors 2, 3, ... by increasingly dark shades of gray. Further examples of colored tilings appear in Figures 8.1.1 and 8.1.2 and many diagrams

* This definition is adopted for simplicity. For some purposes, it is more convenient to define a *k-colored pattern* as one in which *at most* k colors are used.

of k -colored patterns, for small values of k , are given in Sections 8.2, 8.3 and 8.4.

The definition of a symmetry of a colored tiling $\mathcal{T}^\wedge$ given in Section 1.3 extends in an obvious way to colored patterns: a (*color-preserving*) *symmetry* of $\mathcal{P}^\wedge$ is any isometry which maps each colored copy of a motif of $\mathcal{P}^\wedge$ into a copy of the motif of the same color. The group of all such symmetries is called the *color-preserving symmetry group* of $\mathcal{P}^\wedge$ and is denoted by $S(\mathcal{P}^\wedge)$. However, in the analysis of colored patterns, a much more useful concept is the following.

A *color symmetry* of a colored pattern $\mathcal{P}^\wedge$ is a pair $s^\wedge = (s, \theta)$ where $s \in S(\mathcal{P})$ is a symmetry of $\mathcal{P}$, and θ is a permutation of colors which is *compatible* with s . By this we mean that $\theta(\chi(M_i)) = \chi(s(M_i))$ for each copy M_i of the motif M . In other words, whenever s maps a copy of the motif of color k onto one of color j , then it maps the whole color class k onto color class j , and this happens if and only if the permutation θ maps k onto j . We shall say that s and θ are *associated* with each other and with the color symmetry $s^\wedge = (s, \theta)$.

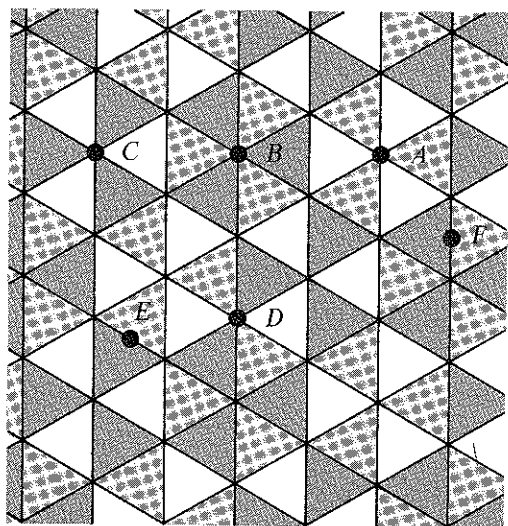


Figure 8.1.1
A 3-chromatic tiling which is not perfectly colored.

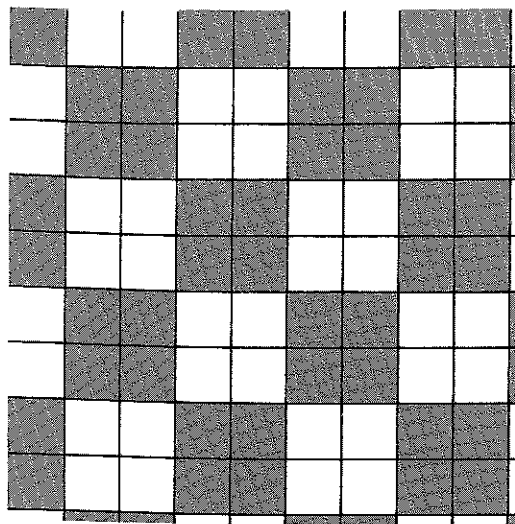


Figure 8.1.2
A 2-chromatic tiling $\mathcal{T}^\wedge$ for which the groups $S(\mathcal{T}^\wedge)$, $S_c(\mathcal{T}^\wedge)$ and $S(\mathcal{T})$ are all distinct, although each is of type $p4m$.

For example, in the 3-colored tiling $\mathcal{T}^\wedge$ of Figure 8.1.1, which has the regular tiling (3^6) as its underlying tiling $\mathcal{T}$, let a, b, c denote clockwise rotations about the points A, B, C by π , and let α, β, γ be the permutations (12) , (23) and (31) , respectively. Then $a^\wedge = (a, \alpha)$, $b^\wedge = (b, \beta)$ and $c^\wedge = (c, \gamma)$ are color symmetries of $\mathcal{T}^\wedge$. Other color symmetries of $\mathcal{T}^\wedge$ are $d^\wedge = (d, \delta)$ and $f^\wedge = (f, \phi)$ where d is the clockwise rotation by $\frac{1}{3}\pi$ about D , f is a rotation by π about E , $\delta = (123)$ and $\phi = (23)$. On the other hand, certain symmetries of $\mathcal{T}$ —such as a half-turn about the point F —are clearly not associated with color symmetries of $\mathcal{T}^\wedge$ since there is no compatible permutation of colors.

The color-preserving symmetries of a pattern $\mathcal{P}^\wedge$ are simply those color symmetries (s, θ) for which θ is the identity permutation. In Figure 8.1.1, rotations by $2\pi/3$ about any of the points A, B or C , or reflection in the line through A, B, C are examples of such symmetries.

If $\mathcal{P}^\wedge$ is a k -colored pattern, the set $C(\mathcal{P}^\wedge)$ of all color symmetries of $\mathcal{P}^\wedge$ clearly forms a group under composition

$$(s_1, \theta_1)(s_2, \theta_2) = (s_1 s_2, \theta_1 \theta_2)$$

and we call $C(\mathcal{P}^\wedge)$ either the *color symmetry group* of $\mathcal{P}^\wedge$ or the *group of color symmetries* of $\mathcal{P}^\wedge$. The k -colored

pattern $\mathcal{P}$ is called *k-chromatic* if the group $C(\mathcal{P})$ acts transitively on the colored copies $M_i = (M, \chi(M_i))$ of the motif of $\mathcal{P}$. It is not hard to verify that the colored tiling of Figure 8.1.1 is 3-chromatic, as are all 3-colored tilings of Figure 5.2.13 except (a), (f) and (g).

If $\mathcal{P}$ is a colored pattern then the subgroup of $S(\mathcal{P})$ which consists of all symmetries of $\mathcal{P}$ associated with color symmetries of $\mathcal{P}$ is called the *underlying group* of $\mathcal{P}$ and is denoted by $S_c(\mathcal{P})$. In other words, $S_c(\mathcal{P})$ consists of all those elements $s \in S(\mathcal{P})$ with which it is possible to associate a permutation θ such that (s, θ) is a color symmetry of $\mathcal{P}$. Clearly $\mathcal{P}$ is *k-chromatic* if and only if $S_c(\mathcal{P})$ is a motif-transitive subgroup of $S(\mathcal{P})$. Further, we shall say that $\mathcal{P}$ is *perfectly colored* if $S_c(\mathcal{P}) = S(\mathcal{P})$, that is, if every symmetry of $\mathcal{P}$ is associated with a color symmetry of $\mathcal{P}$. Thus perfectly *k-colored* patterns and perfectly *k-colored* isohedral tilings are necessarily *k-chromatic*.

The 3-chromatic tiling of Figure 8.1.1 is not perfectly colored since, as we have already remarked, certain symmetries of $\mathcal{P}$ (such as a half-turn about the point F) are not associated with color symmetries. All the 3-chromatic tilings of Figure 5.2.13 (that is, all except (a), (f) and (g)) are perfectly colored. For further examples consider the 2-colored patterns obtained from the tilings of Figure 5.2.13 by ignoring the white tiles and taking the interiors of the black and stippled tiles as colored

copies of the motif. Of these, only the patterns obtained from (a), (b), (c) and (d) are perfectly colored—the pattern obtained from (e), is even *not* 2-chromatic, while the remaining two, (f) and (g), are 2-chromatic but the coloring is not perfect since the rotational symmetries of the patterns are not associated with colored symmetries. Figure 8.1.2 shows a 2-chromatic tiling $\mathcal{T}$ which is not perfectly colored; the three groups $S(\mathcal{T})$, $S_c(\mathcal{T})$ and $S(\mathcal{T})$ are distinct even though each is of type pm .

Although certain patterns, such as those of type PP1, can be *k-chromatically* colored (and even perfectly *k-colored*) for all values of *k*, this is not generally the case. For example it is easy to see that a pattern of type PP21 has no 2-chromatic coloring and one of type PP41 has no perfect 3-coloring. The problem of finding the values of *k* for which perfect *k-colorings* are possible will be discussed in Section 8.7.

If $\mathcal{P}$ is a *k-colored* pattern for which the underlying pattern type is primitive, then the only motif-transitive subgroup of $S(\mathcal{P})$ is $S(\mathcal{P})$ itself and it readily follows that $\mathcal{P}$ is *k-chromatic* if and only if it is perfectly colored. Hence perfect *k-colorings* of primitive patterns can be used to determine and describe all the possible types of groups of color symmetries of patterns. These latter groups, known as *k-color groups*, form the topic of the next section.

EXERCISES 8.1

1. Determine which of the colorings of the three regular tilings shown in Figure 2.9.2 are (a) *k-chromatic*, and (b) perfect.
 2. For each of the three regular tilings and for each integer $k \geq 2$, show that there exist at least two essentially distinct *k-chromatic* colorings of the tiling. (Here “essentially distinct” means that one cannot be transformed into the other by a permutation of colors.)
 3. For each of the three regular tilings and for each *k* satisfying $2 \leq k \leq 10$, decide whether there exists a perfect *k-coloring* of the tiling.
 4. For $k = 2$ and $k = 3$ find all *k-chromatic* colorings of the three regular tilings.
 5. For each *k* satisfying $2 \leq k \leq 6$ and for each of the eight Laves tilings which is not regular, decide whether the tiling admits a *k-chromatic* coloring.
-

8.2 COLOR GROUPS

It is convenient to represent the color symmetry group $C(\mathcal{P})$ of a colored pattern $\mathcal{P}$ by a *color group diagram*. This is analogous to the group diagrams of symmetry groups which we introduced in Section 1.4; the symmetry elements in $S_C(\mathcal{P})$ are denoted by symbols similar to those of Table 1.4.1 and near each element $s \in S_C(\mathcal{P})$ is indicated the permutation θ with which it is associated in a color symmetry $s^\theta = (s, \theta) \in C(\mathcal{P})$. More precisely, near a symbol for the center of a cn or dn rotation we indicate the permutation associated with one counterclockwise step of rotation, near a line of reflection we indicate the permutation associated with the reflection, and near the symbol of a glide-reflection we show the permutation associated with one glide step. The identity permutation ε is indicated in the diagrams by empty parentheses (). Notice that on the diagram we do *not* indicate any symmetries of $\mathcal{P}$ which are not associated with color symmetries of $\mathcal{P}$.

For example, in Figure 8.2.1 we show a 4-chromatic pattern $\mathcal{P}$ (with underlying pattern type PP38) together with part of its color group diagram.

We shall say that the color symmetry groups $C(\mathcal{P}_1)$ and $C(\mathcal{P}_2)$ of two k -colored patterns $\mathcal{P}_1$ and $\mathcal{P}_2$ are *isomorphic* if their color group diagrams can be made to coincide by applying a suitable affine transformation (and also, possibly, altering one's choice of vectors corresponding to the translations—compare Section 1.4) combined with a permutation of colors in one of the patterns. The class of all mutually isomorphic color symmetry groups of k -chromatic strip or periodic discrete patterns is called a (strip or periodic) *k-color group*.

8.2.1 For each $k \geq 2$ there exist only finitely many different strip and periodic k -color groups.

In fact in order to determine these groups we need only choose a set of elements which generate a given strip or periodic group and then associate with each such generator, in every possible way, a permutation of $(1, 2, \dots, k)$.

Our choice of permutations is restricted by the fact that every algebraic relation between the generating symmetries must also be satisfied by the associated permutations. Since there are clearly only a finite number of possibilities for each value of k , Statement 8.2.1 is proved.

For very small values of k , the above procedure gives a practical method of enumerating the k -color groups. The only difficulty is to discover and eliminate isomorphic groups. However, this becomes rapidly more difficult as k increases and it seems necessary to use computers to carry out the enumeration. In Table 8.2.1 (which is based on results of Wieting [1981]) we list the known values for the number of k -color groups for $k \leq 12$ specified by the underlying strip or periodic group.

Using algebraic methods, the enumeration of k -color groups has been attempted by several authors. For example Senechal [1979a] gives an enumeration for various infinite classes of integers k . Additional references and historical details will be given in Section 8.8. For a complete account of Senechal's results the reader should consult the original paper; here we shall only quote an easily stated partial result:

8.2.2 The number of periodic k -color groups, where k is a prime number greater than 3, equals 16 if $k \equiv 1 \pmod{12}$, 15 if $k \equiv 7 \pmod{12}$, 14 if $k \equiv 5 \pmod{12}$ and 13 if $k \equiv 11 \pmod{12}$.

An analogous enumeration can easily be carried out for the strip k -color groups and we obtain the following results (communicated to us by M. Senechal); this enumeration has been carried out independently by Jarratt & Schwarzenberger [1981].

8.2.3 For each odd integer k there are precisely 7 different strip k -color groups. For even values of k there are 17 groups if $k \equiv 2 \pmod{4}$ and 19 groups if $k \equiv 0 \pmod{4}$.

In Figures 8.2.2 and 8.2.3 we illustrate the periodic 2-color and 3-color groups by perfectly colored primitive patterns. Illustrations analogous to those in Figure 8.2.2

can be found in many places (see, for example, Weber [1929], Belov & Tarhova [1956a], Šubnikov & Kopcik [1972]); the representations in Figure 8.2.3 seem to be new. In Tables 8.2.2 and 8.2.3 we present further information about the 2-color and 3-color groups.

As stated in the previous section, although the representation of the color groups by perfectly colored primitive patterns (instead of by their color group diagrams) is convenient, it can be misleading in the sense that two

such representations of the same color group can “appear” very different. Examples are given in Figure 8.2.4. Another example of two “apparently” different colorings corresponding to the same color group, this time of a non-primitive pattern, will be given in Figure 8.3.1. In the following sections we shall present finer classifications of colorings of patterns that overcome, in part, this shortcoming of classification by color group alone.

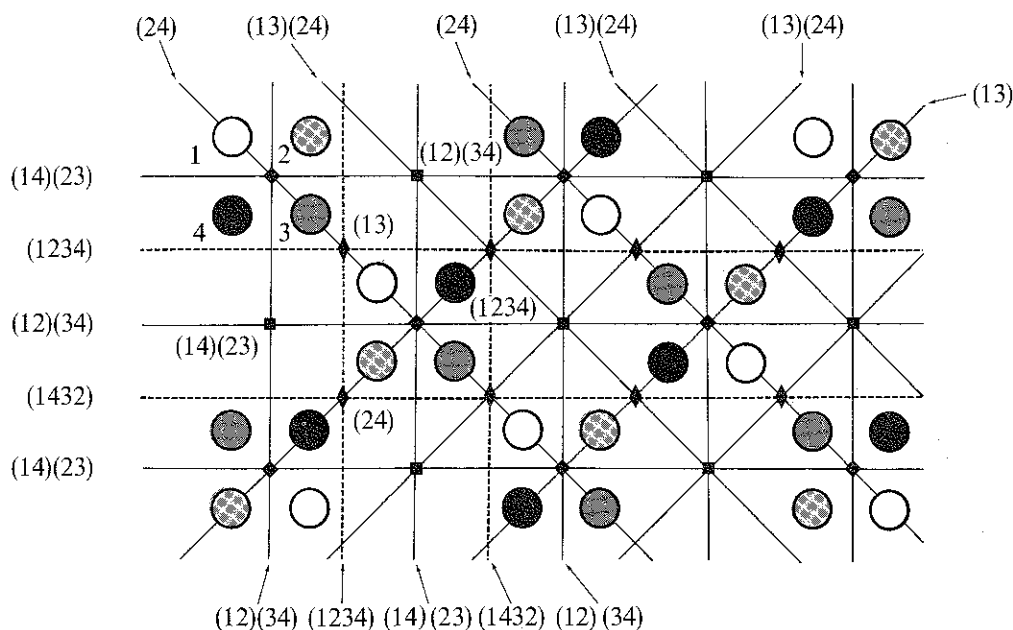


Figure 8.2.1

A 4-chromatic tiling and the corresponding color group diagram. For clarity, only a few of the symmetry elements have been indicated; in particular, no translations are shown.

Table 8.2.1 THE NUMBERS OF STRIP AND PERIODIC k -COLOR GROUPS FOR $k \leq 12$

The underlying group is indicated in the left column. The entries for the periodic groups are from Wieting [1981].

Underlying group	Number of colors										
	2	3	4	5	6	7	8	9	10	11	12
<i>p111</i>	1	1	1	1	1	1	1	1	1	1	1
<i>plal</i>	1	1	1	1	1	1	1	1	1	1	1
<i>plm1</i>	3	1	3	1	3	1	3	1	3	1	3
<i>pm11</i>	2	1	2	1	2	1	2	1	2	1	2
<i>pl12</i>	2	1	2	1	2	1	2	1	2	1	2
<i>pma2</i>	3	1	3	1	3	1	3	1	3	1	3
<i>pmm2</i>	5	1	7	1	5	1	7	1	5	1	7
Total strip groups	17	7	19	7	17	7	19	7	17	7	19
<i>p1</i>	1	1	2	1	1	1	2	2	1	1	2
<i>pg</i>	2	2	4	2	5	2	7	3	6	2	11
<i>pm</i>	5	2	10	2	11	2	16	3	12	2	23
<i>cm</i>	3	2	7	2	7	2	13	3	8	2	17
<i>p2</i>	2	1	3	1	2	1	4	2	2	1	3
<i>pgg</i>	2	1	4	1	4	1	7	2	5	1	9
<i>pmg</i>	5	2	11	2	11	2	19	3	12	2	26
<i>pmm</i>	5	1	13	1	9	1	21	2	10	1	25
<i>cmm</i>	5	1	11	1	8	1	21	2	9	1	22
<i>p3</i>	—	2	1	—	1	1	—	3	—	—	4
<i>p31m</i>	1	2	1	—	5	—	1	3	—	—	7
<i>p3m1</i>	1	2	1	—	4	—	1	3	—	—	7
<i>p4</i>	2	—	5	1	2	—	9	1	4	—	9
<i>p4g</i>	3	—	7	—	2	—	13	1	3	—	10
<i>p4m</i>	5	—	13	—	2	—	28	1	3	—	16
<i>p6</i>	1	2	1	—	5	1	1	3	—	—	8
<i>p6m</i>	3	2	2	—	11	—	3	3	—	—	20
Total periodic groups	46	23	96	14	90	15	166	40	75	13	219

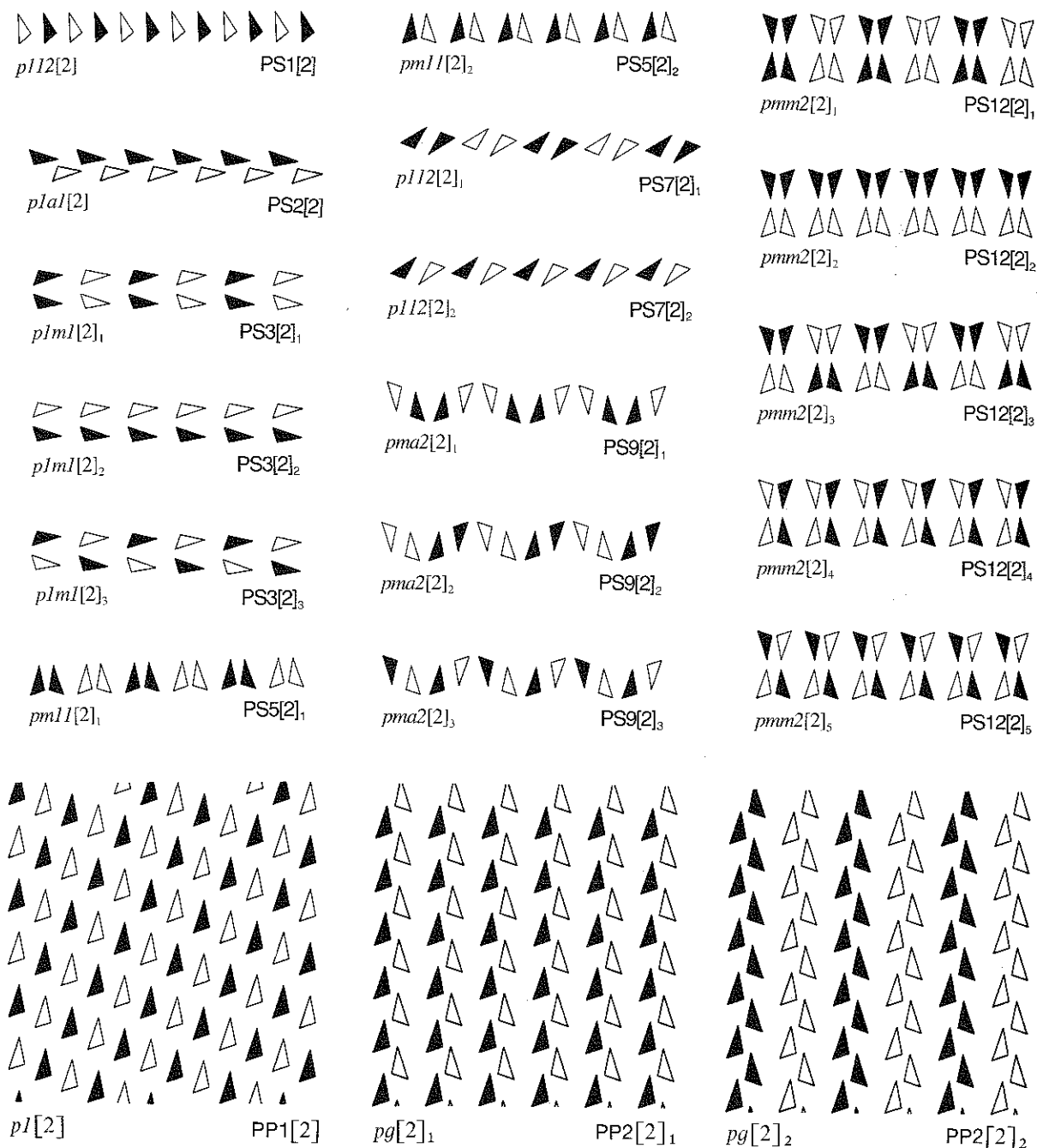
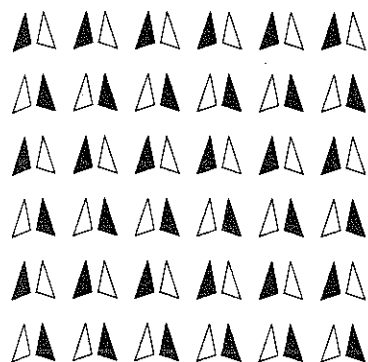
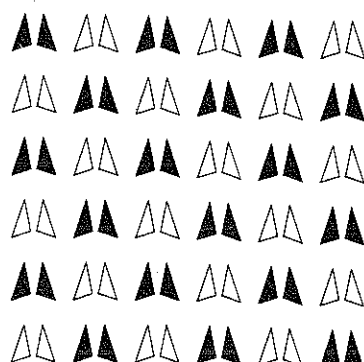


Figure 8.2.2
 Perfectly 2-colored primitive patterns that illustrate the 17 strip and 46 periodic 2-color groups. Near each pattern we have indicated a symbol for its 2-color symmetry group and its color pattern type (see Section 8.3 for details).



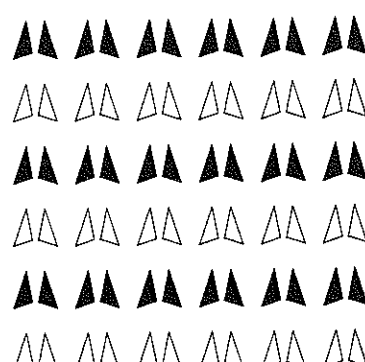
$pm[2]_1$

$PP3[2]_1$



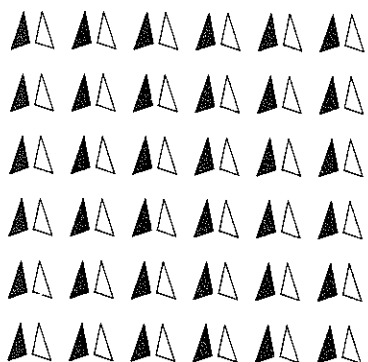
$pm[2]_2$

$PP3[2]_2$



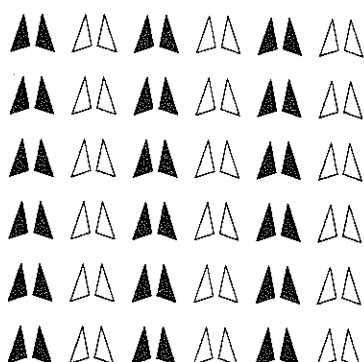
$pm[2]_3$

$PP3[2]_3$



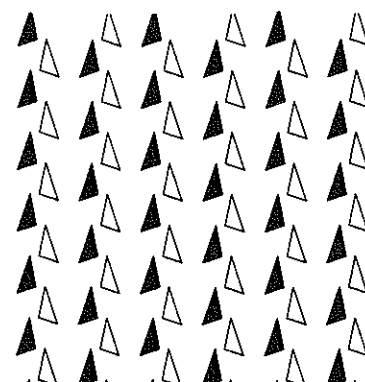
$pm[2]_4$

$PP3[2]_4$



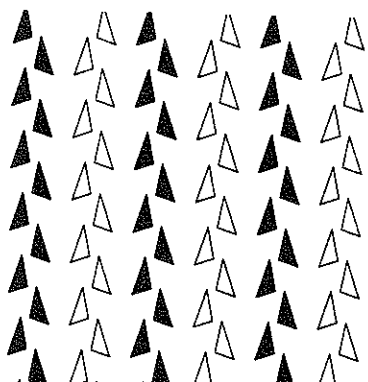
$pm[2]_5$

$PP3[2]_5$



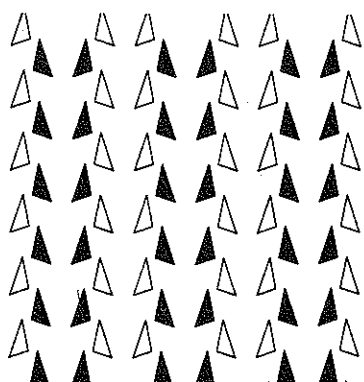
$cm[2]_1$

$PP5[2]_1$



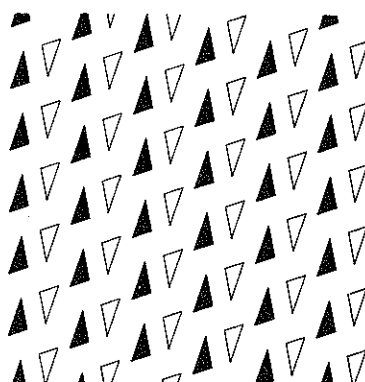
$cm[2]_2$

$PP5[2]_2$



$cm[2]_3$

$PP5[2]_3$



$p2[2]_1$

$PP7[2]_1$

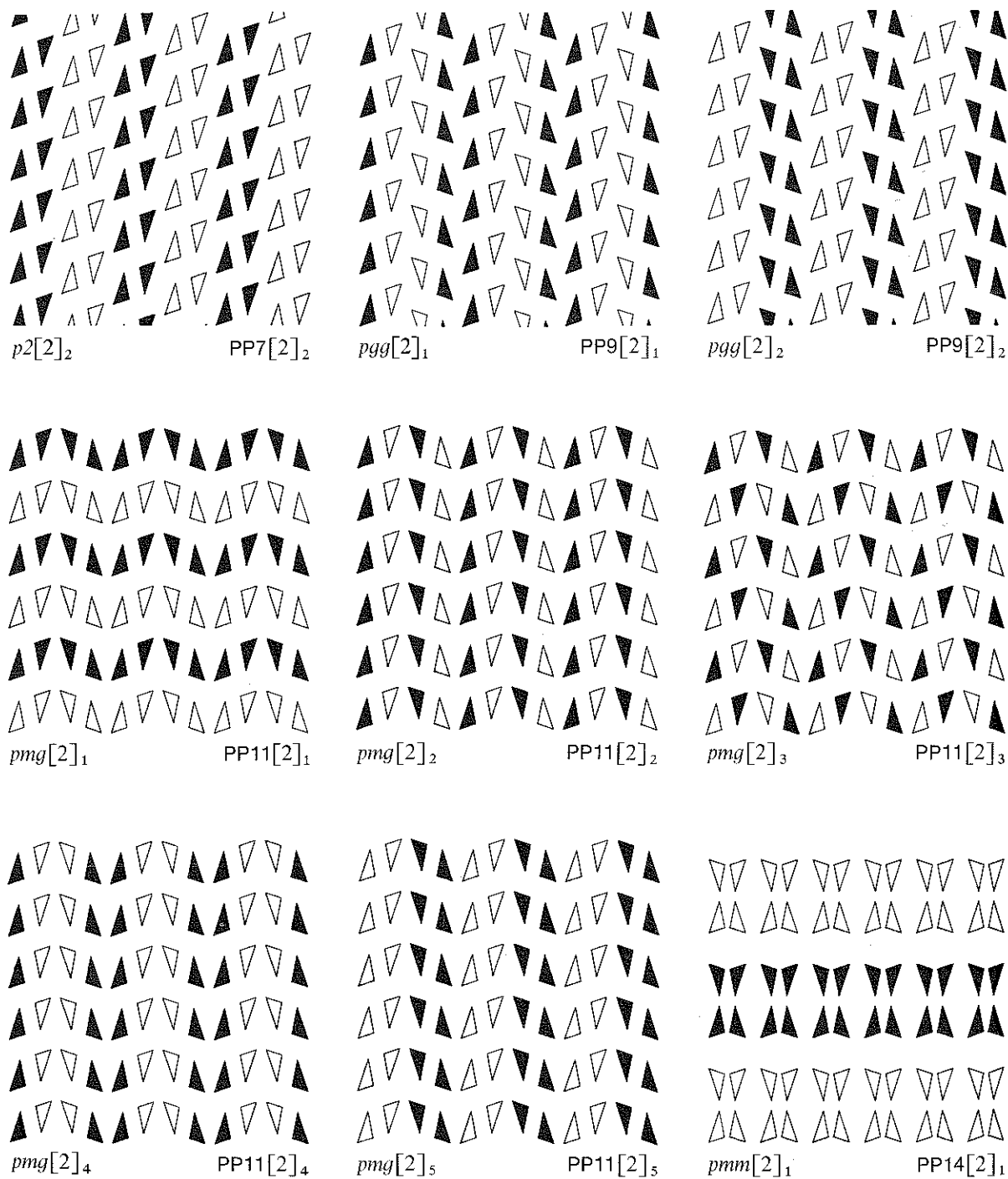


Figure 8.2.2 (continued)



$pmm[2]_2$

PP14[2]₂



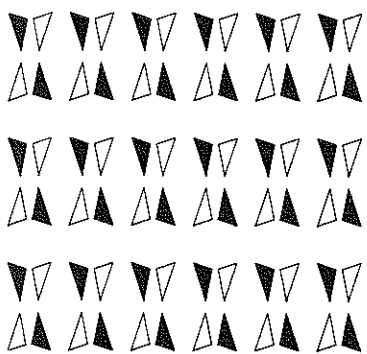
$pmm[2]_3$

PP14[2]₃



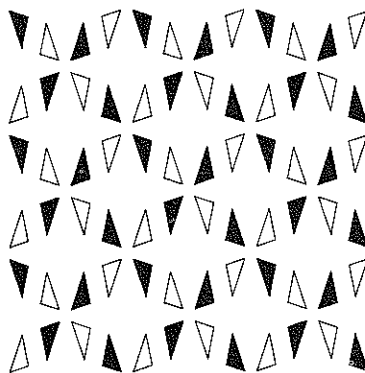
$pmm[2]_4$

PP14[2]₄



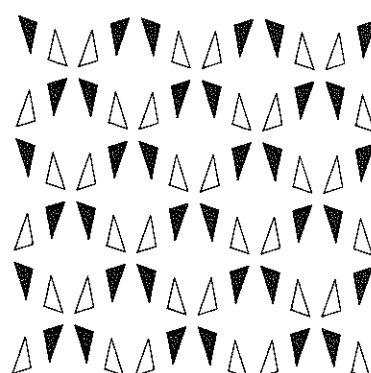
$pmm[2]_5$

PP14[2]₅



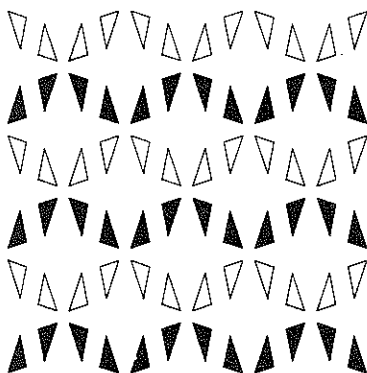
$cmm[2]_1$

PP17[2]₁



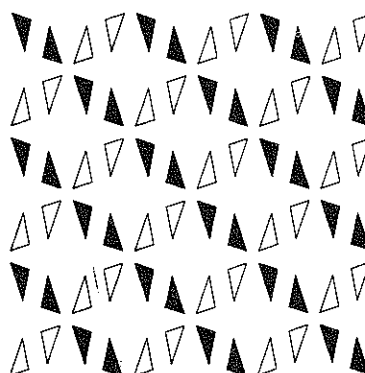
$cmm[2]_2$

PP17[2]₂



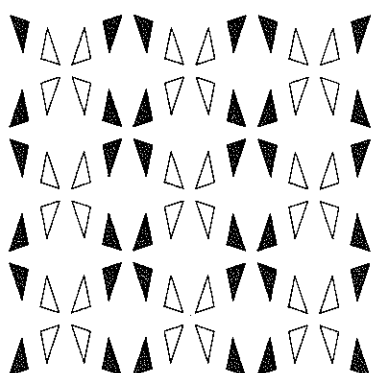
$cmm[2]_3$

PP17[2]₃



$cmm[2]_4$

PP17[2]₄



$cmm[2]_5$

PP17[2]₅

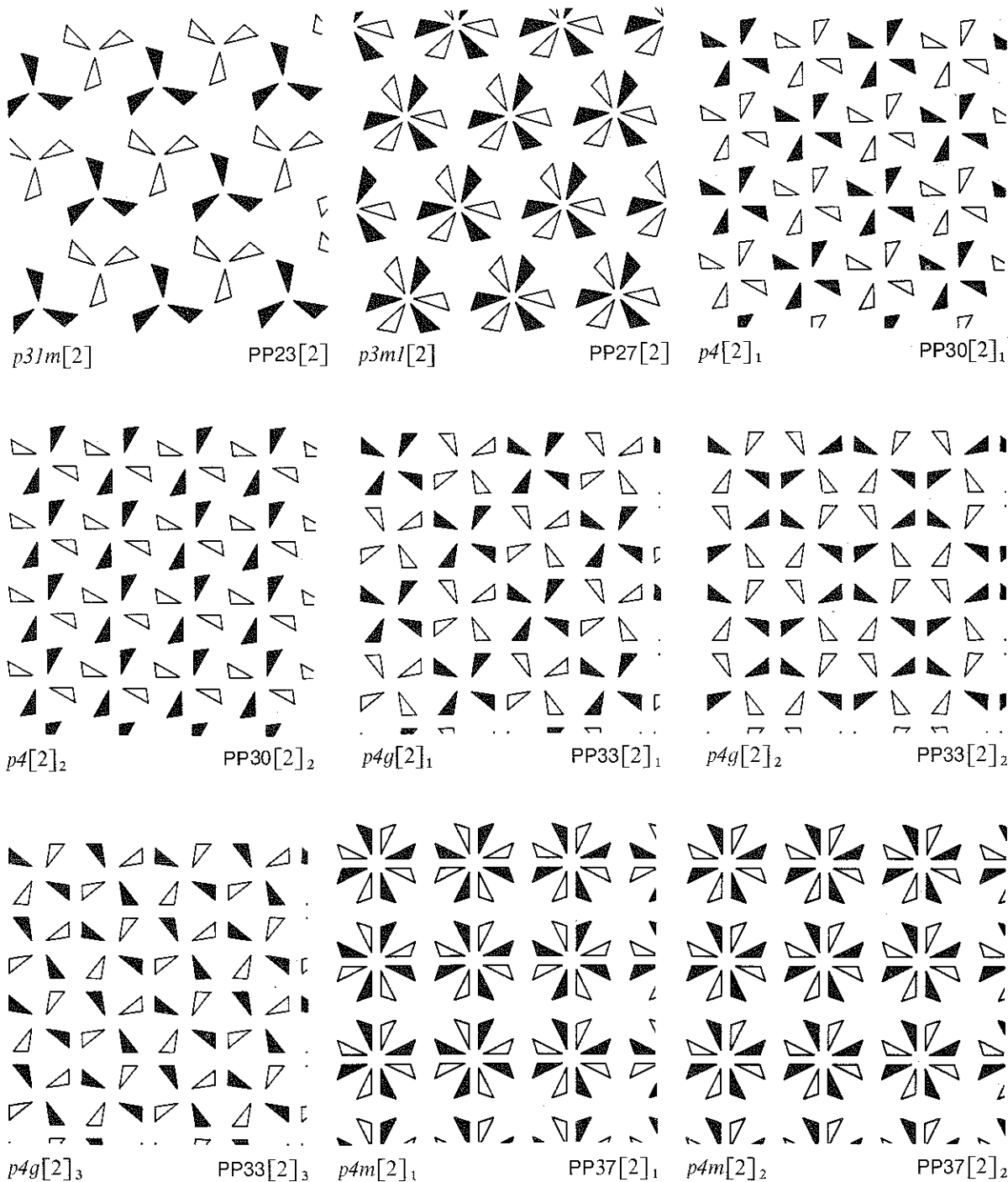
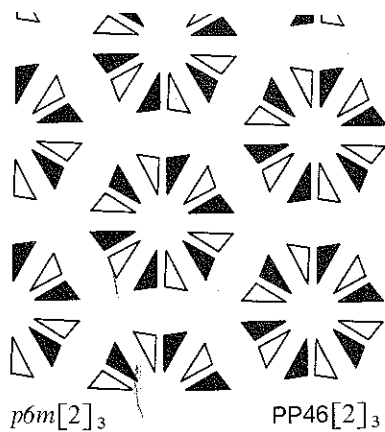
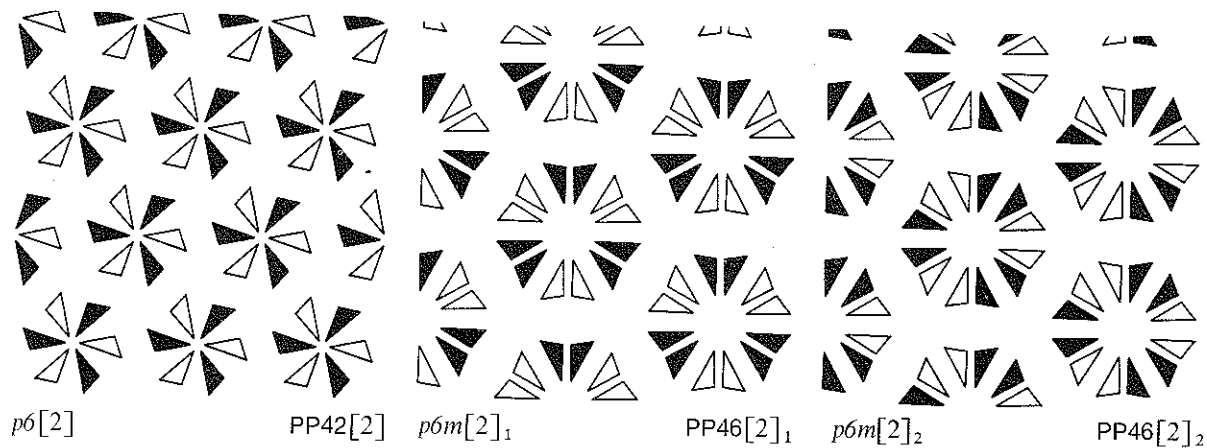
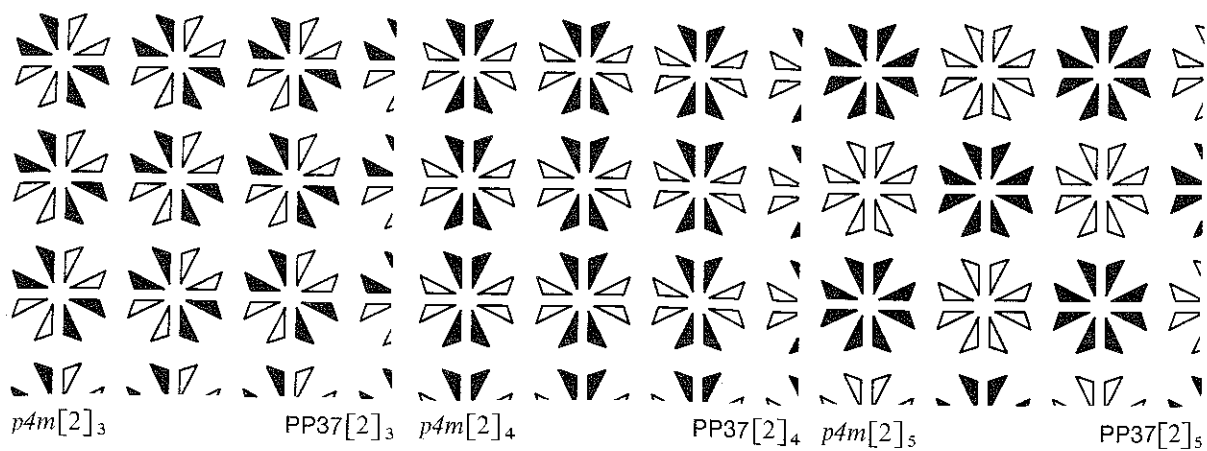


Figure 8.2.2 (continued)



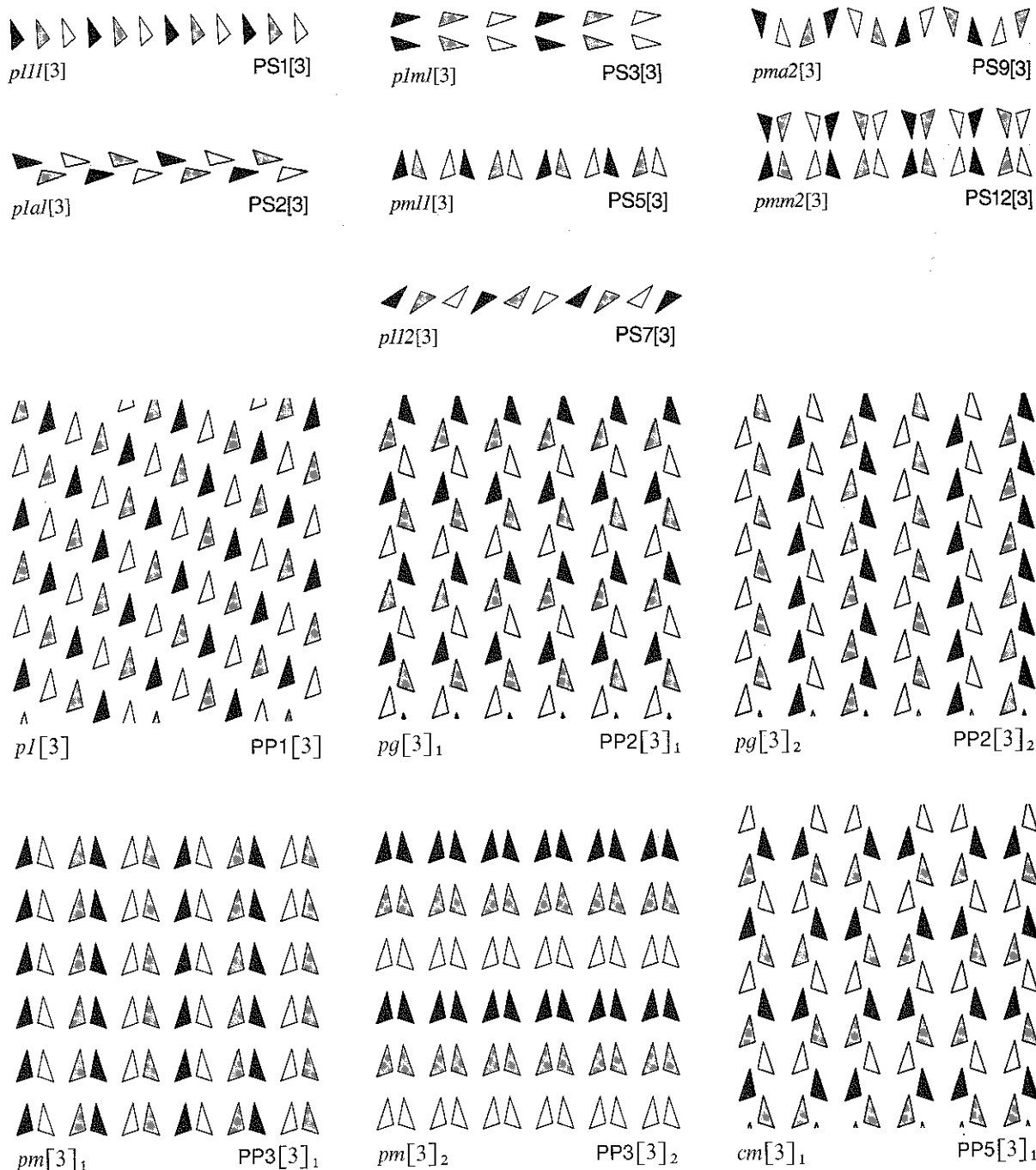
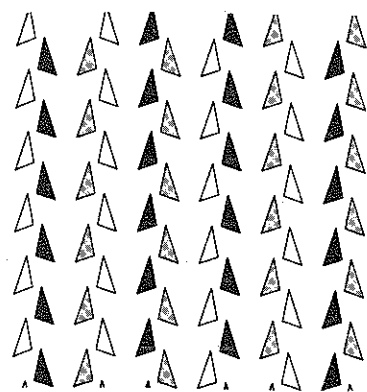
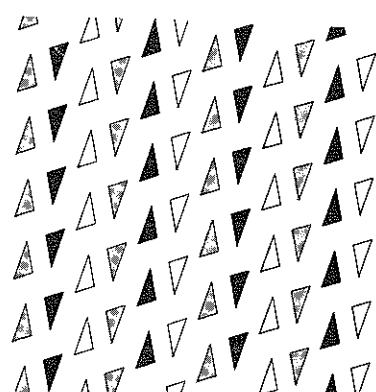


Figure 8.2.3
Perfectly 3-colored primitive patterns that illustrate the 7 strip and 23 periodic 3-color groups. Near each pattern we have indicated a symbol for its 3-color symmetry group and its color pattern type.



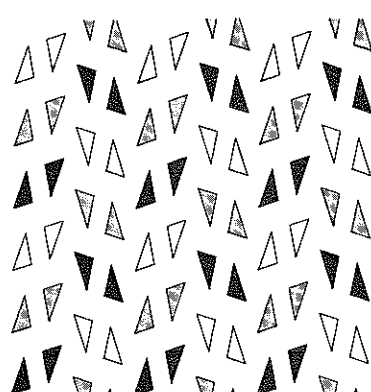
$cm[3]_2$

PP5[3]₂



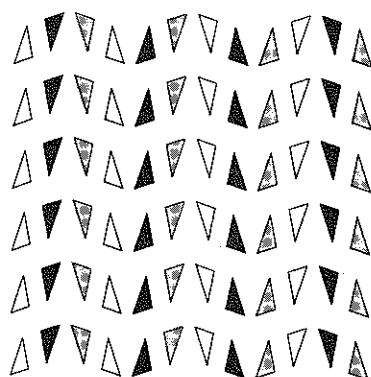
$p2[3]$

PP7[3]



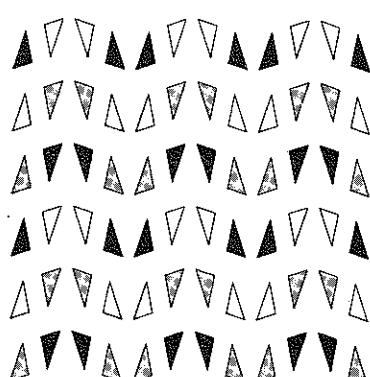
$pgg[3]$

PP9[3]



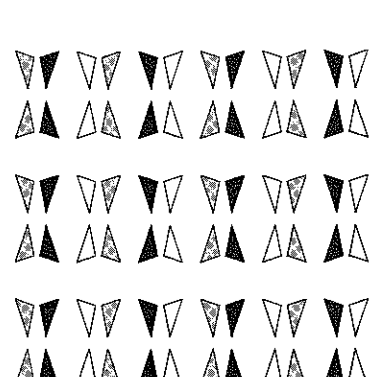
$pmg[3]_1$

PP11[3]₁



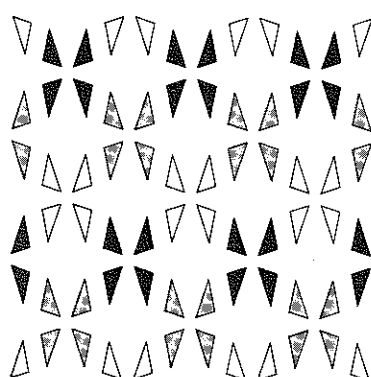
$pmg[3]_2$

PP11[3]₂



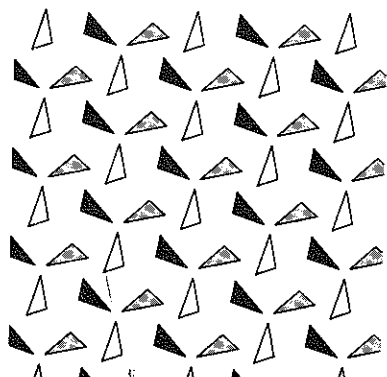
$pmm[3]$

PP14[3]



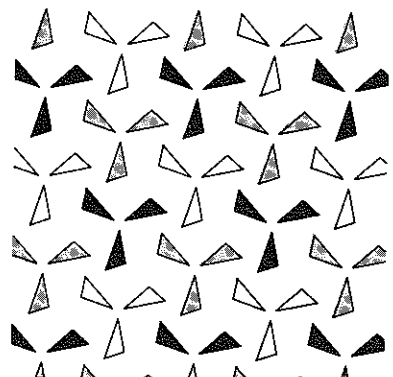
$cmm[3]$

PP17[3]



$p3[3]_1$

PP21[3]₁



$p3[3]_2$

PP21[3]₂

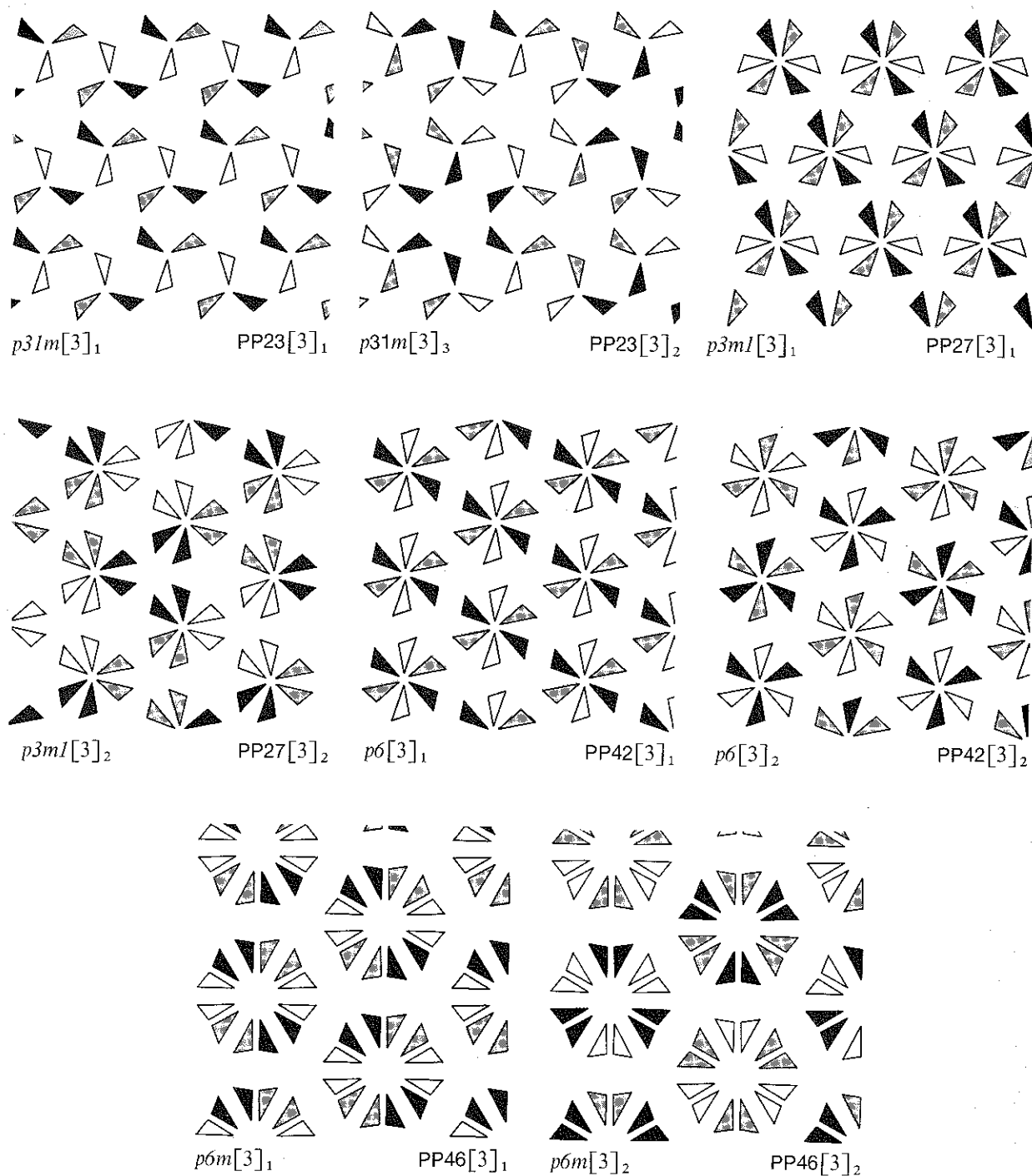


Figure 8.2.3 (continued)

Table 8.2.2 THE 46 PERIODIC 2-COLOR GROUPS

A symbol of each group is given in **column (1)**; these correspond to the symbols used in Figure 8.2.2, and were introduced in Grünbaum [1976] and Grünbaum & Shephard [1977e]. Symbols used by other authors (see, for example, Shubnikov, Belov & others [1964] or MacGillavry [1965]) are shown in **column (5)**. In **column (2)** we indicate the isohedral type of tilings which admit 2-chromatic colorings with each 2-color group. A color-tile symbol and the corresponding color-adjacency symbol of the 2-chromatic tiling are shown in **columns (3) and (4)**; these symbols will be explained in Section 8.6. The color of the starting tile is not indicated in the color-tile symbol since the color group does not depend on it. An asterisk * in **column (2)** signals that only marked 2-chromatic tilings can have this 2-color group. Another set of symbols for color groups has been proposed by Coxeter [1985]; see Crowe [1985] for a comparison of the symbols for 2-color groups used by various authors.

Symbol of the group	Type of tiling	Color-tile symbol	Color-adjacency symbol	Alternative symbol
(1)	(2)	(3)	(4)	(5)
$p1[2]$	IH1	$a^+b^+c^+d^+e^+f^+$	$d_*^+e_*^+f_*^+a_*^+b_*^+c^+$	p_6^1l
$pg[2]_1$ $pg[2]_2$	IH3	$a^+b^+c^+d^+e^+f^+$	$c_*^-e_*^+a_*^-f_*^-b_*^+d_*^-$ $c^-e_*^+a^-f_*^-b_*^+d_*^-$	pg' p_6^1lg
$pm[2]_1$ $pm[2]_2$ $pm[2]_3$ $pm[2]_4$ $pm[2]_5$	IH42	$a^+b^+c^+d^+$	$c_*^+b_*^-a_*^+d_*^-$ $c_*^+b_*^-a_*^+d^-$ $c_*^+b^-a_*^+d^-$ $c^+b_*^-a^+d_*^-$ $c^+b_*^-a^+d^-$	p_6^1g $c'm$ p_6^1m pm' p_6^1lm
$cm[2]_1$ $cm[2]_2$ $cm[2]_3$	IH22	$a^+b^+c^+d^+e^+$	$a_*^-e^+d_*^-c_*^-b^+$ $a_*^-e^+d^-c^-b^+$ $a^-e^+d_*^-c_*^-b^+$	cm' p_6^1g p_6^1m
$p2[2]_1$ $p2[2]_2$	IH4	$a^+b^+c^+d^+e^+f^+$	$a_*^+e^+c_*^+d_*^+b^+f^+$ $a_*^+e^+c^+d^+b^+f_*^+$	$p2'$ p_6^12
$pgg[2]_1$ $pgg[2]_2$	IH6	$a^+b^+c^+d^+e^+f^+$	$a_*^+e_*^-c_*^+f^-b_*^-d^-$ $a^+e_*^-c^+f_*^-b_*^-d_*^-$	pgg' $pg'g'$
$pmg[2]_1$ $pmg[2]_2$ $pmg[2]_3$ $pmg[2]_4$ $pmg[2]_5$	IH50	$a^+b^+c^+d^+$	$c_*^+b^-a_*^+d^+$ $c^+b_*^-a^+d_*^+$ $c_*^+b_*^-a_*^+d_*^+$ $c^+b^-a^+d^+$ $c^+b_*^-a^+d^+$	p_6^1mg $pm'g$ p_6^1gg pmg' $pm'g'$

Table 8.2.2 (continued)

Symbol of the group	Type of tiling	Color-tile symbol	Color-adjacency symbol	Alternative symbol
(1)	(2)	(3)	(4)	(5)
$pmm[2]_1$	IH48	$a^+b^+c^+d^+$	$a_-^-b_-^-c_-^-d_-^-$	p'_6mm
$pmm[2]_2$	*		$a_*^-b_*^-c_*^-d_*^-$	pmm'
$pmm[2]_3$			$a_*^-b_*^-c_*^-d_*^-$	$c'mm$
$pmm[2]_4$	*		$a_*^-b_*^-c_*^-d_*^-$	p'_6gm
$pmm[2]_5$	*		$a_*^-b_*^-c_*^-d_*^-$	$pm'm'$
$cmm[2]_1$	IH54	$a^+b^+c^+d^+$	$a_*^-b_*^-c_*^-d_*^+$	p'_6gg
$cmm[2]_2$			$a^-b_*^-c_*^-d_*^+$	cmm'
$cmm[2]_3$			$a_*^-b_*^-c_*^-d_*^+$	p'_6mg
$cmm[2]_4$			$a_*^-b_*^-c_*^-d_*^+$	$cm'm'$
$cmm[2]_5$			$a^-b^-c^-d_*^+$	p'_6mm
$p31m[2]$	IH38	$a^+b^+c^+$	$a_*^-c^+b^+$	$p31m'$
$p3m1[2]$	IH87*	$a^+b^+c^+$	$a_*^-b_*^-c_*^-$	$p3m'$
$p4[2]_1$	IH28	$a^+b^+c^+d^+e^+$	$a_*^+c_*^+b_*^+e_*^+d_*^+$	p'_4
$p4[2]_2$			$a_*^+c_*^+b_*^+e_*^+d_*^+$	$p4'$
$p4g[2]_1$	IH56	$a^+b^+c^+d^+$	$b^+a^+c_*^-d_*^-$	$p4g'm'$
$p4g[2]_2$			$b_*^+a_*^+c_*^-d_*^-$	$p4'g'm$
$p4g[2]_3$			$b_*^+a_*^+c_*^-d_*^-$	$p4'gm'$
$p4m[2]_1$	IH80	$a^+b^+c^+$	$a_*^-b_*^-c_*^-$	p'_4gm
$p4m[2]_2$	*		$a_*^-b_*^-c_*^-$	$p4m'm'$
$p4m[2]_3$	*		$a^-b_*^-c_*^-$	$p4'm'm$
$p4m[2]_4$			$a_*^-b^-c^-$	$p4'mm'$
$p4m[2]_5$			$a^-b^-c_*^-$	p_64mm
$p6[2]$	IH21	$a^+b^+c^+d^+e^+$	$e_*^+c^+b^+d_*^+a_*^+$	$p6'$
$p6m[2]_1$	IH77	$a^+b^+c^+$	$a^-b_*^-c_*^-$	$p6'm'm$
$p6m[2]_2$			$a_*^-b^-c^-$	$p6'mm'$
$p6m[2]_3$			$a_*^-b_*^-c_*^-$	$p6m'm'$

Table 8.2.3 THE 23 PERIODIC 3-COLOR GROUPS

A symbol of each group is given in **column (1)**; these correspond to symbols used in Figure 8.2.3, and were used in Grünbaum [1976] and Grünbaum & Shephard [1977e], [1983b]. In **column (2)** we indicate the isohedral type of tilings which admit 3-chromatic colorings with each 3-color group. A color-tile symbol and the corresponding color-adjacency symbol of the 3-chromatic tiling are shown in **columns (3) and (4)**; these symbols will be explained in Section 8.6. The color of the starting tile is not indicated in the color-tile symbol since the color group does not depend on it. An asterisk * in column (2) signals that only marked 3-chromatic tilings can have this 3-color group.

Symbol of the group	Type of tiling	Color-tile symbol	Color-adjacency symbol
(1)	(2)	(3)	(4)
$pl[3]$	IH1	$a^+b^+c^+d^+e^+f^+$	$d^+(123)e^+()f^+(321)a^+(321)b^+()c^+(123)$
$pg[3]_1$ $pg[3]_2$	IH3	$a^+b^+c^+d^+e^+f^+$	$c^-(321)e^+()a^-(123)f^-(123)b^+()d^-(321)$ $c^-(12)e^+(321)a^-(12)f^-(13)b^+(123)d^-(13)$
$pm[3]_1$ $pm[3]_2$	IH42	$a^+b^+c^+d^+$	$c^+()b^-(12)a^+()d^-(13)$ $c^+(123)b^-()a^+(321)d^-()$
$cm[3]_1$ $cm[3]_2$	IH22	$a^+b^+c^+d^+e^+$	$a^-()e^+(123)d^-(321)c^-(123)b^+(321)$ $a^-(12)e^+()d^-(13)c^-(13)b^+()$
$p2[2]$	IH4	$a^+b^+c^+d^+e^+f^+$	$a^+(12)e^+(321)c^+(12)d^+(13)b^+(123)f^+(13)$
$pgg[3]$	IH6	$a^+b^+c^+d^+e^+f^+$	$a^+(23)e^-(123)c^+(23)f^-(13)b^-(321)d^-(13)$
$pmg[3]_1$ $pmg[3]_2$	IH50	$a^+b^+c^+d^+$	$c^+()b^-(13)a^+()d^+(12)$ $c^+(123)b^-()a^+(321)d^+(13)$
$pmm[3]$	IH48*	$a^+b^+c^+d^+$	$a^-()b^-(12)c^-()d^-(13)$
$cmm[3]$	IH54	$a^+b^+c^+d^+$	$a^-(23)b^-()c^-(12)d^+(13)$
$p3[3]_1$ $p3[3]_2$	IH7	$a^+b^+c^+d^+e^+f^+$	$b^+(321)a^+(123)d^+(321)c^+(123)f^+(321)e^+(123)$ $b^+(123)a^+(321)d^+(321)c^+(123)f^+()e^+()$
$p31m[3]_1$ $p31m[3]_2$	IH38	$a^+b^+c^+$	$a^-(23)c^+(123)b^+(321)$ $a^-()c^+(123)b^+(321)$
$p3ml[3]_1$ $p3ml[3]_2$	IH87* *	$a^+b^+c^+$	$a^-(12)b^-(23)c^-(13)$ $a^-(12)b^-(23)c^-(12)$
$p6[3]_1$ $p6[3]_2$	IH21	$a^+b^+c^+d^+e^+$	$e^+(123)c^+(123)b^+(321)d^+()a^+(321)$ $e^+(12)c^+(321)b^+(123)d^+(13)a^+(12)$
$p6m[3]_1$ $p6m[3]_2$	IH77	$a^+b^+c^+$	$a^-(12)b^-(12)c^-(13)$ $a^-()b^-(12)c^-(13)$

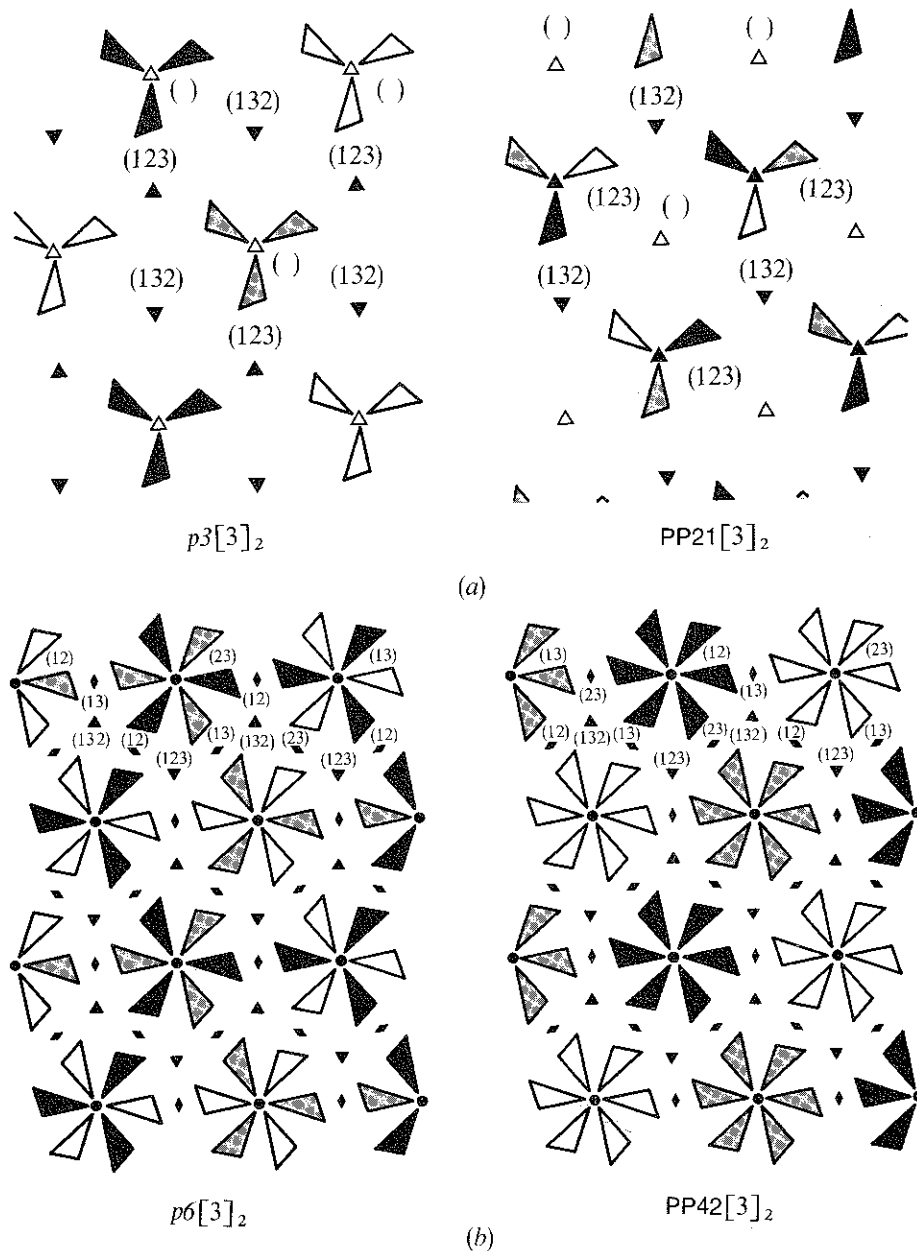


Figure 8.2.4
Examples of primitive 3-color patterns which differ in appearance and yet illustrate the same 3-color groups. In each part of the figure we show the diagram of a color group and a pattern for which this color group is the color symmetry group. In (a) the motifs are differently related to the three kinds of 3-fold rotational symmetries; in (b) the patterns differ in the choice of motif that bears color 1.

EXERCISES 8.2

1. Decide which of the k -colored patterns or tiling patterns in Figures 5.2.13 and 5.3.5 (d) (e) (f) are k -chromatic, $k = 2$ or 3 ; for those that are, draw a color-group diagram and determine its k -color group in the notation of Figures 8.2.2 and 8.2.3.
 2. For each k satisfying $4 \leq k \leq 6$ find which of the seven primitive strip patterns admits a perfect k -coloring. Hence check the numbers of k -color strip groups ($k = 4, 5, 6$) given in Table 8.2.1.
 3. (a) For each k and n satisfying $2 \leq k \leq 6$ and $2 \leq n \leq 12$ determine all the k -chromatic colorings of patterns of types $PF1_n$, $PF2_n$ and $PF3_n$.
(b) Determine all 2-chromatic and 3-chromatic colorings of patterns $PF1_n$ for all n .
 4. Determine the k -color group of each of the k -chromatic colorings of the regular tilings found in Exercise 8.1.4 (for $k = 2$ or 3).
-

8.3 COLOR PATTERN TYPES

In Chapter 5, in the context of uncolored patterns, we showed that the notion of pattern type can be used to refine the classification of patterns by symmetry groups alone. In the case of colored patterns the situation is completely analogous; to obtain the appropriate definitions we need only consider color symmetries instead of symmetries.

More precisely we shall say that two k -chromatic patterns $\mathcal{P}_1$ and $\mathcal{P}_2$ are of the same k -color pattern type if—possibly after an appropriate renaming (permutation) of the colors in one of the patterns—they have the same k -color symmetry groups, the same stabilizers of their colored motifs (that is, induced k -color groups), and the same set of colored-motif transitive subgroups of their color symmetry groups. An alternative, and equivalent, definition which follows the general procedure for classifying objects with respect to their symmetries (see Chapter 7) is as follows: two colored patterns $\mathcal{P}_1$ and $\mathcal{P}_2$ are of the same color pattern type if, possibly after an appropriate permutation of the colors in one of the patterns,

there exists a one-to-one color-preserving mapping $\Psi: M(\mathcal{P}_1) \rightarrow M(\mathcal{P}_2)$ from the set $M(\mathcal{P}_1)$ of colored copies of the motif of $\mathcal{P}_1$ to the corresponding set $M(\mathcal{P}_2)$ for $\mathcal{P}_2$, such that Ψ is compatible with the underlying group $S_c(\mathcal{P}_1)$ and Ψ^{-1} is compatible with the underlying group $S_c(\mathcal{P}_2)$.

To illustrate this definition consider the three 3-chromatic colorings of the pattern $\mathcal{P}$ of type PP15 shown in Figure 8.3.1. All are perfect 3-colorings, and the color symmetry group of each is the 3-color group $pmm[3]$. In Figures 8.3.1(a) and (b) the induced color group (that is, the subgroup that leaves one copy of the motif invariant) is dI with no changes of color. In Figure 8.3.1(c) it is dI but associated with a transposition of two of the colors as indicated on the diagram*. It follows that the 3-color pattern type of the pattern in Figure 8.3.1(c) is different from the 3-color pattern type of those in Figures 8.3.1(a) and (b). To show that the patterns in Figures 8.3.1(a) and (b) are of the same 3-color pattern type, we can either verify that the motif-transitive color subgroups

* Since we shall not discuss the possibilities in detail, it is not appropriate to assign special symbols for these groups.

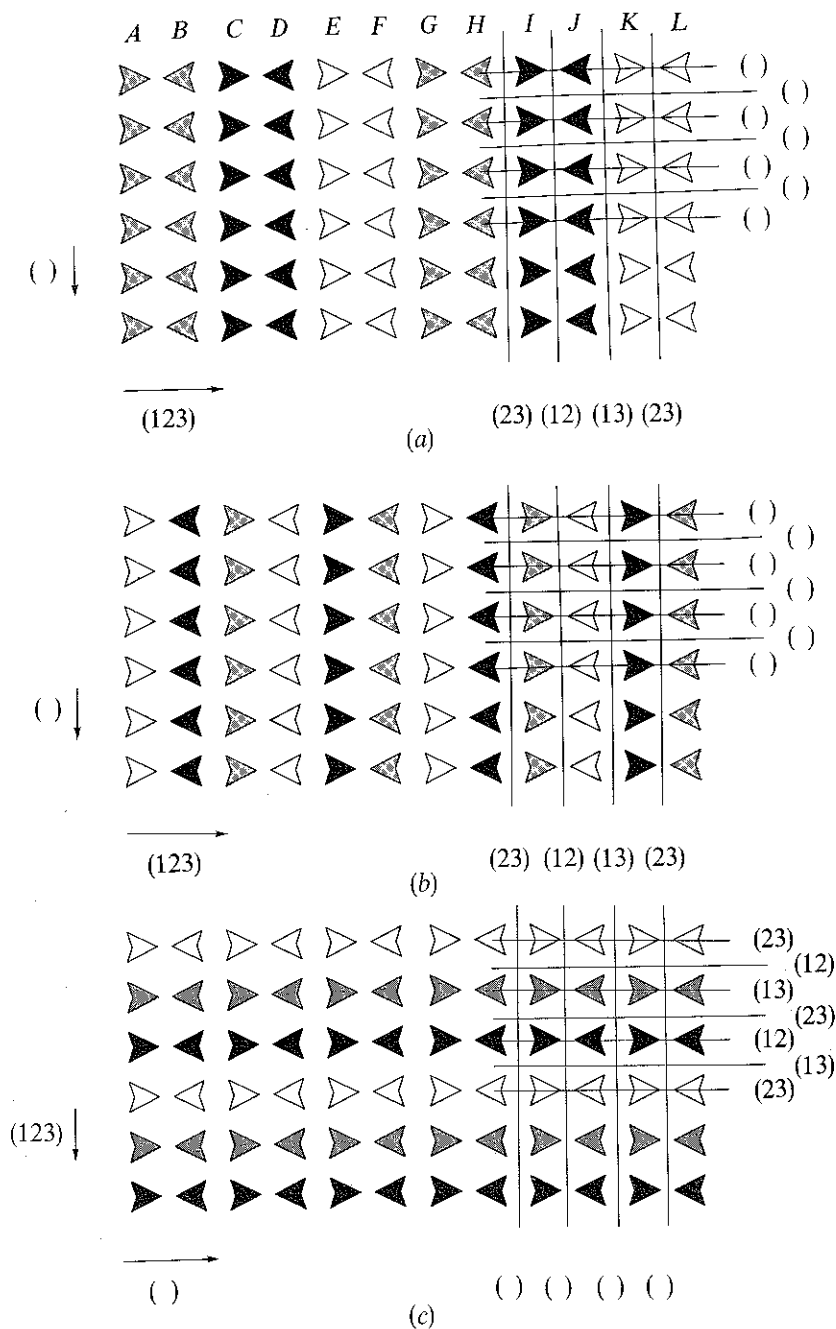


Figure 8.3.1
Examples of perfectly colored 3-chromatic patterns with color group $pmm[3]$ and underlying pattern of type PP15. The patterns of (a) and (b) have the same 3-color pattern type PP15[3], while that of (c) has a different 3-color pattern type, namely PP15[3]*. The meaning of the letters above the patterns in parts (a) and (b) is explained in the text.

of $C(\mathcal{P}_1)$ and $C(\mathcal{P}_2)$ are the same, or else use the alternative definition given above; we define Ψ to be the one-to-one correspondence that maps the motifs in each column of Figure 8.3.1(a) into the motifs in the column of Figure 8.3.1(b) that carries the same letter $A, B, C, \dots$ at its head. It is easy to verify that Ψ and its inverse are compatible with the groups $S_C(\mathcal{P}_1)$ and $S_C(\mathcal{P}_2)$, respectively. The fact that the colorings in Figures 8.3.1(a) and (b) "look different" is a motivation for some of the finer classifications to be introduced later.

As a second example consider the 17 strip and 46 periodic perfectly 2-colored patterns in Figure 8.2.2 and

the 7 strip and 23 periodic perfectly 3-colored patterns in Figure 8.2.3. The fact that the color symmetry groups of all these are different implies that they are also of different color pattern types. We shall refer to these patterns by the symbols marked near each. It is immediate that these are the only k -color pattern types for $k = 2$ or $k = 3$ with discrete and primitive underlying patterns.

Consider now the two 2-chromatic patterns in Figure 8.3.2. They have the same underlying pattern of type PP15, their 2-color symmetry groups are both $pmm[2]_4$ and their induced groups are both $d1$ with no color

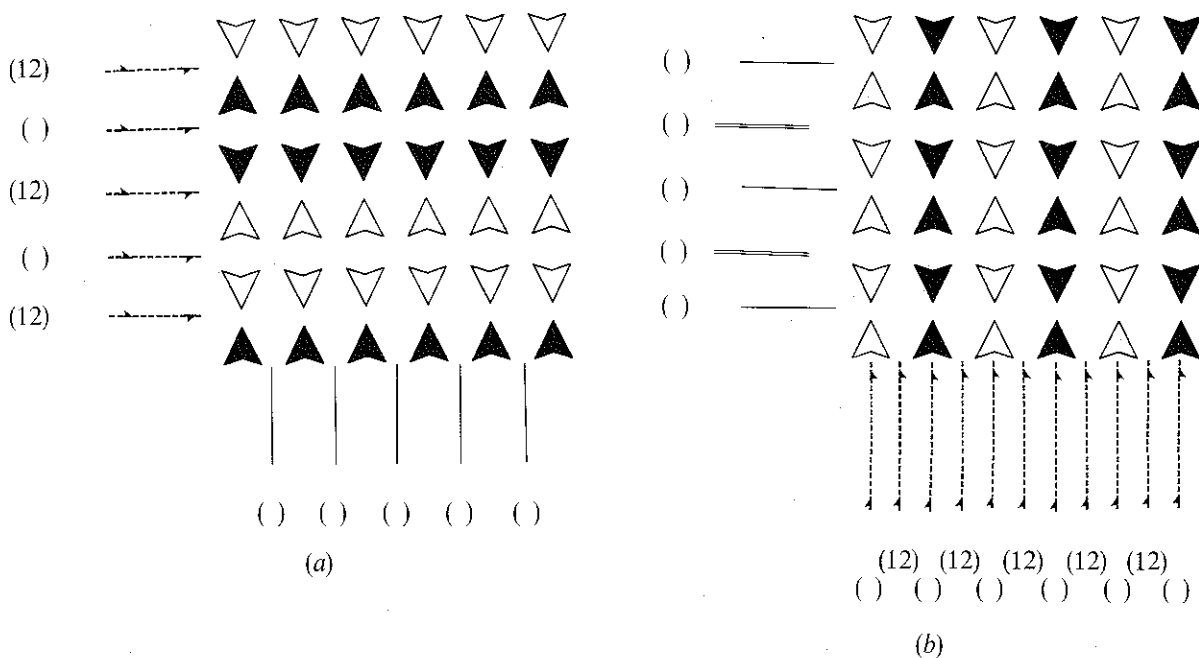


Figure 8.3.2

Two 2-chromatic colorings of a pattern of type PP15. They have the same color symmetry group $pmm[2]_4$ and the same induced group ($d1$ with no color changes), but are of different 2-color types since the sets of color subgroups that act transitively on the motifs are unequal. For example the pattern in (a) admits only one such subgroup of type $pmg[2]_1$ (indicated on the diagram by the lines of reflection and glide-reflection) while the pattern in (b) admits two such motif-transitive subgroups. (The lines of reflection of one of these are indicated by single lines and of the other by double lines; the glide-reflections are the same for both.)

changes. However they are of different 2-color pattern types since the sets of motif-transitive subgroups of their 2-color symmetry groups are different. To see this it is enough to verify that although both admit motif-transitive subgroups of type $pmq[2]_1$, the pattern in Figure 8.3.2(a) admits just one such group, while that in Figure 8.3.2(b) admits two.

With non-perfect colorings (so that $S_C(\mathcal{P}) \neq S(\mathcal{P})$) new considerations arise. For example consider the five 2-colored patterns with underlying pattern of type PP4

shown in Figure 8.3.3. The 2-color symmetry groups of the (perfectly colored) patterns shown in Figures 8.3.3(a) (c) and (d) are $pm[2]_2$, $pm[2]_3$ and $pm[2]_5$, so they are of different color pattern types, namely $PP4[2]_2$, $PP4[2]_3$ and $PP4[2]_5$, respectively. They are also different from the 2-color pattern types shown in Figure 8.2.2 since here the induced groups are non-trivial. On the other hand the 2-colored patterns in Figures 8.3.3(b) and (e) are 2-chromatic with color symmetry groups $pm[2]_2$ and $pm[2]_5$, but consideration of their color symmetries

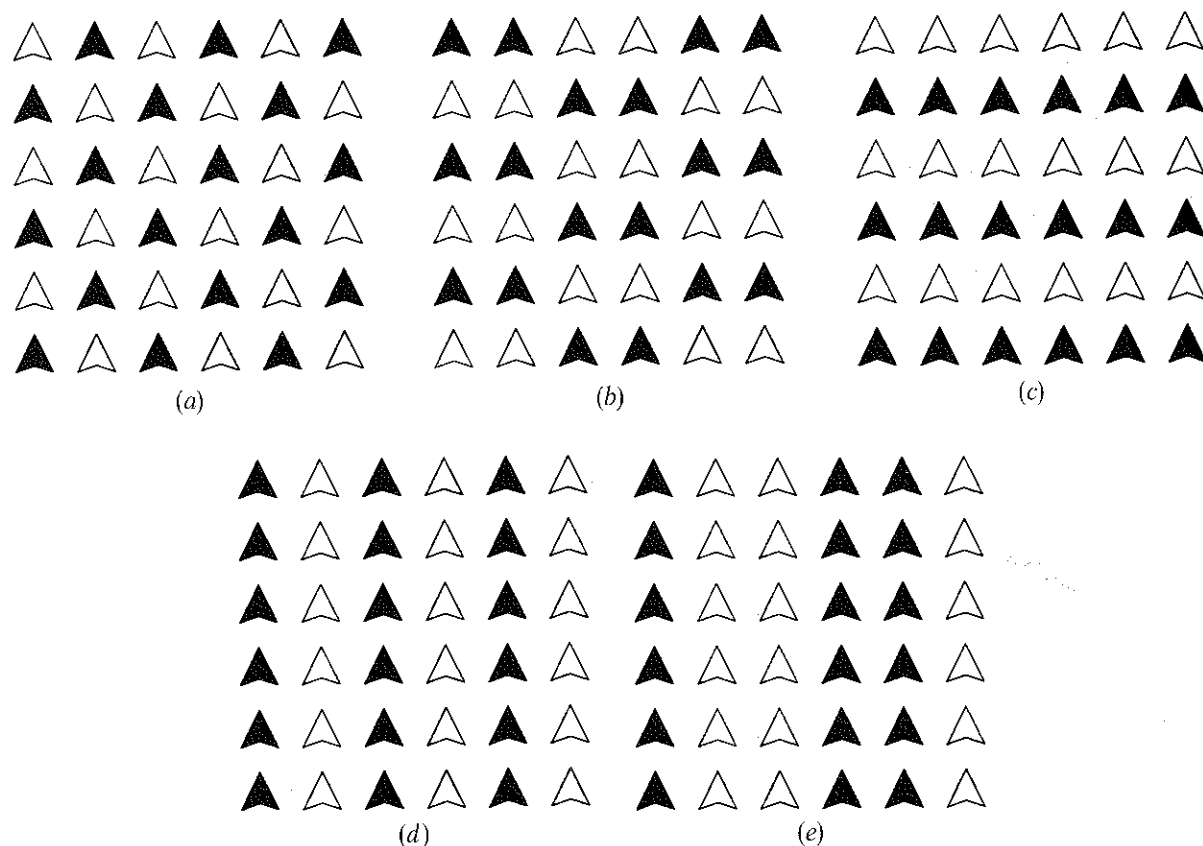


Figure 8.3.3

Examples of 2-chromatic patterns with underlying pattern of type PP4. The 2-color pattern types are (a) $PP4[2]_2$, (b) $PP3[2]_2$, (c) $PP4[2]_3$, (d) $PP4[2]_5$ and (e) $PP3[2]_5$.

reveals that they are of 2-color pattern types $PP3[2]_2$ and $PP3[2]_5$ —the types denoted by the same symbols in Figure 8.2.2. The reason for this phenomenon is that in Figures 8.3.3(b) and (e) the colorings are not perfect and in each case the underlying group $S_C(\mathcal{P})$ induces on the copies of the motif only the trivial group. In other words, although PP4 is not primitive, it behaves as if it were primitive with respect to the color symmetry groups of these two colorings.

Another example of a similar nature is shown in Figure 8.3.4. Here we show two patterns of the same 2-color pattern type whose underlying patterns are of different pattern types, namely PP16 and PP41. This situation can only arise if at least one of the patterns is not perfectly colored, and for this reason it is convenient, when enumerating k -color pattern types, to restrict attention to perfectly colored patterns.

One way to carry out such an enumeration is analogous to that mentioned in Section 5.2. We consider each of the primitive k -colored patterns of Figures 8.2.2 (for

$k = 2$) and 8.2.3 (for $k = 3$), and see whether it is possible to “move” the colored copies of the motif (in such a way that the color symmetry group is unchanged) until two or more copies “merge”. The various possibilities can be deduced from those for uncolored patterns (see Figure 5.2.5), though of course here we may merge two copies of the motif only if they are of the same color. If this procedure is carried out systematically we obtain the following:

8.3.1 *There exist 28 strip and 88 periodic 2-color pattern types.*

8.3.2 *There exist 15 strip and 59 periodic 3-color pattern types.*

Examples of patterns of all these types not already illustrated in Figures 8.2.2 and 8.2.3 appear in Figures 8.3.5 and 8.3.6. It seems that the number of k -color pattern types with $k \geq 4$ is unknown.

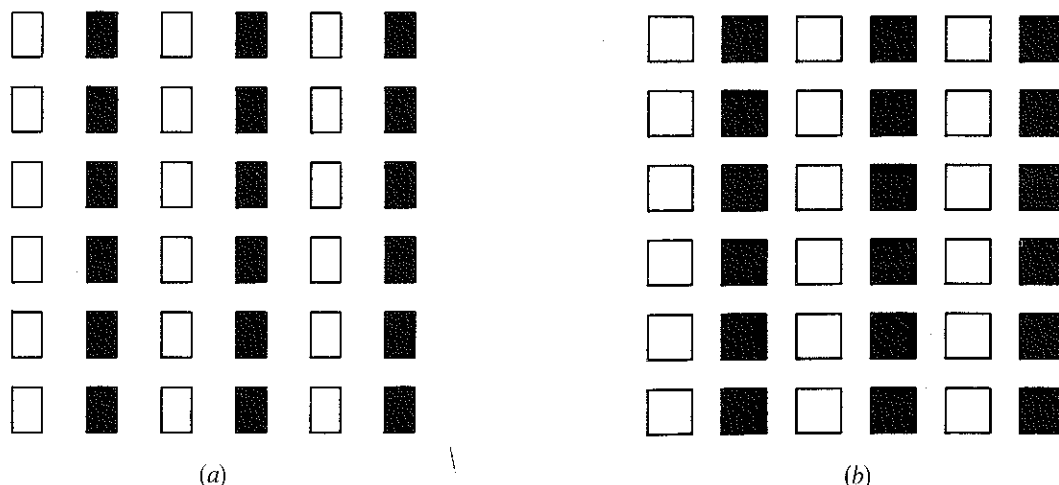


Figure 8.3.4

Two examples of 2-chromatic patterns of the same 2-color pattern type $PP16[2]_1$ with underlying discrete patterns of different types (PP16 in (a) and PP41 in (b)). The pattern in (a) is perfectly colored, that in (b) is not.

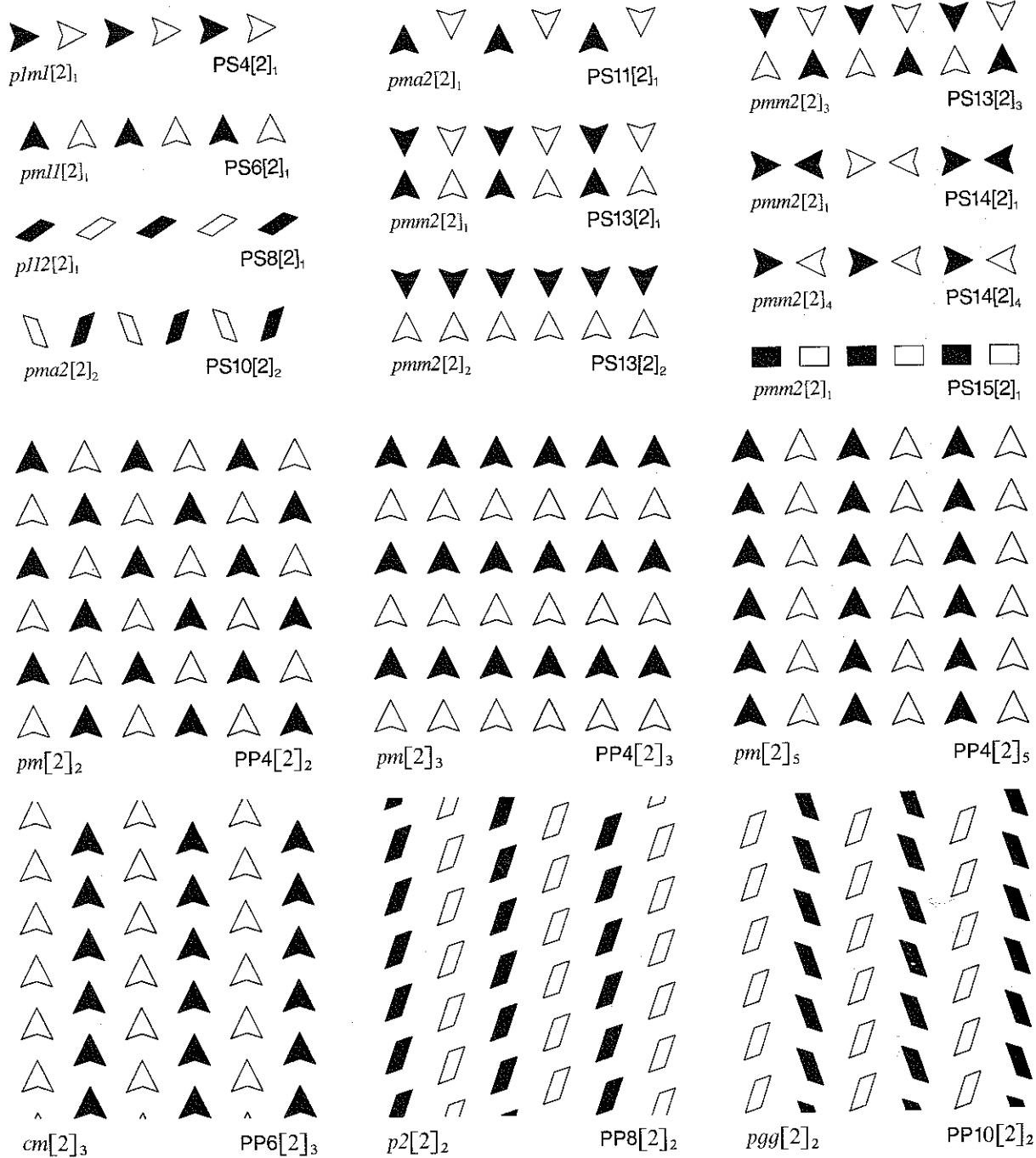
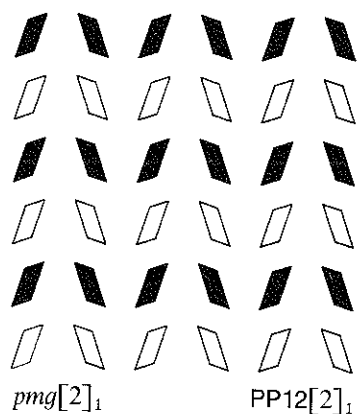
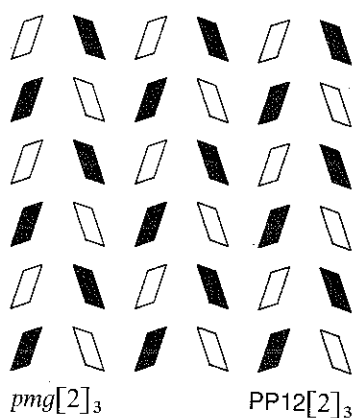


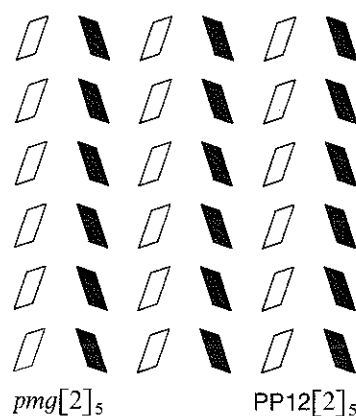
Figure 8.3.5
Examples of all the different 2-color pattern types (11 strip and 42 periodic) whose underlying patterns are non-primitive.



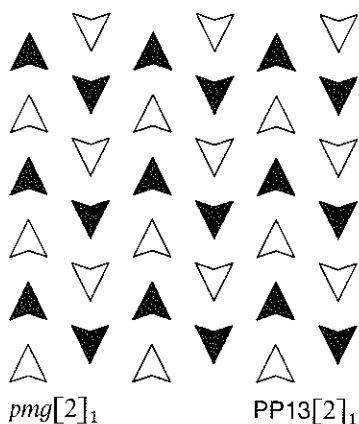
PP12[2]₁



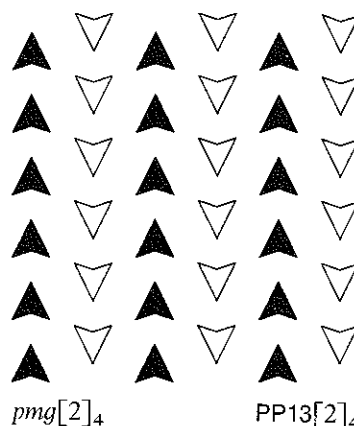
PP12[2]₃



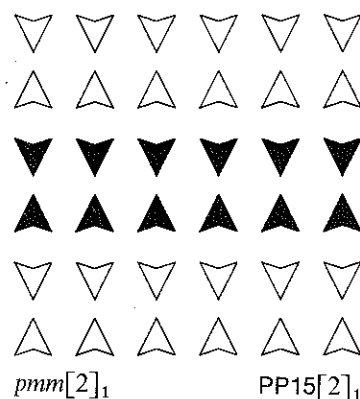
PP12[2]₅



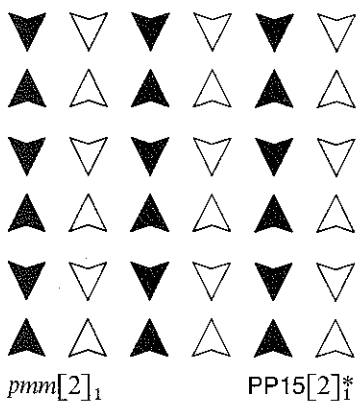
PP13[2]₁



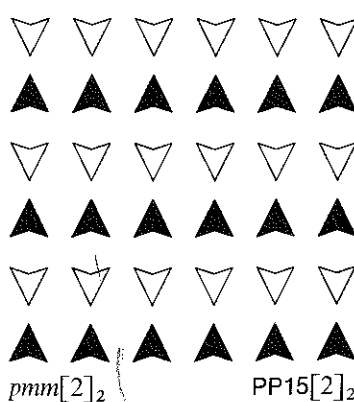
PP13[2]₄



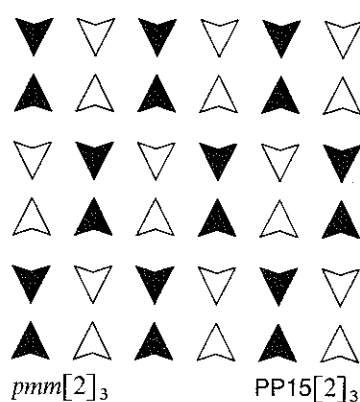
PP15[2]₁



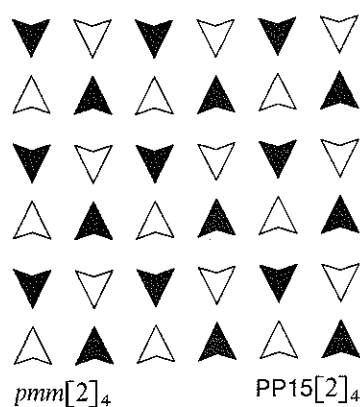
PP15[2]₁*



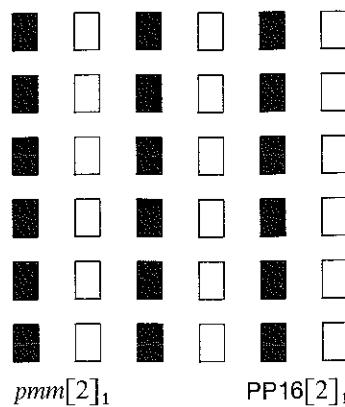
PP15[2]₂



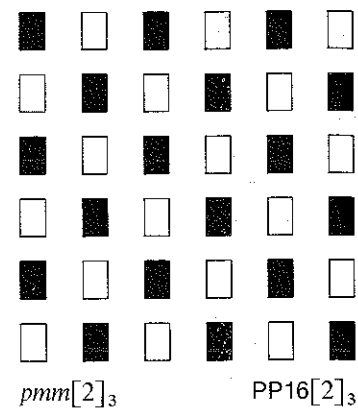
PP15[2]₃



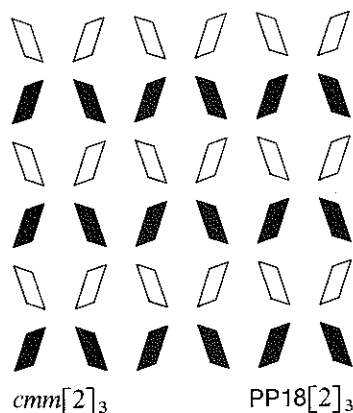
PP15[2]₄



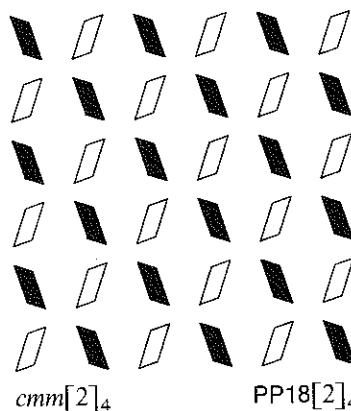
PP16[2]₁



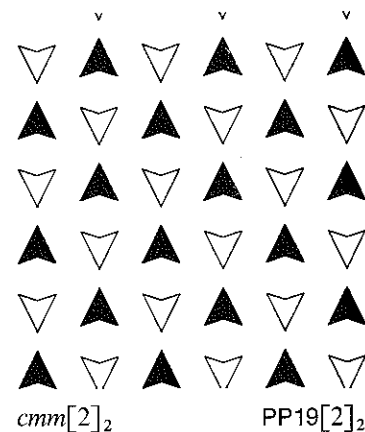
PP16[2]₃



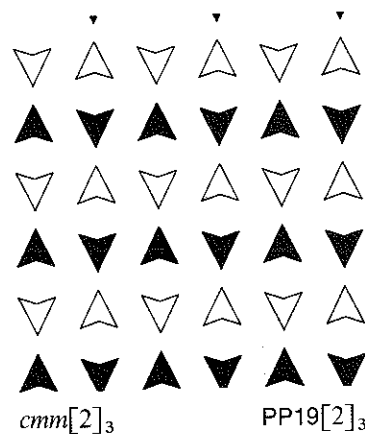
PP18[2]₃



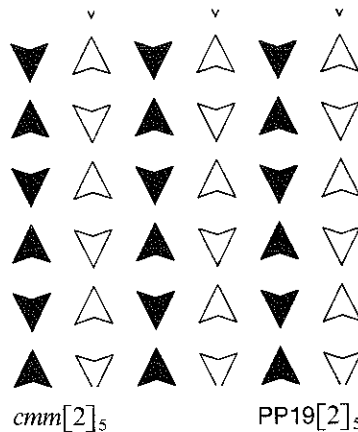
PP18[2]₄



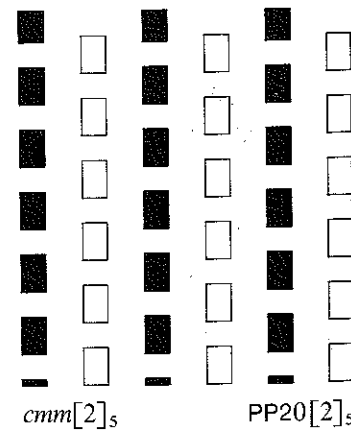
PP19[2]₂



PP19[2]₃

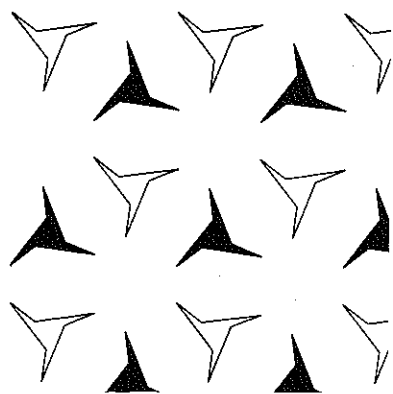


PP19[2]₅



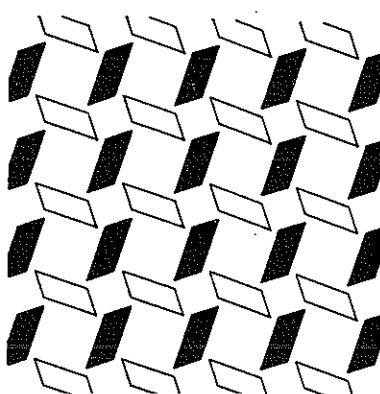
PP20[2]₅

Figure 8.3.5 (continued)



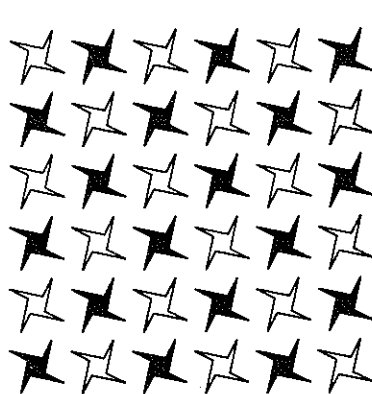
$p31m[2]$

PP24[2]



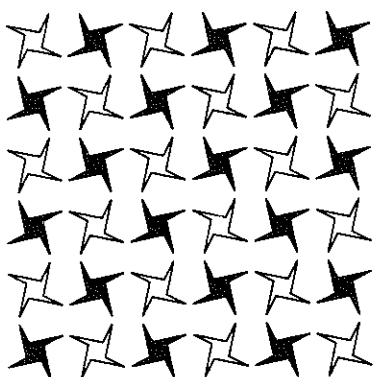
$p4[2]_2$

PP31[2]₂



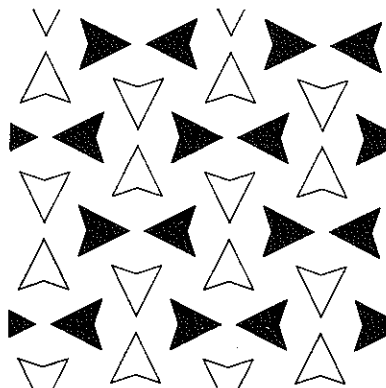
$p4[2]_1$

PP32[2]₁



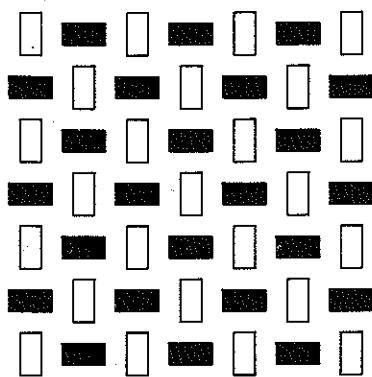
$p4g[2]_1$

PP34[2]₁



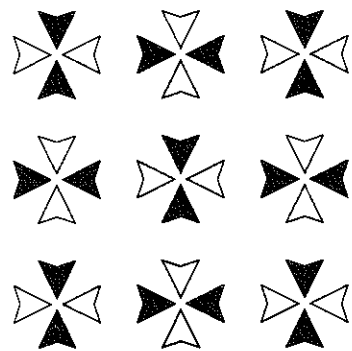
$p4g[2]_2$

PP35[2]₂



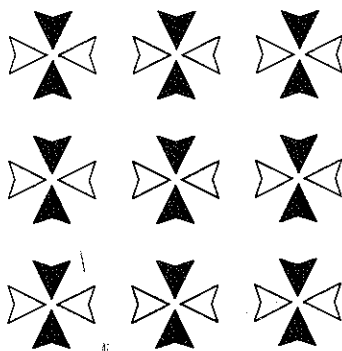
$p4g[2]_2$

PP36[2]₂



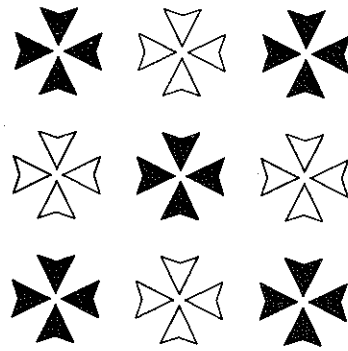
$p4m[2]_1$

PP38[2]₁



$p4m[2]_4$

PP38[2]₄



$p4m[2]_5$

PP38[2]₅

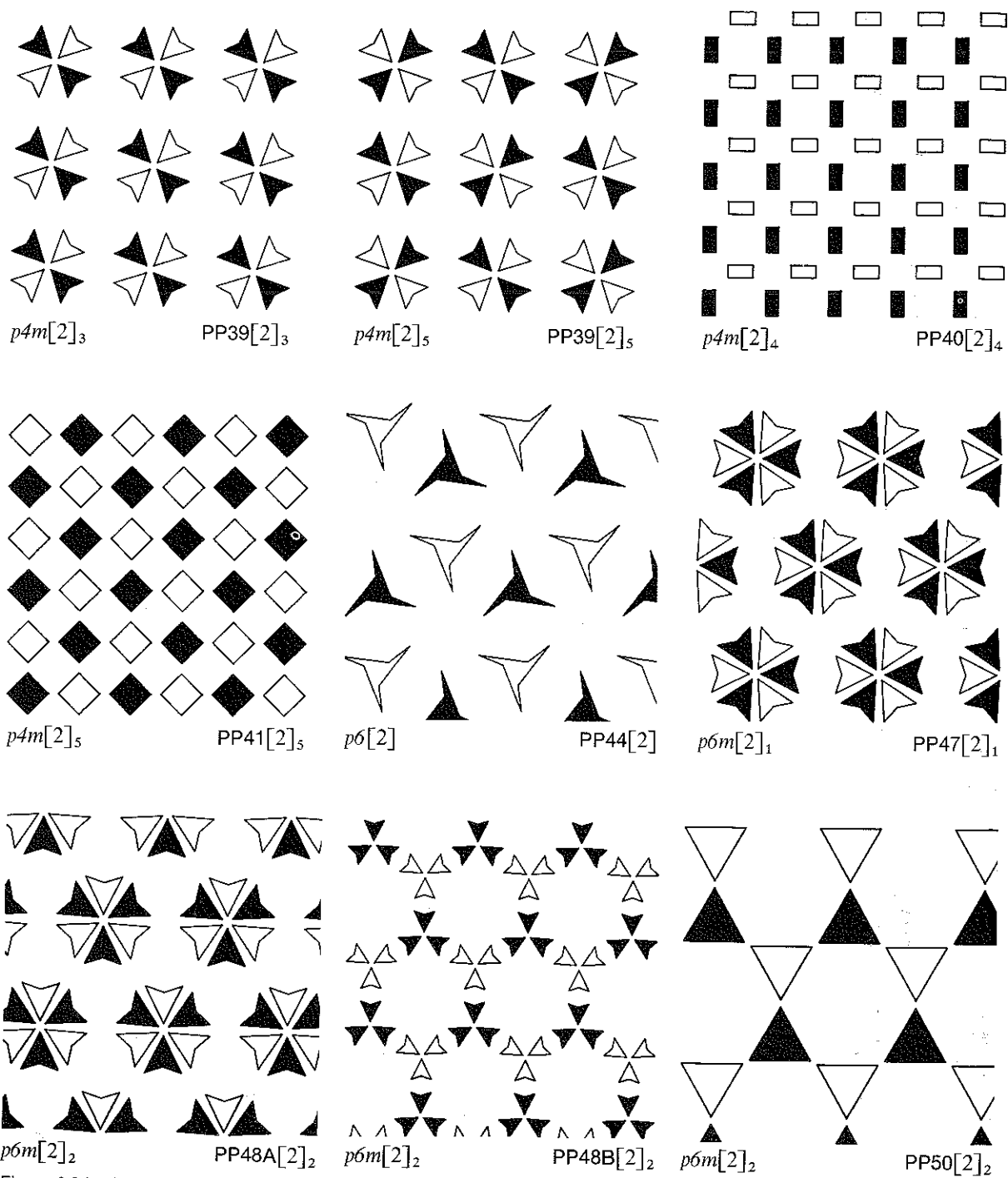


Figure 8.3.5 (continued)

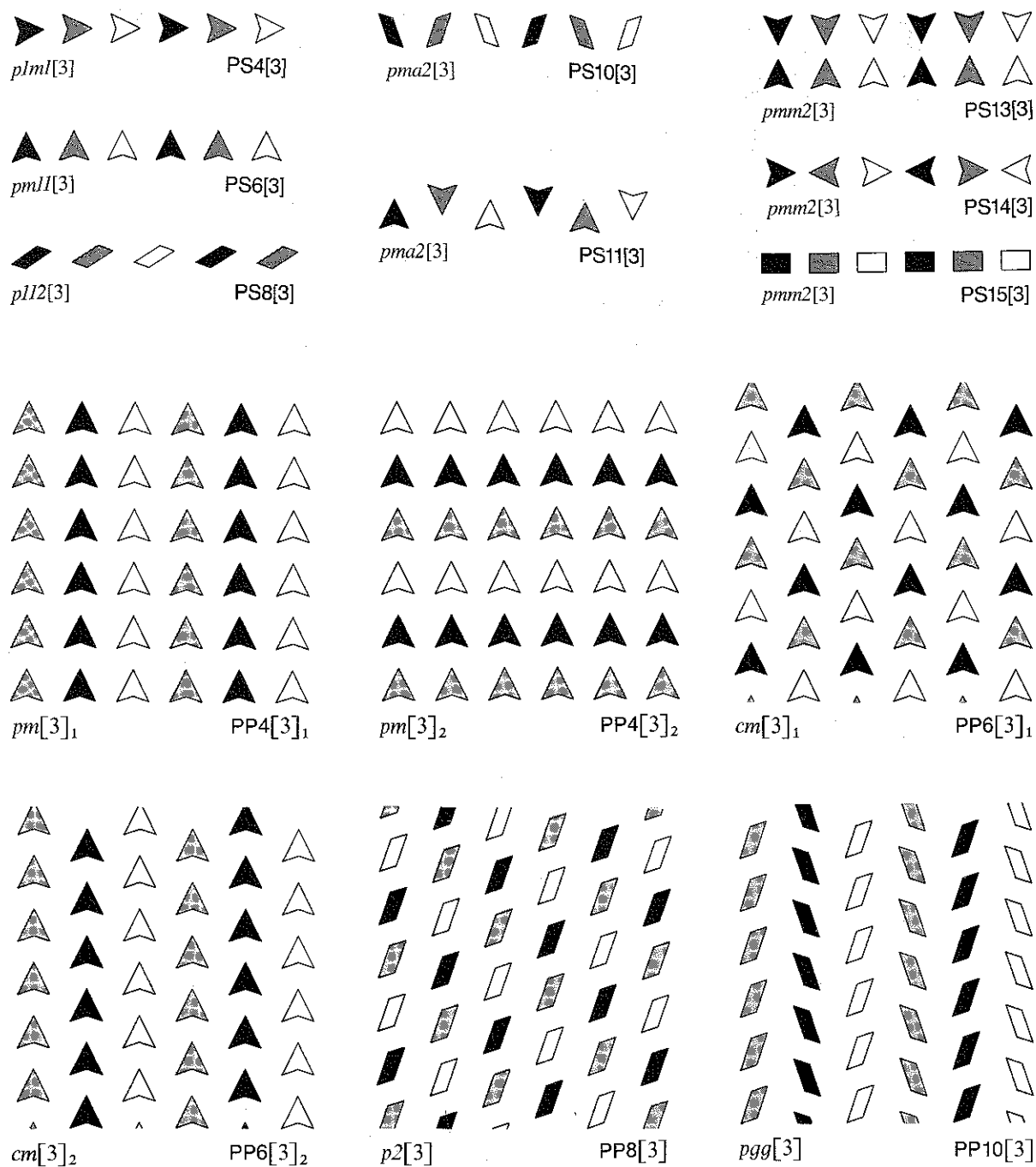
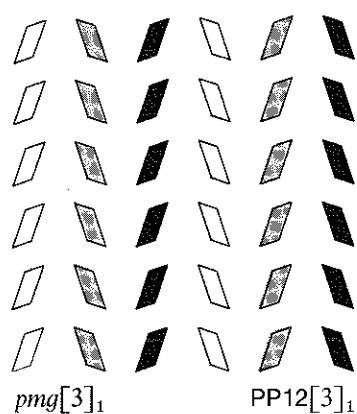
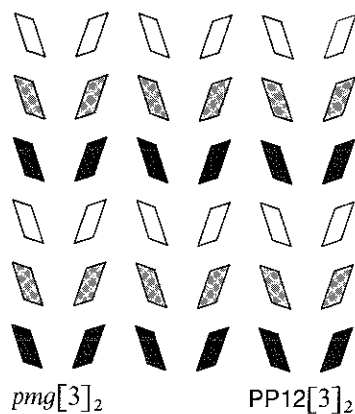


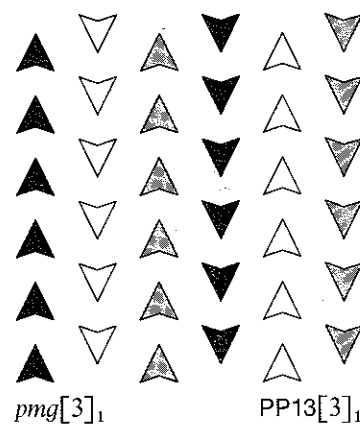
Figure 8.3.6
Examples of all the different 3-color pattern types (8 strip and 36 periodic) whose underlying patterns are non-primitive.



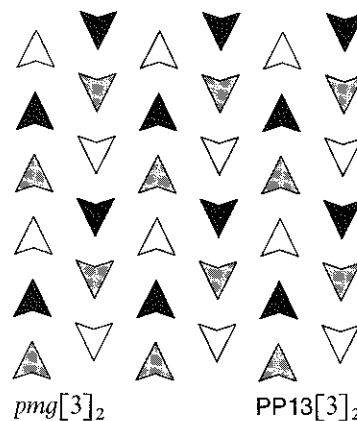
PP12[3]₁



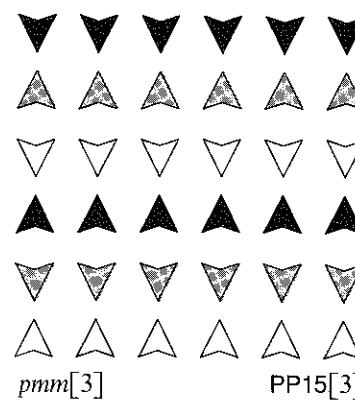
PP12[3]₂



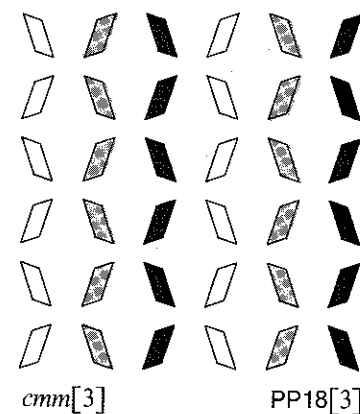
PP13[3]₁



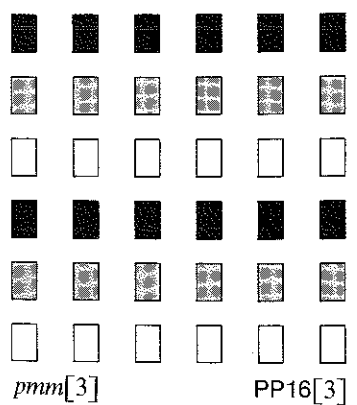
PP13[3]₂



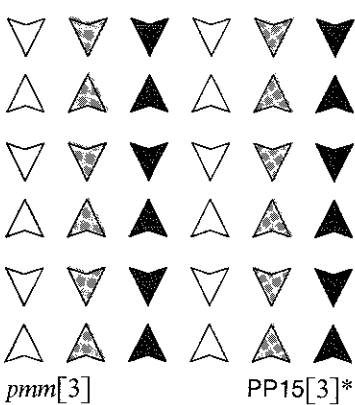
PP15[3]



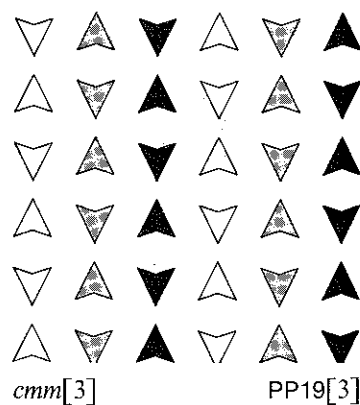
PP18[3]



PP16[3]

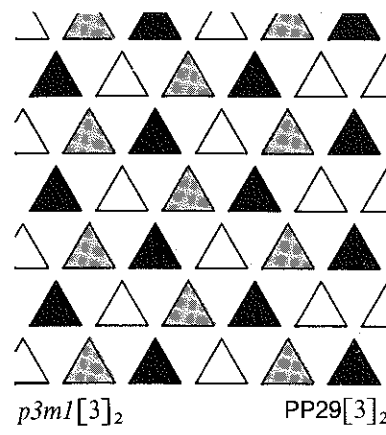
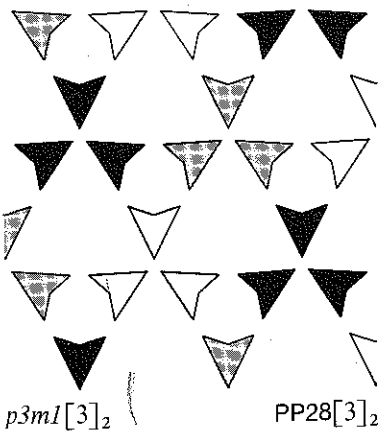
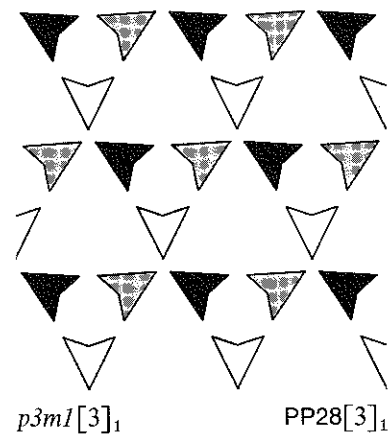
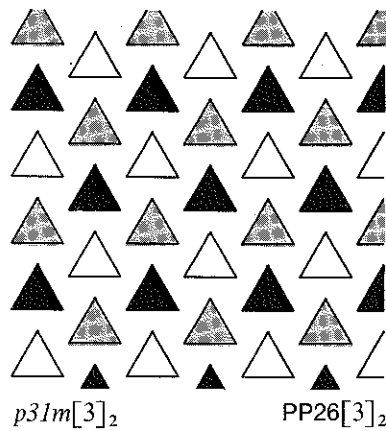
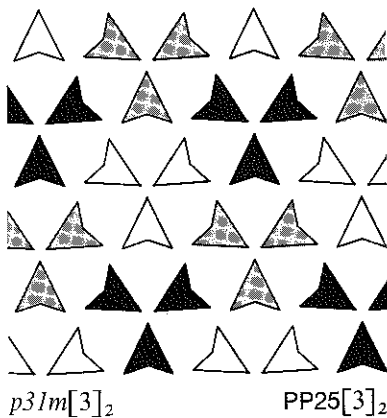
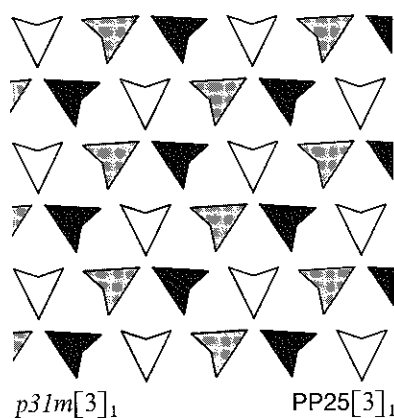
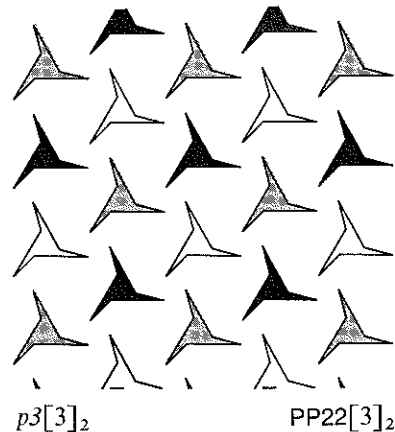
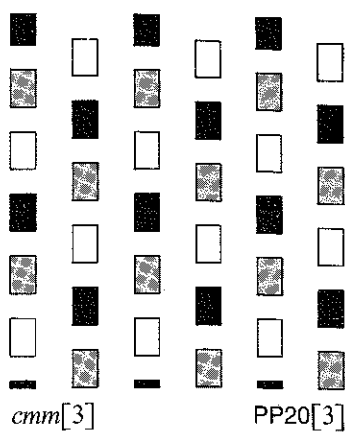
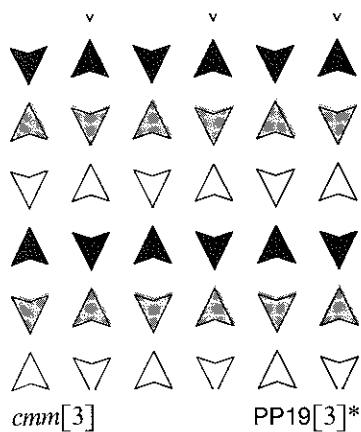


PP15[3]*



PP19[3]

Figure 8.3.6 (continued)



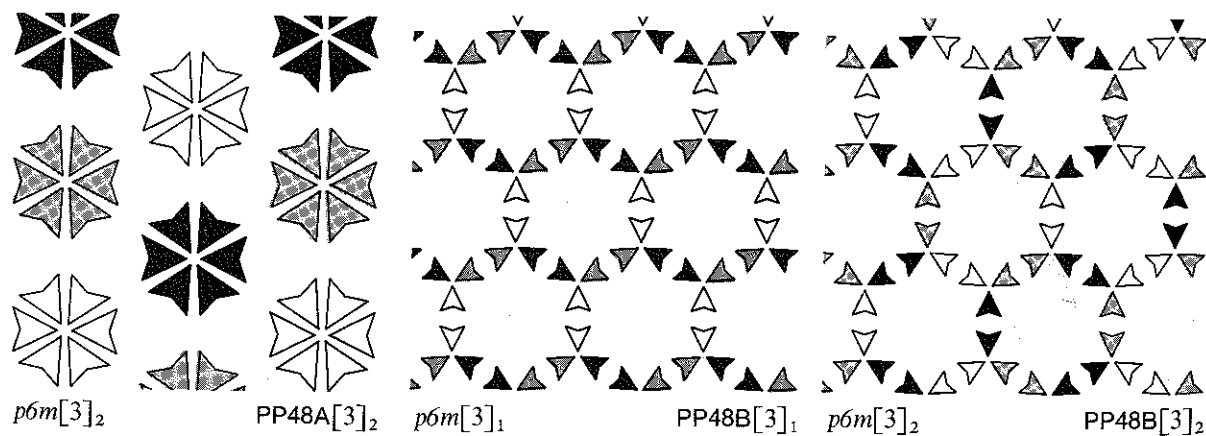
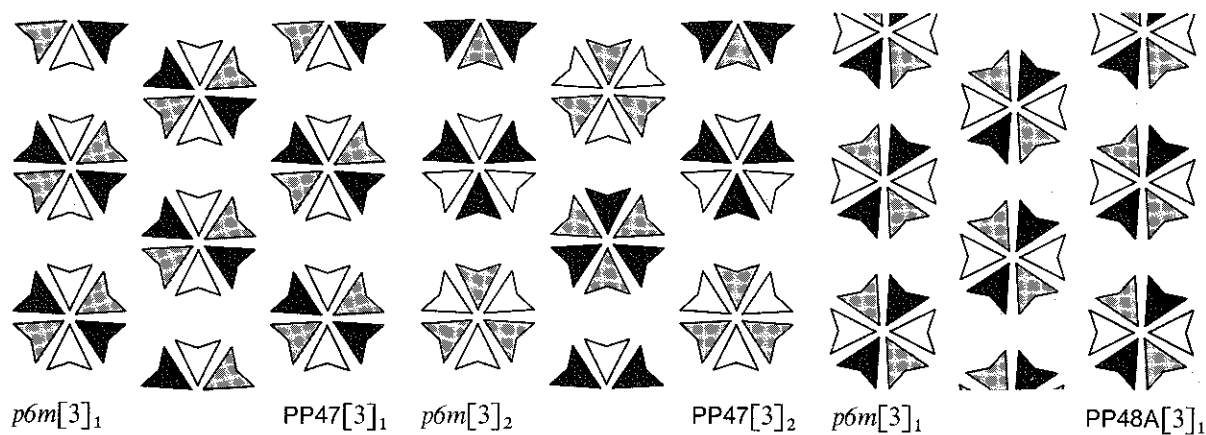
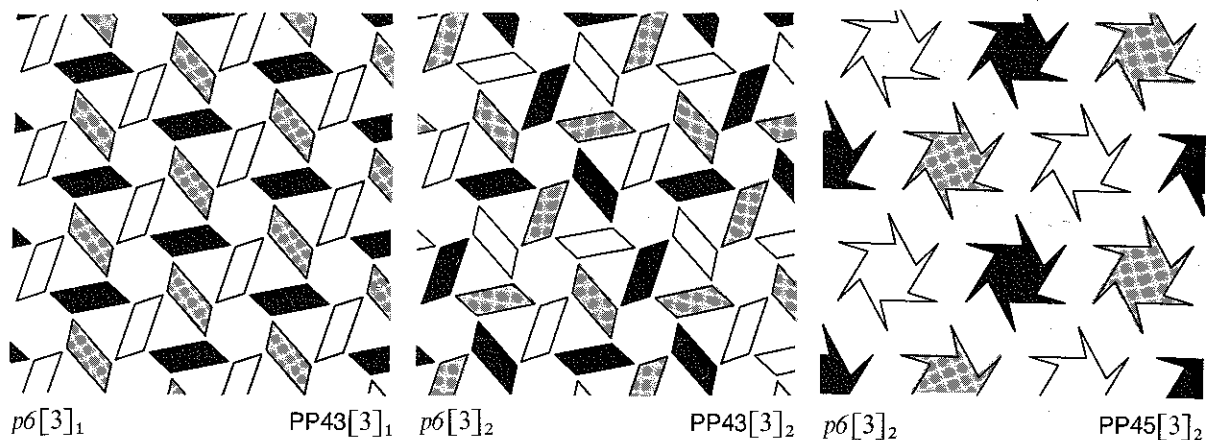
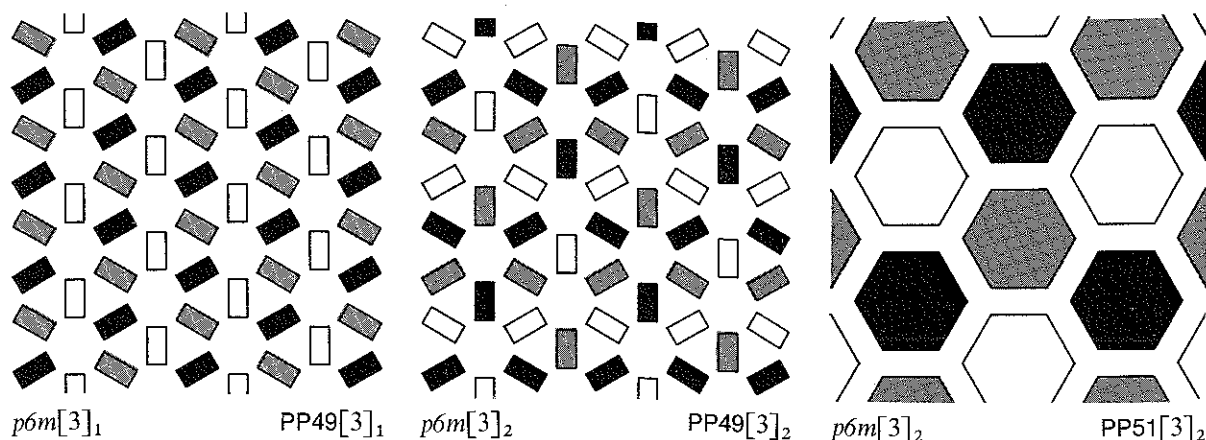


Figure 8.3.6 (continued)



EXERCISES 8.3

1. (a) Determine the 3-color pattern types of those tiling patterns in Figure 5.2.13 which are 3-chromatic.
 (b) For the tilings of Figure 5.2.13 consider the 2-colored patterns formed by the black and stippled tiles. For those which are 2-chromatic determine the 2-color pattern types.
2. For each of the 2-chromatic patterns of types PP38[2]₁, PP38[2]₄, PP38[2]₅, PP39[2]₃ and PP39[2]₅ (see Figure 8.3.5) determine all the colored-motif transitive subgroups of their color symmetry groups.
- *3. (a) Determine all the 4-color pattern types of strip patterns.
 (b) Determine all the k -color pattern types of strip patterns, where k is an odd integer, $k \geq 3$.

8.4 CHROMATIC COLORINGS OF GIVEN PATTERNS AND EQUICOLOR TYPES

The classification criteria we have considered in the previous sections have depended entirely on color groups—the patterns that appeared played a subordinate role. But there is both a practical and theoretical interest in taking the opposite point of view and asking the question “Given a pattern, in how many different ways can it be colored?”. For such a question to be meaningful we must restrict the sort of colorings allowed and also say precisely what we mean by “different ways”. The most

appropriate restriction on colorings appears to be to consider only those which are k -chromatic, and to interpret “different ways” as meaning “of different equicolor types”.

The definition of an *equicolor type* is similar to that of color pattern type given in Section 8.3 except that we ask for the color-preserving one-to-one correspondence Ψ to be compatible with *every* symmetry of $\mathcal{P}_1$ (and not just with the symmetries in the underlying group $S_C(\mathcal{P}_1)$) and for Ψ^{-1} to be compatible with every symmetry of $\mathcal{P}_2$. The classification by equicolor types is, of course, finer than that by color pattern types, as we can see from the examples in Figure 8.3.4. These are of different equi-

color types even though they are of the same color pattern type. In fact, it follows directly from the definition that if two colored patterns are of the same equicolor type then the underlying patterns must be of the same pattern type. In the case of perfectly colored patterns, since $S_C(\mathcal{P}) = S(\mathcal{P})$ the classifications by color pattern types and by equicolor types coincide.

To illustrate the above definition, consider the pattern $\mathcal{P}$ of type PP1 in Figure 8.4.1(a), and the 2-chromatic colorings of $\mathcal{P}$ shown in Figures 8.4.1(b), (c) and (d). These are equicolored as can be seen by taking for Ψ the color-preserving one-to-one correspondence indicated by the labels A, B, C on the copies of the motif. (In other words Ψ maps each copy of the motif onto one bearing the same label; it should be clear how to extend Ψ to the unlabelled copies of the motif.) The color symmetry groups of all these patterns are isomorphic; this isomorphism is defined by specifying that each vector labelled $\circ$ (which represents the identity so that the corresponding translation preserves colors) is mapped onto the vector with the same label, and similarly for the vectors labelled $\star$ (for which the corresponding translations interchange the colors). These pairs of vectors generate, and therefore completely define, the color symmetry groups of the patterns. Notice that differences between

these three colored patterns that arise from considering the acute and obtuse angles of the polygonal motifs are irrelevant as far as the classification into equicolor types is concerned.

Another example is given in Figure 8.4.2. Here we show five 2-chromatic colorings of a rectangle pattern of type PP20. As indicated in the caption, these are of four equicolor types. Here the classification by equicolor types coincides with that by color pattern type in spite of the fact that some of the patterns are not perfectly colored.

In order to determine the different equicolor types possible in a k -chromatic coloring of a given pattern $\mathcal{P}$ we can use the information on motif-transitive subgroups of $S(\mathcal{P})$ given in Section 5.2 together with (for $k = 2$ or $k = 3$) the results on k -color pattern types given in Sections 8.2 and 8.3. To carry out a similar determination for $k \geq 4$ it would be necessary to list the 4-color pattern types first. We illustrate the procedure by a typical example.

Let $k = 2$ and let the pattern $\mathcal{P}$ be of type PP32, see Figure 8.4.3(a). From Table 5.2.3 we see that $S(\mathcal{P})$ has six different motif-transitive proper subgroups. Five of these are indicated by their symmetry elements in Figures 8.4.3(b) (c) (d) (e) and (f). With respect to these

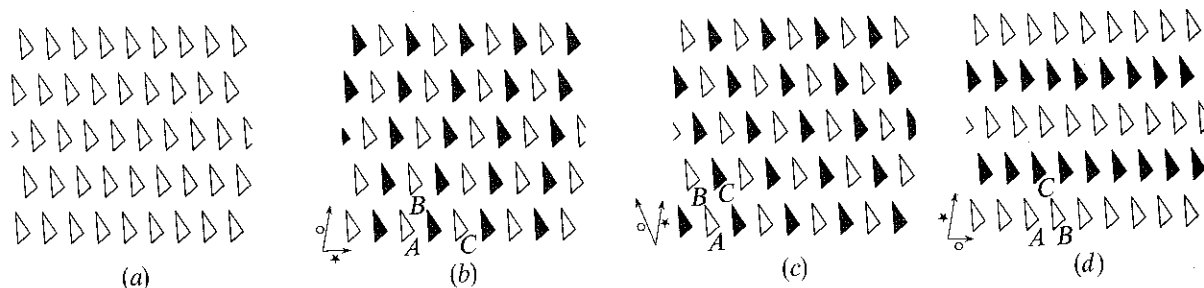


Figure 8.4.1

The pattern $\mathcal{P}$ in (a) is of pattern type PP1, and the three 2-chromatic colorings of $\mathcal{P}$ shown in (b), (c) and (d) are of 2-color pattern type PP1[2]. These three are distinguishable by the properties of their motifs such as the positions of their acute and obtuse angles relative to the translational color symmetries. The color symmetry group in each case is generated by a color-preserving translation, marked $\circ$, and a color-reversing translation marked $\star$. Consequently all three 2-color patterns are of the same equicolor type.

groups, $\mathcal{P}$ may be considered to be a pattern of types PP31, PP30, PP8, PP7 and PP7, respectively. The sixth motif-transitive proper subgroup of $S(\mathcal{P})$, namely $p1$, does not lead to any additional types since the motif of $\mathcal{P}$ has symmetry group $c4$ which is a forbidden supergroup for PP1; hence it will not be considered in the remainder of the discussion.

The next step is to consider all the perfect 2-colorings

of each of the pattern types PP32, PP31, PP30, PP8 and PP7 and to find all the ways in which each can be applied to the patterns of Figure 8.4.3(a) to (f). From Figure 8.4.3(a) we are thus led to the unique 2-coloring shown in Figure 8.4.3(g) which is of 2-color pattern type PP32[2]₁. The pattern in Figure 8.4.3(b) can be colored as in 2-color pattern type PP31[2]₂, but the resultant colored pattern has additional symmetries and coincides with

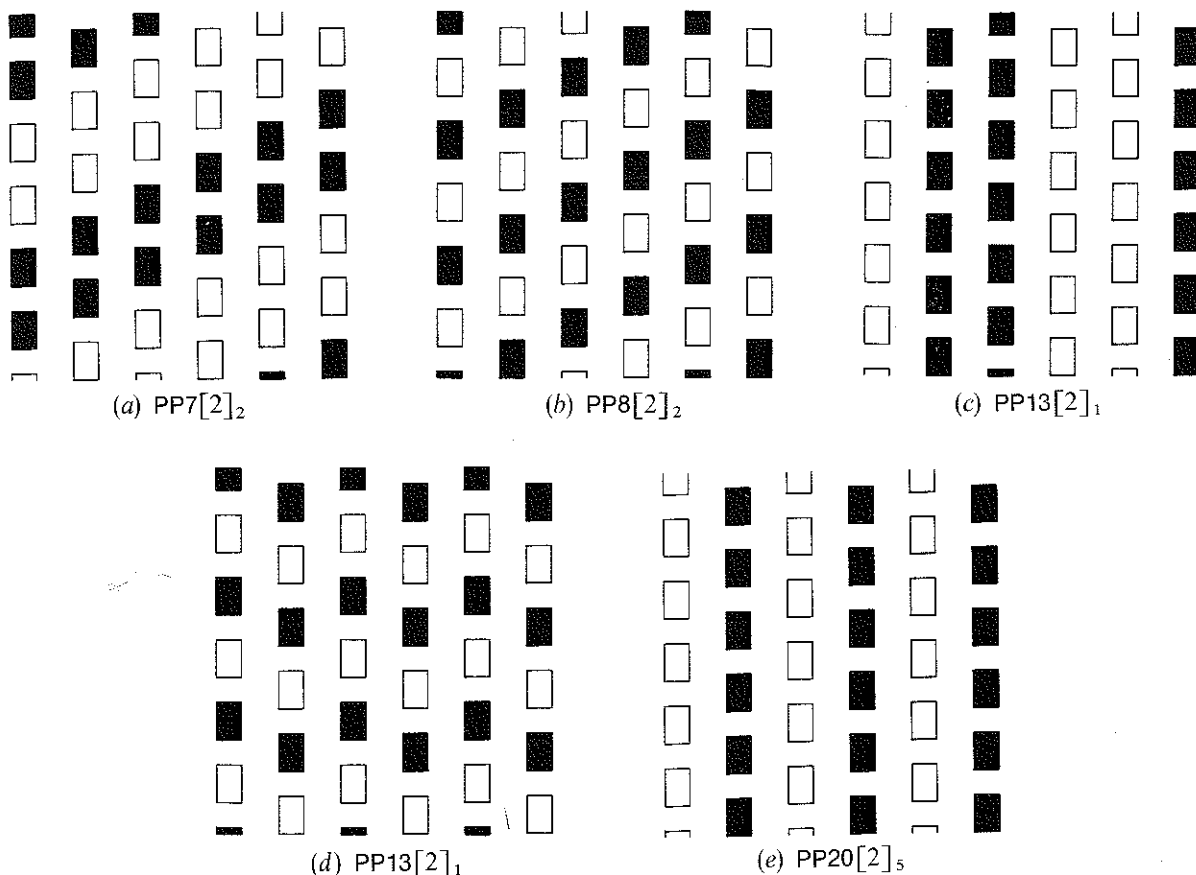


Figure 8.4.2

Five 2-chromatic colorings of a pattern of type PP20. The 2-color pattern type is indicated near each. All are of different equicolor types except for (c) and (d) which are equicolored.

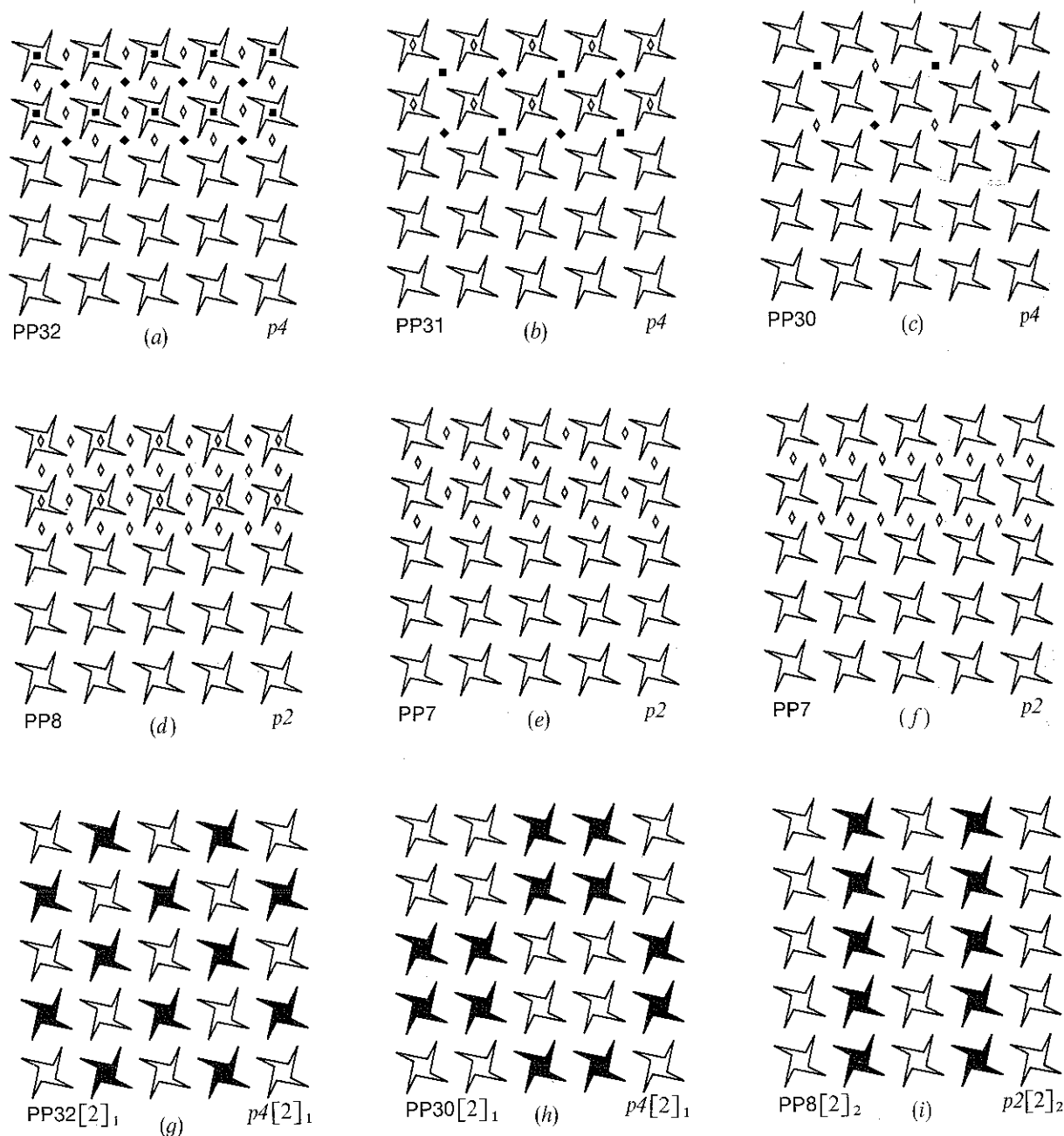
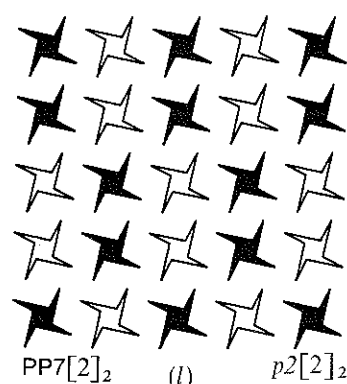
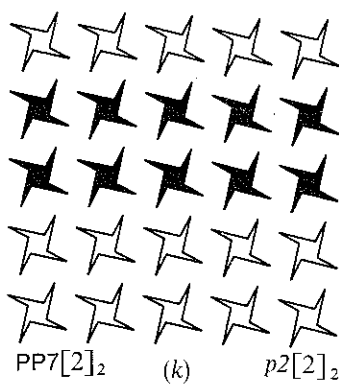
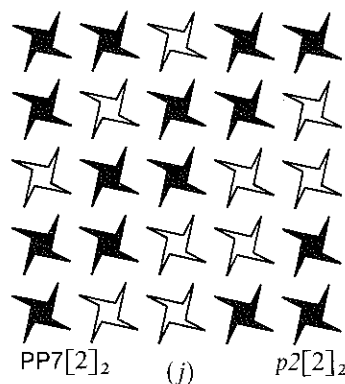


Figure 8.4.3

A pattern $\mathcal{P}$ of type PP32 is shown in (a) and the motif-transitive proper subgroups of $S(\mathcal{P})$ are shown in (b), (c), (d), (e) and (f). Near each we indicate both the subgroup and the pattern type that results when we restrict the symmetry group of the pattern to the subgroup under consideration. The last six parts of the figure (parts (g) to (i)) show the equicolor types of 2-colorings of $\mathcal{P}$. Near each we have indicated its 2-color pattern type and its 2-color group.



that of Figure 8.4.3(g). The pattern in Figure 8.4.3(c) can be colored either as $PP30[2]_1$ or $PP30[2]_2$; the first leads to the colored pattern of Figure 8.4.3(h), while the second again yields that of Figure 8.4.3(g). For the pattern in Figure 8.4.3(d) we can apply only the coloring of type $PP8[2]_2$ —but this can be done in two ways that are inequivalent; either the centers of 2-fold rotation at A and B are color-reversing, or else those at A and C are. In the former case we obtain the pattern of Figure 8.4.3(i) while in the latter we again obtain that of Figure 8.4.3(g). For the patterns in Figures 8.4.3(e) and (f) colorings of types $PP7[2]_1$ and $PP7[2]_2$ are applicable. For Figure 8.4.3(e) the coloring $PP7[2]_1$ again leads to the pattern shown in Figure 8.4.3(g), while coloring $PP7[2]_2$ can be applied in two distinct ways. One leads to the pattern of Figure 8.4.3(j) and the other leads to the pattern $PP8[2]_2$

of Figure 8.4.3(i). Coloring the pattern of Figure 8.4.3(f) by type $PP7[2]_1$ leads to the pattern $PP8[2]_2$ of Figure 8.4.3(i), while of the four distinct colorings by type $PP7[2]_2$ two lead again to patterns $PP32[2]_1$ and $PP8[2]_2$ of Figures 8.4.3(g) and (i) and the last two yield two new patterns shown in Figures 8.4.3(k) and (l).

Thus we obtain in all six different equicolor types of 2-chromatic colorings of patterns of type $PP32$. Contrast this with the fact that there is only one 2-color pattern type with underlying pattern $PP32$, namely $PP32[2]_1$.

Repeating a similar process for each pattern type one could obtain a complete list of equicolor types of k -chromatic colorings of discrete strip and periodic patterns. So far as we are aware, this enumeration of equicolor types, while completely straightforward, has never been carried out even for $k = 2$.

EXERCISES 8.4

1. For the 2-chromatic patterns considered in Exercise 8.3.1(b) determine which pairs are of the same equicolor type.
2. Determine all equicolor types of 2-chromatic and 3-chromatic colorings of dot patterns of types $DPP47$ and $DPP48$.
- *3. Determine all equicolor types of 2-chromatic and 3-chromatic colorings of strip patterns.
- *4. Find all equicolor types of 4-chromatic colorings of the coherent strip patterns that have an open line-segment as motif.
- *5. Find all equicolor types of 4-chromatic colorings of the regular square tiling.

8.5 COLOR-HOMEOMERISM AND HOMECHROMATIC TYPES

In Chapter 7 we explained how the classification of (uncolored) patterns by "pattern types" can be refined using homeomerism. In the case of colored patterns similar considerations apply. In this section we shall introduce two new methods of classification—into color-homeomorphic types and into homeochromatic types. Each of these is finer than the methods described in the previous sections of this chapter since it takes into account the topological structure of the patterns. Unlike the methods discussed earlier, the classifications to be described here yield an infinite number of types. For this reason they are too fine for many purposes. However, they seem to be most appropriate (in that they fit in with our intuitive ideas of when two patterns are "the same" or not) in two situations: when investigating colored patterns whose motifs are simple geometrical shapes (such as circular or elliptical disks or line segments) and in the study of colored tilings. The first of these topics will be referred to in the exercises—the number of types is large but finite and, so far as we are aware, has in no case been determined. The second topic, that of colored tilings, will be considered in some detail in the next section.

Let $\mathcal{P}_1$ and $\mathcal{P}_2$ be two colored patterns. If, possibly after permuting the colors in one of the patterns, we can find a color-preserving homeomorphism $h: E_1 \rightarrow E_2$ which maps $\mathcal{P}_1$ into $\mathcal{P}_2$, is compatible with every symmetry in $S_C(\mathcal{P}_1)$ and is such that h^{-1} is compatible with every symmetry in $S_C(\mathcal{P}_2)$, then $\mathcal{P}_1$ and $\mathcal{P}_2$ are said to be *color-homeomorphic*.

The definition of the term *homeochromatic* is analogous: we specify that h must be compatible with every symmetry of $S(\mathcal{P}_1)$ and h^{-1} compatible with every symmetry of $S(\mathcal{P}_2)$ (that is, with the symmetry groups of the underlying patterns $\mathcal{P}_1$ and $\mathcal{P}_2$). Intuitively* we may think of two patterns as being of the same homeochromatic type if h , besides mapping $\mathcal{P}_1$ onto $\mathcal{P}_2$ also maps a group diagram for $S(\mathcal{P}_1)$ onto a group diagram for $S(\mathcal{P}_2)$ (and vice versa for h^{-1}). They are of the same color-homeomorphic type if the analogous condition holds for the color-group diagrams of $C(\mathcal{P}_1)$ and $C(\mathcal{P}_2)$.

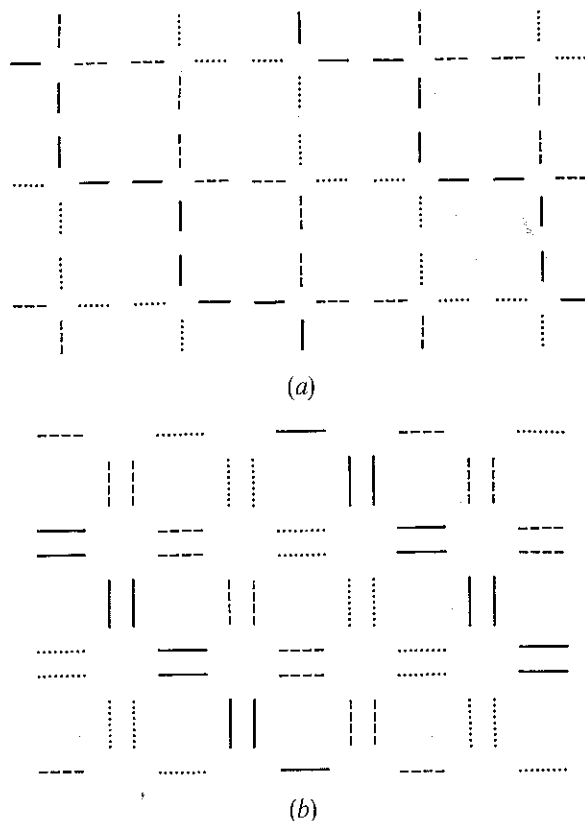


Figure 8.5.1
Two 3-chromatic colorings of line-segment patterns of type PP38. The colored patterns are of the same 3-color pattern type PP17[3] and are even color homeomorphic. However the underlying patterns are not homeomorphic.

These definitions have two immediate consequences. The first is that the classifications by color homeomerism and by homeochromatic type coincide when perfectly colored patterns are under consideration for then $S_C(\mathcal{P}) = S(\mathcal{P})$. The second is that when two patterns are homeochromatic then the existence of a color-preserving homeomorphism, as required in the definition, implies that the underlying patterns are homeomorphic. On the other hand, the corresponding statement is not true for color-homeomorphic patterns. In Figure 8.5.1 we show two patterns which are of the same color-pattern type PP17[3], are equicolored and are even color-

* This interpretation is subject to the same qualifications as those expressed on page 336.

homeomeric, yet the underlying patterns are not homeomeric, being of types LPP38-1 and LPP38-2 in Table 7.4.1 (see Figures 7.4.1(a) and (b)). This kind of situation can only arise when at least one of the patterns is not perfectly colored; here $S_c(\mathcal{P}_1)$ and $S_c(\mathcal{P}_2)$ are of type cmm while $S(\mathcal{P}_1)$ and $S(\mathcal{P}_2)$ are of type $p4m$. Another example where color-homeomeric patterns have underlying patterns of different homeomeric types will be given in Figure 8.6.1 in the context of tiling patterns.

The above definitions are further illustrated in Figure 8.5.2. Fourteen patterns are shown, each with the interior of a quadrangle as motif, and each is perfectly colored so the classifications by color-homeomeric types and by homeochromatic types coincide. By considering the homeomeric types of the underlying (uncolored) patterns we can divide the fourteen patterns into five sets as follows:

The pattern of Figure 8.5.2(a).

The patterns of Figures 8.5.2(b), (c) and (d).

The patterns of Figures 8.5.2(e), (f), (g) and (h).

The patterns of Figures 8.5.2(i), (j) and (k).

The patterns of Figures 8.5.2(l), (m) and (n).

Clearly no pattern in one of these sets is homeochromatic to any pattern in another set.

The pattern in the first set (Figure 8.5.2(a)) is of color-pattern type $PP6[3]_2$ while the thirteen patterns in the other four sets are all of type $PP2[3]_2$.

In the second set it is easy to see that the patterns of Figures 8.5.2(b) and (c) are homeochromatic since a slight distortion of the motif suffices to convert one into the other and this indicates how a color-preserving homeomorphism, as required in the definition of homeochromatic type, can be constructed. It is not quite so easy to see that the patterns of Figures 8.5.2(c) and (d) are homeochromatic—here the necessary color-preserving homeomorphism h_0 can be described as one that translates the “rows” of copies of the motif alternately to the right and to the left through a distance equal to the bar at the lower left corner of the diagram. In between the rows, h_0 can be completed linearly.

In the third set, the patterns of Figures 8.5.2(e), (f) and (h) are of the same homeochromatic type, while that of (g) is of a different type. The latter assertion follows because in (g) each copy of the motif is contiguous to two of the same color, while in the other three patterns this does not occur. The precise definitions of the color-preserving homeomorphisms needed to prove that the patterns Figures 8.5.2(e), (f) and (h) are homeochromatic are slightly complicated. They can be described as similar to the homeomorphism h_0 mentioned above, shifting the “rows” of copies of the motif alternately to the right and to the left. Notice however that the homeomorphisms do not preserve the shapes of the motifs—the mappings required are indicated by the similar labelling of corresponding points of the motifs as shown to the right of the three diagrams.

The patterns of the fourth set (Figures 8.5.2(i), (j) and (k)) can be thought of as arising from the patterns of Figures 8.5.2(f), (g) and (h) by moving the “columns” nearer together until the copies of the motifs touch at their corners. This modification affects the classification in that the first two patterns (i) and (j) are not homeochromatic, while the third (k) is homeochromatic to (j) (and not to (i) as might be expected from the above discussion of the corresponding patterns of the third set). In fact it is quite easy to describe a suitable color-preserving homeomorphism that enables one to show that patterns (k) and (j) are homeochromatic.

In the fifth and final set, we notice that in each pattern (Figures 8.5.2(l), (m) and (n)) some copies of the motif touch others at single points while others are contiguous along line-segments. This rather more complicated topological structure is sufficient to ensure that all three patterns are of different homeochromatic types.

In the above examples, the topology of each pattern played a crucial role—different homeochromatic types were distinguished by the contiguity of copies of the motif with others of the same or different colors, and whether these contiguities were at single points or line-segments. Our final example (Figure 8.5.3) shows two pairs of patterns; the patterns of each pair are of the same equicolor type yet are not color-homeomeric in spite of the fact that the underlying patterns are of the same topological

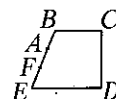
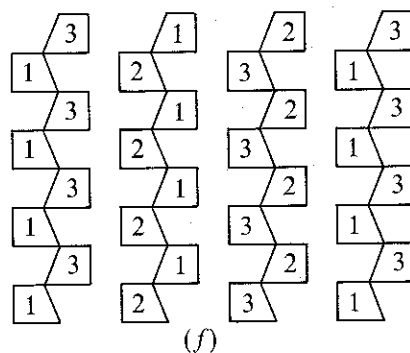
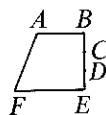
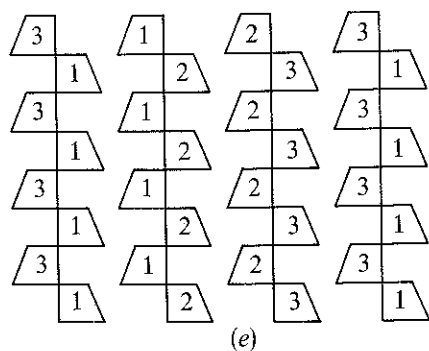
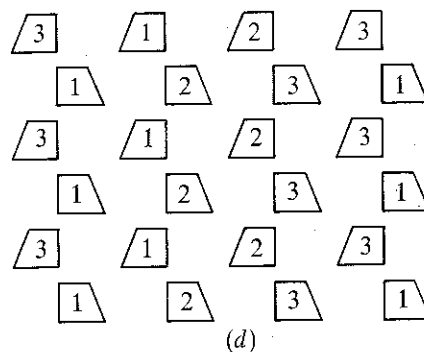
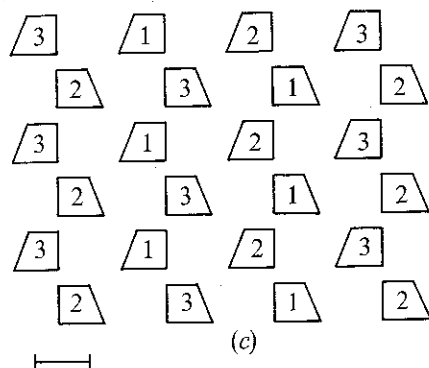
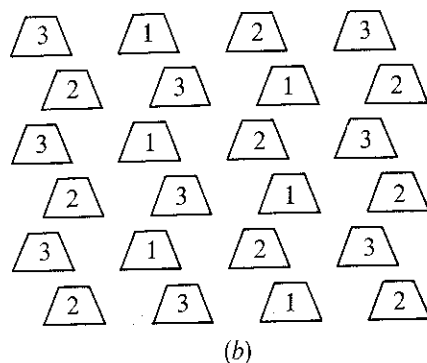
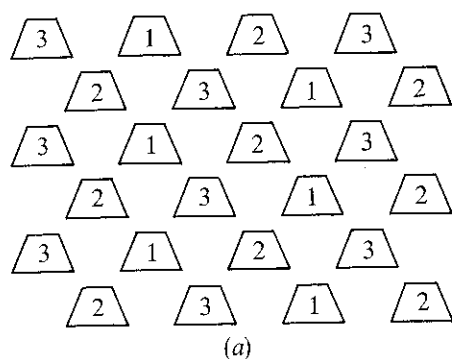
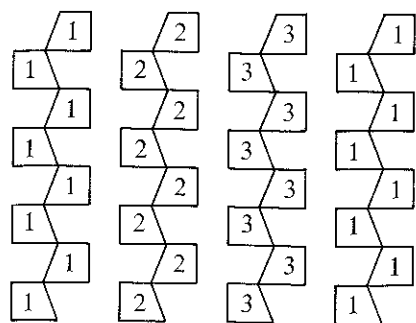
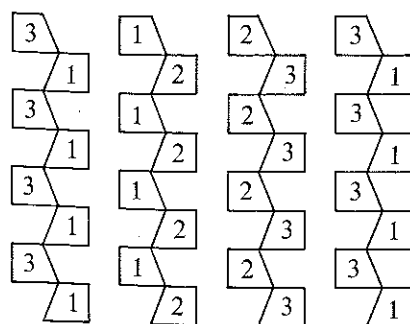


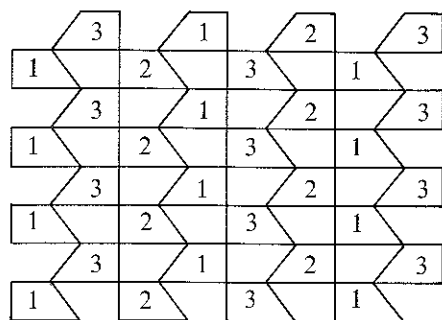
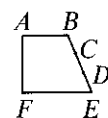
Figure 8.5.2
Fourteen perfectly colored patterns. The classification of these patterns into homeochromatic types is discussed in the text.



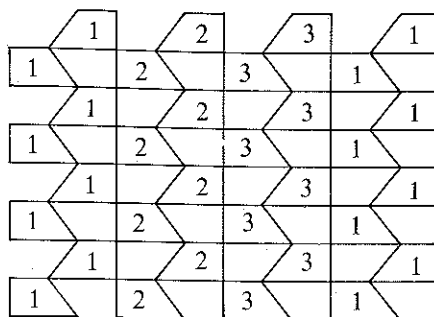
(g)



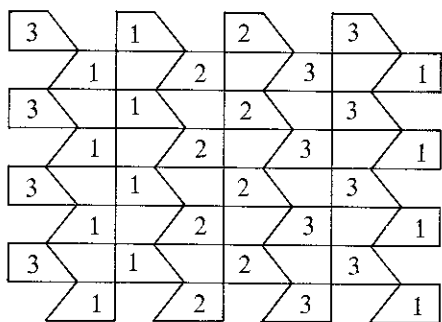
(h)



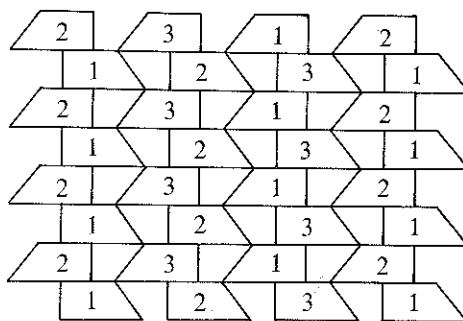
(i)



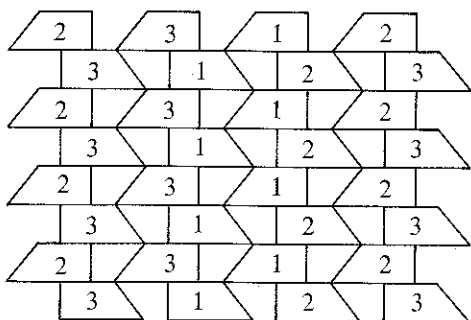
(j)



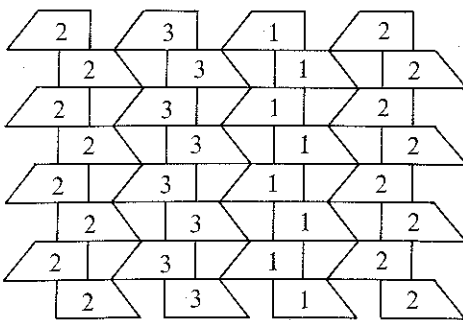
(k)



(l)



(m)



(n)

type and in Figure 8.5.3(a) are even homeomeric. We can explain this phenomenon by saying that in the first pattern of Figure 8.5.3(a) similarly colored copies of the motif are separated by a single line of reflection (of $S_C(\mathcal{P})$) while in the second they are not. In the remaining patterns (Figure 8.5.3(b)) the colored line-segments bear different relations to the lines of reflection of $S_C(\mathcal{P})$.

In Chapter 7 we discussed three methods of classifying (uncolored) patterns—by symmetry group, by pattern type, and by homeomeric type. In this chapter we have introduced five methods of classifying colored patterns—by color symmetry group, by color-pattern type, by equicolor type, by color-homeomeric type and by homeochromatic type. We conclude by summarizing the relationship between all these methods (see Figure 8.5.4).

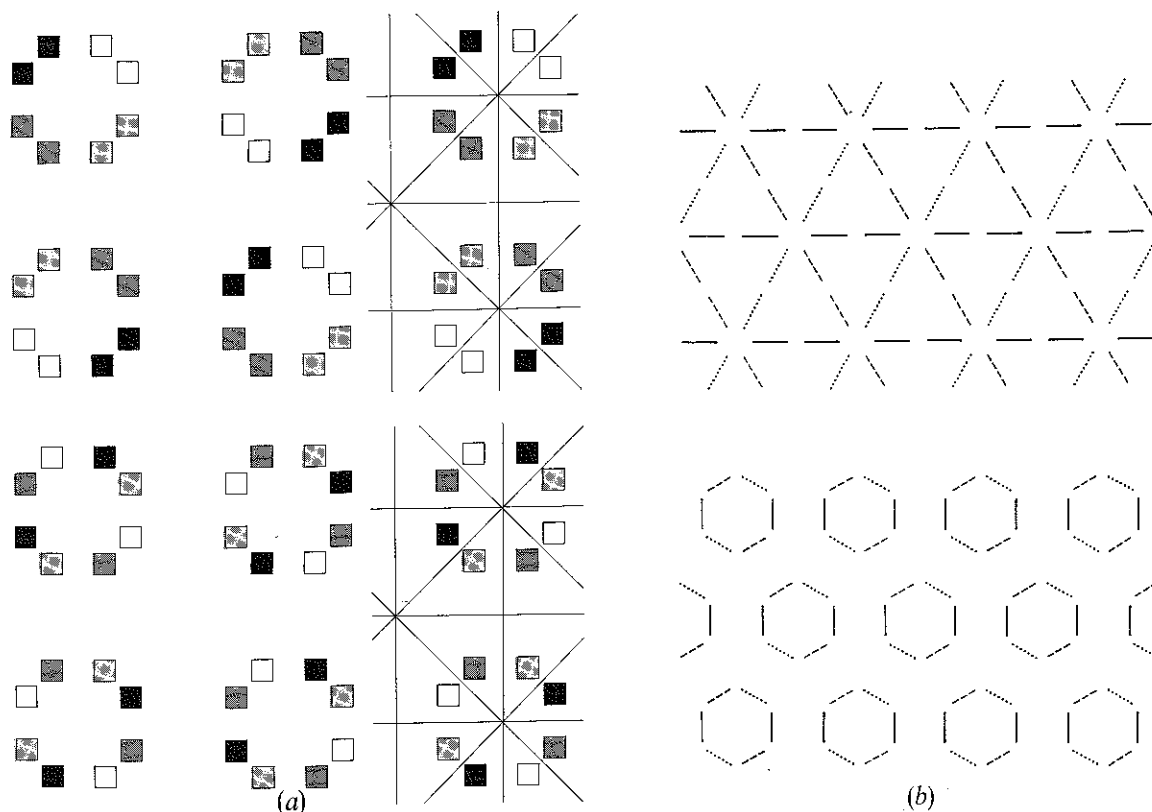


Figure 8.5.3

(a) Two perfect 4-colorings of a discrete pattern of type PP37; they have the same 4-color pattern type and no copy of the motif is contiguous to another, but they are not homeochromatic. Notice that corresponding lines on the right of each diagram correspond to reflective color symmetries of the same kind. (b) Two 3-chromatic segment patterns (with closed line-segments as motifs) of color pattern type PP47[3]₁. They are equicolored but not homeochromatic.

With the arrows representing implications (and so pointing from a finer classification to a coarser one) the diagram is self-explanatory. The examples given here and in the previous sections enable us to establish the following result.

8.5.1 *The relationships between the eight methods of clas-*

sification are completely described by Figure 8.5.4. There are no relationships other than those indicated on the diagram.

The finest of the classifications, that by homeochromatic type, is the most appropriate for our discussion of colored tilings which will be given in the next section.

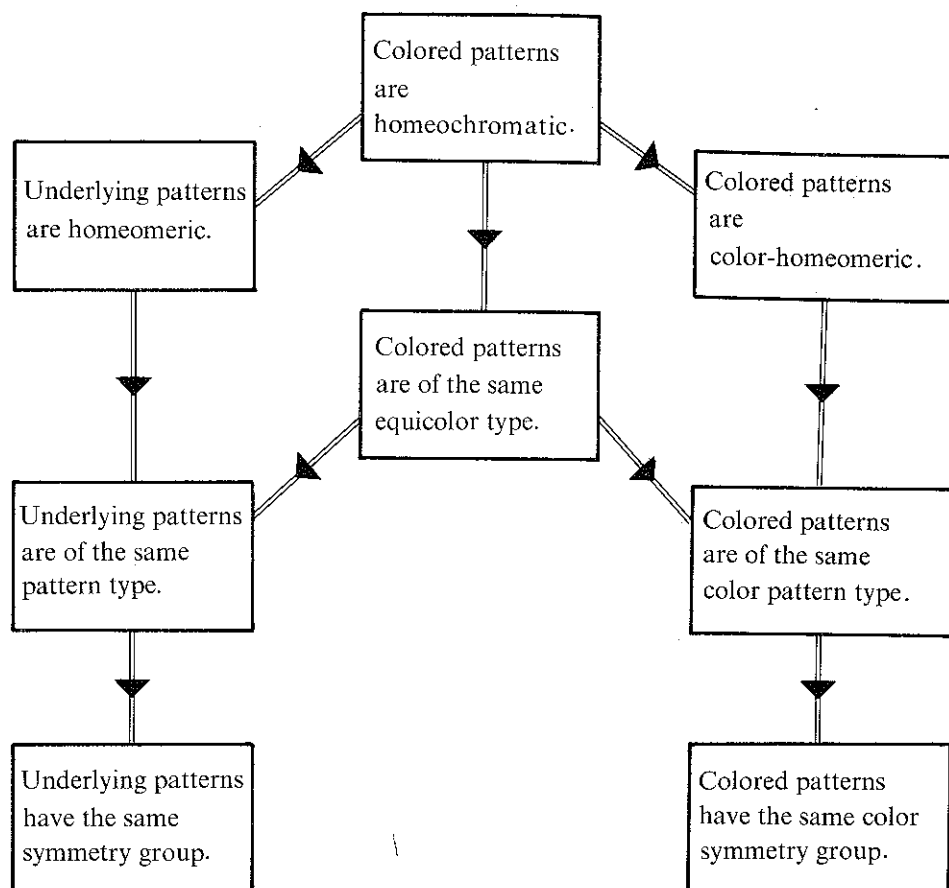


Figure 8.5.4
The relationships between the eight methods of classifying patterns and colored patterns described in this chapter and Chapter 7.

EXERCISES 8.5

1. (a) Show that dot patterns of type DPP41 admit 2-chromatic colorings of six different homeochromatic types.
 - (b) Investigate which of the homeochromatic types in part (a) have the same (i) equicolor type, (ii) color-homeomeric type, (iii) color pattern type, or (iv) color symmetry group.
 - (c) Consider the homeochromatic types of 2-chromatic colorings of the closed circular disk patterns of type CPP41-1 which correspond to the homeochromatic types of dot patterns considered in part (a). Determine which of them have the same (i) homeochromatic type, (ii) color-homeomeric type, or (iii) color pattern type, as a suitably chosen colored ellipse pattern.
- *2. (a) Determine all the homeochromatic types of
- 2-chromatic colorings of each of the coherent strip patterns with an open line-segment as motif.
- (b) Determine which of the homeochromatic types found in part (a) have the same (i) equicolor type, (ii) color-homeomeric type, (iii) color pattern type, or (iv) color symmetry group.
 - (c) Decide which of the homeochromatic types found in part (a) have the same (i) homeochromatic type, (ii) color-homeomeric type, or (iii) color pattern type, as a suitably chosen colored pattern in which the motif is an open semi-circle.
3. For each of the six coherent open circular disk strip patterns find all homeochromatic types of
- *(a) 3-chromatic colorings, and
- **(b) k -chromatic colorings for all $k \geq 4$.
-

8.6 COLORED TILINGS

We turn now to a detailed investigation of k -chromatic tilings. Since we wish to take into account the topology of the tilings, it is clearly appropriate to use one of the classifications described in the previous section. Also, as we wish to distinguish between colored tilings whose underlying tilings are of different isohedral types, the homeochromatic classification is the more appropriate (see Figure 8.6.1).

The procedure we shall follow is a variant of that used in Chapter 6—the incidence symbol for an isohedral

tiling will be adapted so as to include information about the way in which the tiling is colored. In order to simplify the exposition we need to introduce the concept of a primitively-colored tiling.

Let $\mathcal{T}^\wedge$ be a k -chromatic tiling. In Section 8.1 we considered three groups of symmetries associated with $\mathcal{T}^\wedge$, namely $S(\mathcal{T})$ the symmetry group of the underlying tiling $\mathcal{T}$, $S_c(\mathcal{T}^\wedge)$ the group of symmetries of $\mathcal{T}$ associated with color symmetries of $\mathcal{T}^\wedge$, and $S(\mathcal{T}^\wedge)$ the group of color-preserving symmetries of $\mathcal{T}^\wedge$ (that is to say, the group of those color symmetries which map each tile onto one of the same color). From these we obtain three induced groups, as follows:

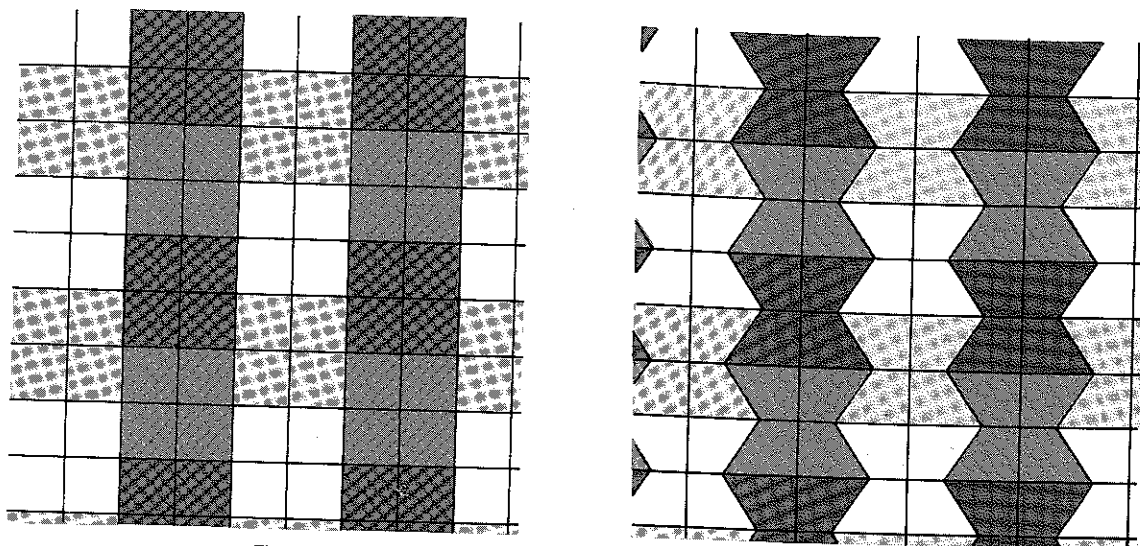


Figure 8.6.1
Two tilings which are color homeomorphic but are not homeochromatic. Since the underlying tilings are not homeomorphic it seems preferable to distinguish between tilings such as these. Hence the classification by homeochromatic type is the more appropriate.

- (a) $S(\mathcal{T}|T)$ is the *induced tile group* of $\mathcal{T}$ consisting of those elements of $S(\mathcal{T})$ that leave a given tile T fixed;
- (b) $S_c(\mathcal{T}^{\wedge}|T^{\wedge})$ is the *induced colored tile group* of $\mathcal{T}^{\wedge}$ consisting of those elements of $S_c(\mathcal{T}^{\wedge})$ that leave a given colored tile $T^{\wedge}$ fixed; and
- (c) $S(\mathcal{T}^{\wedge}|T^{\wedge})$ is the *induced tile group of the colored tiling* $\mathcal{T}^{\wedge}$ consisting of those elements of $S(\mathcal{T}^{\wedge})$ that leave a given colored tile $T^{\wedge}$ fixed. Thus isometries in $S(\mathcal{T}^{\wedge}|T^{\wedge})$ not only map $T^{\wedge}$ onto itself, but also map every tile of $\mathcal{T}^{\wedge}$ onto one of the same color.

Each of these groups is a subgroup of the preceding one. It is possible for two of the groups, or for all three of them, to be equal (see Figures 8.6.2 and 8.6.3), but in

general the three will be distinct (see Figure 8.6.4). A tiling is said to be *primitively colored* if $S_c(\mathcal{T}^{\wedge}|T^{\wedge}) = S(\mathcal{T}^{\wedge}|T^{\wedge})$, that is to say, if every color symmetry of $\mathcal{T}^{\wedge}$ that maps $T^{\wedge}$ onto itself is associated with the identity permutation of colors. An example of a primitively-colored 3-chromatic tiling is given in Figure 8.6.3(a). The following basic result relates the k -chromatic tilings to those which are primitively colored.

8.6.1 Every k -chromatic tiling can be converted into a primitively colored k -chromatic tiling by adding a suitable uncolored mark to each tile.

We begin by adding a mark M to a chosen tile T of the underlying tiling $\mathcal{T}$ of the k -chromatic tiling $\mathcal{T}^{\wedge}$ in such a way that the symmetry group of the resulting

marked tile T_0 coincides with $S(\mathcal{T}^\wedge|T^\wedge)$. The tiling $\mathcal{T}$ determines a tiling pattern $\mathcal{P}_h(\mathcal{T})$ (see Section 5.4) and the underlying group $S_C(\mathcal{T}^\wedge)$ acts transitively on the copies of the motif of $\mathcal{P}_h(\mathcal{T})$, that is, on the interiors of the tiles of $\mathcal{T}$. Applying Statement 5.2.3 with $\mathcal{P} = \mathcal{P}_h(\mathcal{T})$, $G = S_C(\mathcal{T}^\wedge)$, $F = S(\mathcal{T}^\wedge|T^\wedge)$ we find a subgroup H of G that acts transitively on the tiles of $\mathcal{T}$ and has F as its

stabilizer. Use H to transfer copies of the mark M on T_0 onto all the other tiles of $\mathcal{T}$, and so obtain a marked tiling $\mathcal{T}_0$. If each tile of $\mathcal{T}_0$ is colored in the same way as the corresponding tile of $\mathcal{T}^\wedge$, we obtain a colored marked tiling $\mathcal{T}_0^\wedge$ which is primitively colored. Thus Statement 8.6.1 is established. It will be observed that the symmetry group $S(\mathcal{T}_0)$ of the marked tiling $\mathcal{T}_0$ is a

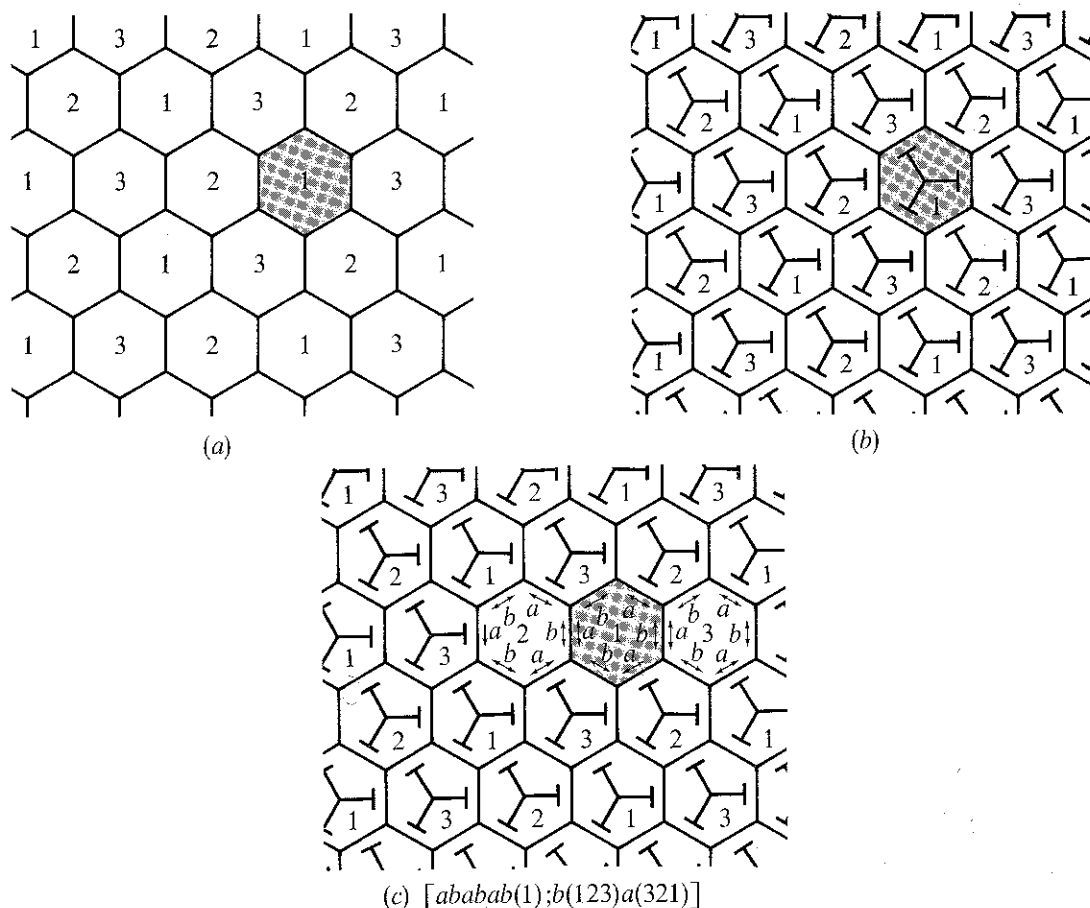
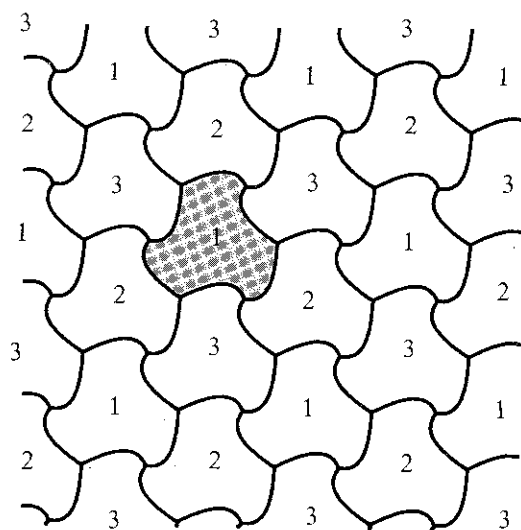
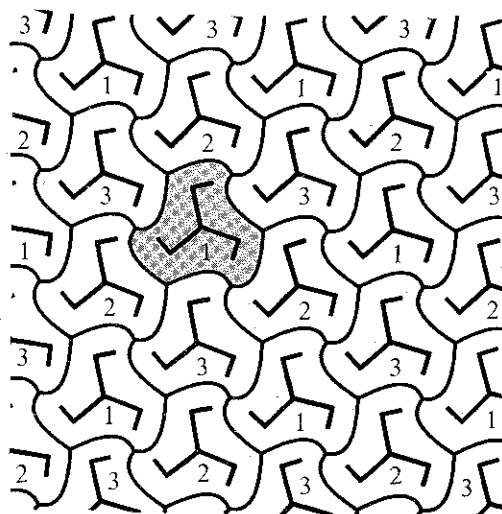


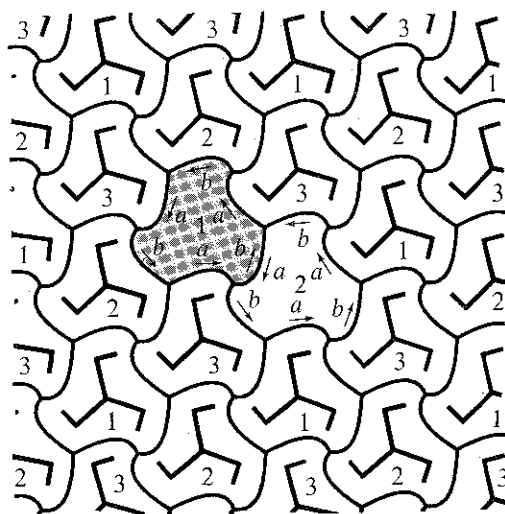
Figure 8.6.2
(a) An example of a 3-chromatic tiling for which $S(\mathcal{T}|T) = S_C(\mathcal{T}^\wedge|T^\wedge) =$
but $S(\mathcal{T}^\wedge|T^\wedge)$ is not equal to this group. (b) A marking of $\mathcal{T}$ that
reduces $\mathcal{T}^\wedge$ to a primitively colored tiling. (c) The derivation of the
corresponding color-incidence symbol.



(a)



(b)

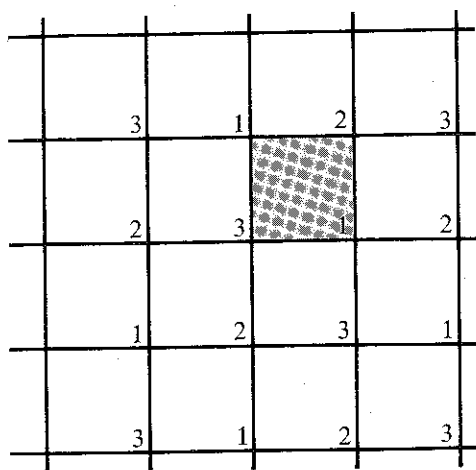


$$[a^+b^+a^+b^+a^+b^+(1);b^+(321)a^+(123)]$$

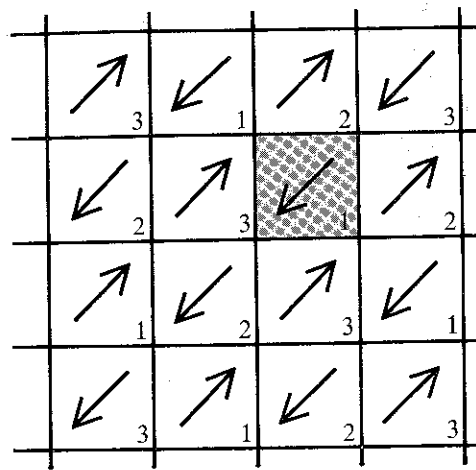
(c)

Figure 8.6.3

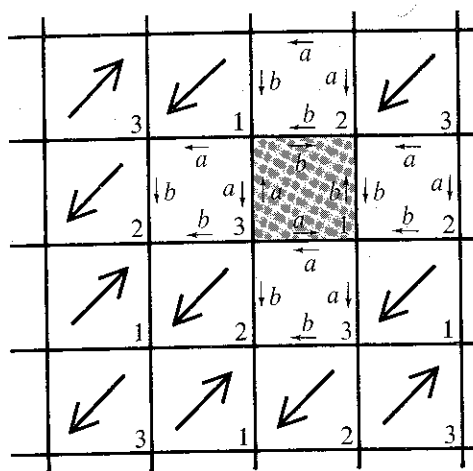
- (a) An example of a 3-chromatic tiling for which $S(\mathcal{T}|T) = S_c(\mathcal{T}^+|T^+) = S(\mathcal{T}^+|T^+) = c3$.
 (b) A marking of $\mathcal{T}$ that arises by the procedure described in the proof of Statement 7.6.1.
 (c) The derivation of the corresponding color-incidence symbol.



(a)



(b)



$$[a^+b^+b^-a^-(1); a^+(13)b^+(12)]$$

(c)

Figure 8.6.4

(a) A 3-chromatic tiling for which the groups $S(\mathcal{T}|T) = d4$, $S_a(\mathcal{T}^+|T^+) = d2$ and $S(\mathcal{T}^+|T^+) = d1$ are distinct. (b) A marking of $\mathcal{T}$ as required in Statement 8.6.1 to ensure color primitivity. (c) The derivation of the corresponding color-incidence symbol.

subgroup of $S_c(\mathcal{T})$ and moreover this group $S(\mathcal{T}_0)$ is tile-transitive.

The procedure just described is illustrated in Figures 8.6.2 to 8.6.5. In each figure part (b) shows a suitable marking $\mathcal{T}_0$ of the underlying tiling of the 3-chromatic tiling $\mathcal{T}$ shown in (a). The colored tiling $\mathcal{T}_0^\wedge$ obtained from $\mathcal{T}_0$ by assigning the colors as in (a) is primitively colored, whereas (except in Figure 8.6.3) the tiling shown in (a) is not. Thus, in an obvious sense, the markings of Figure 8.6.3(b) are redundant, though that need not concern us here.

Even after the mark M on the first tile T has been chosen, in some cases $\mathcal{T}_0$ is not uniquely determined; there may be some choice in the marking of the other tiles. In Figure 8.6.5(b), (d) we show two markings of the tiling underlying that shown in (a), each of which corresponds to a primitive coloring. The reason for this phenomenon is that the choice of the subgroup H , whose existence is asserted by Statement 5.2.3, is not unique—in this case H can be either of type pl or of type pg .

Our next task is to explain the construction of the “color-incidence symbol” of a primitively colored k -chromatic tiling $\mathcal{T}_0^\wedge$. A *color-tile symbol* L of $\mathcal{T}_0^\wedge$ begins with the tile symbol of the underlying tiling $\mathcal{T}_0$. In other words we label the sides of each tile in $\mathcal{T}_0$ as described in Section 6.2 and then read off the labels in cyclic order as we go round a chosen tile T , adding to each label an appropriate superscript if required. The symbol L is completed by adding to the tile symbol of $\mathcal{T}_0$ the color of the tile T in parentheses.

The corresponding *color-adjacency symbol* of $\mathcal{T}_0$ is defined to be the same as an adjacency symbol for $\mathcal{T}_0$ except that after each label we insert the corresponding permutation θ of colors. Thus the letter in the adjacency symbol of $\mathcal{T}_0$ which corresponds to a tile T_i adjacent to T is followed by the permutation θ_i associated with the color symmetry which maps T onto T_i . We illustrate the procedure by reference to Figure 8.6.4(c). A tile symbol of $\mathcal{T}_0$ is $a^+b^+b^-a^-$ and so, if we choose T to be of color 1, the color-tile symbol of $\mathcal{T}_0^\wedge$ is $a^+b^+b^-a^-(1)$. The corresponding adjacency symbol of $\mathcal{T}_0$ is a^+b^+ . The tile T_1 adjacent to T along a side a of the latter bears color 3, and it is easily checked that the color symmetry mapping

T onto T_1 interchanges colors 1 and 3 and leaves color 2 unchanged. Hence the first part of the color-adjacency symbol is $a^+(13)$. The second part is determined similarly; it is $b^+(12)$.

It will be recalled from Chapter 6 that the tile and adjacency symbols are not sufficient to determine the isohedral type of a tiling uniquely—we must also know the topological type. Exactly the same is true for colored tilings and we remark that further ambiguities would arise if the color of the tile T (or *defining tile* as we shall call it) were omitted from the tile symbol. The situation is analogous to that for patterns shown in Figure 8.2.4(b). Here, not only does a different choice of color of the defining tile T affect the “appearance” of the tiling, it can also change its homeochromatic type.

The *color-incidence symbol* of a primitively colored k -chromatic tiling $\mathcal{T}_0^\wedge$ is defined to be an expression of the form $[L; A]$ where L is a color-tile symbol and A is a color-adjacency symbol of $\mathcal{T}_0^\wedge$. Examples of color-incidence symbols are given in Figures 8.6.2 to 8.6.5.

For the sake of uniformity we shall usually take the color of the defining tile T to be $i = 1$. For a different defining tile T' , of color j , the corresponding incidence symbol is easily determined. Let (s, π) be a color symmetry which maps tile T onto T' (so that $\pi(i) = j$). If the incidence symbol corresponding to T is

$$[x_1x_2 \dots x_n(i); y_1(\theta_1)y_2(\theta_2) \dots y_m(\theta_m)],$$

then that corresponding to T' is

$$[x_1x_2 \dots x_n(j); y_1(\theta'_1)y_2(\theta'_2) \dots y_m(\theta'_m)]$$

with $\theta'_r = \pi \circ \theta_r \circ \pi^{-1}$ for $r = 1, 2, \dots, m$. This holds also when $i = j$. Thus the incidence symbol depends essentially on the choice of defining tile and not only on its color.

Corresponding to Statement 7.2.1 we have the following result, which is proved in a similar manner.

8.6.2 *Two primitively colored k -chromatic tilings are homeochromatic if and only if they are of the same topological type and their color-incidence symbols differ trivially.*

The phrase "differ trivially" means that one can be obtained from the other by a change of labels, colors or defining tile, possibly permuting the letters in the tile symbol cyclically or reversing them, and making the corresponding modifications to the adjacency symbol. Examples will be given near the end of this section.

The use of color-incidence symbols makes possible,

for each k , an enumeration of the possible homeochromatic types of primitively colored k -chromatic tilings. We take each (marked) isohedral tiling $\mathcal{T}$ in turn and consider a tile T of $\mathcal{T}$ together with all its adjacents. The incidence symbol of $\mathcal{T}$ is known so we need only list possible candidates for the permutations θ_i ($i = 1, 2, \dots, m$) which are to follow the various letters

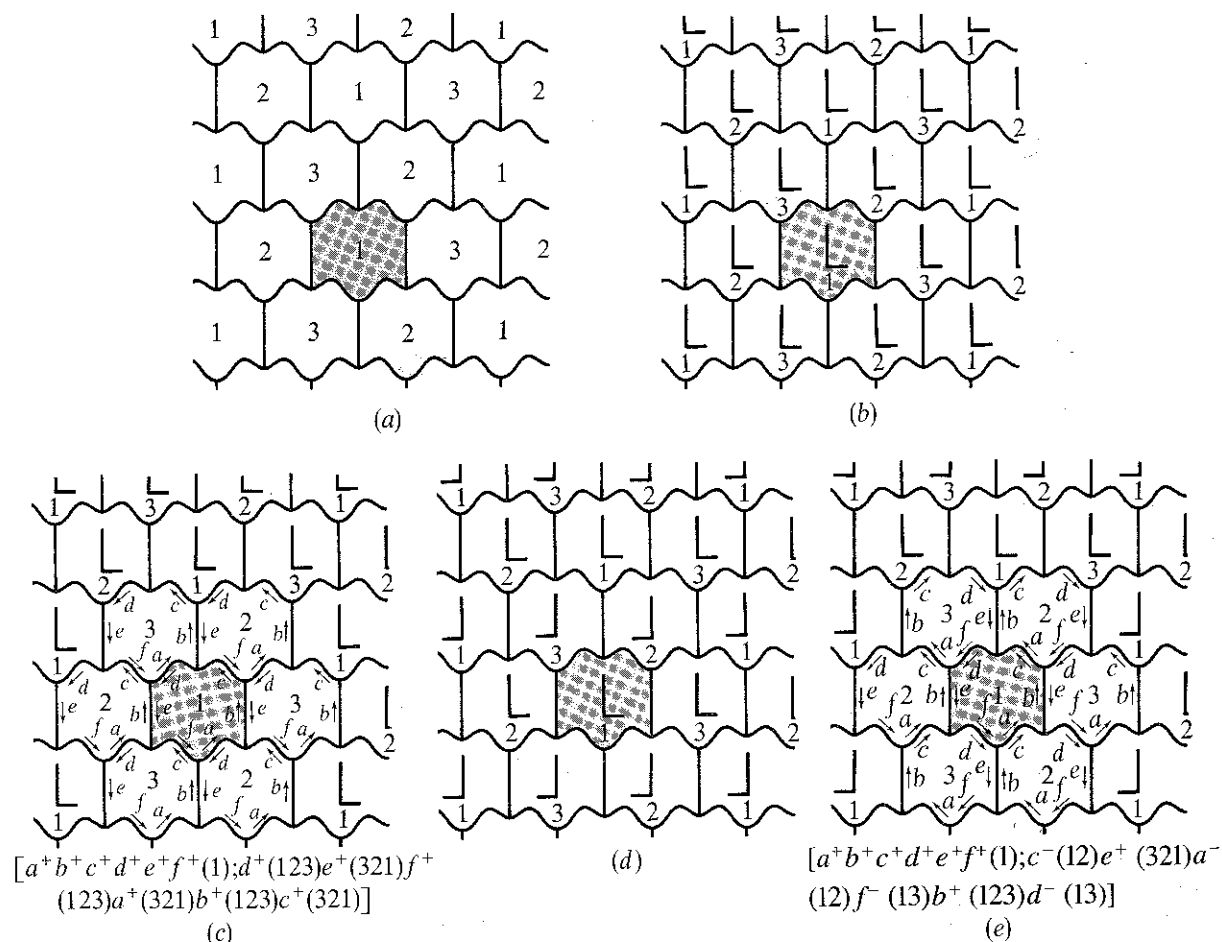


Figure 8.6.5

Parts (b) and (d) show two different markings of the tiling shown in (a) that reduce it to a primitively colored tiling. The derivation of the corresponding color-incidence symbols is shown in (c) and (e).

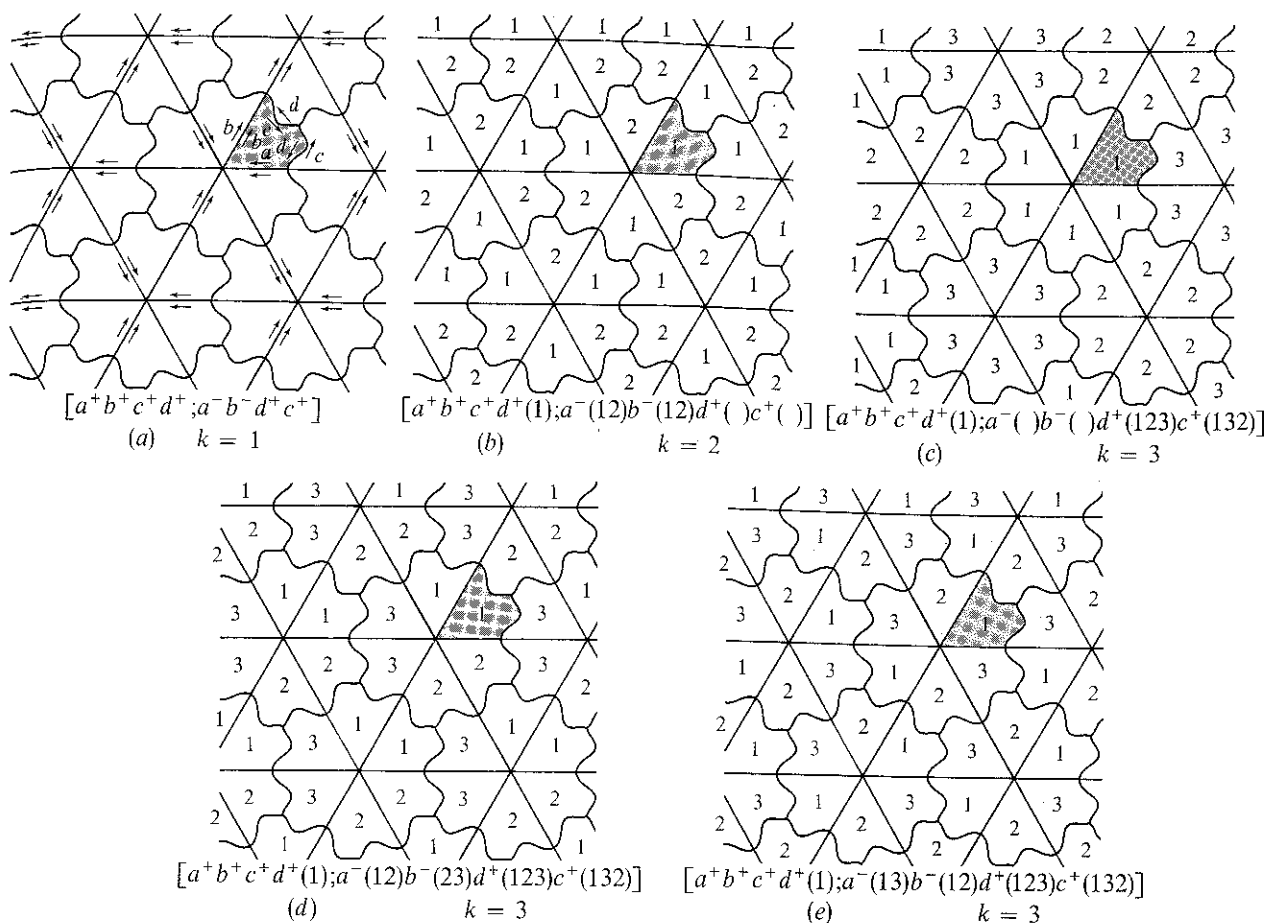


Figure 8.6.6

The primitive pattern IH30 and its color-primitive k -chromatic colorings for $k = 2$ and 3. In each case the corresponding color-incidence symbol is shown.

in the adjacency part of the color-incidence symbol. We pick out those sets of permutations which are consistent in the sense that they lead to a unique coloring of the tiles in $\mathcal{T}$, each by one of k colors. From each such set of permutations we obtain a coloring, and after eliminating duplicates, a complete list of homeo-chromatic types is obtained.

As an example, consider the tiling of type IH30 shown in Figure 8.6.6(a). The incidence symbol is $[a^+b^+c^+d^+; a^-b^-d^+c^+]$ so the color-incidence symbols of the colorings are of the form

$$[a^+b^+c^+d^+(1); a^-(\theta_1)b^-(\theta_2)d^+(\theta_3)c^+(\theta_4)]$$

where the θ_i are permutations of $1, 2, \dots, k$.

For a given k , the number of cases to be considered can be reduced considerably by various means—not every possible set of permutations has to be considered. For example, when $k = 2$, the 3-fold rotation about a vertex where the edges labelled c and d meet can only correspond to the identity, so $\theta_3 = \theta_4 = ()$. Clearly then $\theta_1 = \theta_2 = (12)$ since the only other possibility, when either corresponds to the identity, cannot lead to a 2-colored tiling (see Figure 8.6.6(b)).

For $k = 3$, the 3-fold rotation mentioned above can only correspond to a permutation of colors (123) or (132). (The identity permutation is easily seen to be inadmissible.) Thus $\theta_3 = \theta_4^{-1} = (123)$ or (132). From reflections in the edges labelled a or b we see that $\theta_1^2 = \theta_2^2 = ()$, the

identity, and so each of θ_1 and θ_2 must be either a 2-cycle (ij) or the identity. The possibilities are easily tested and we are led to the three primitively colored 3-chromatic tilings shown in Figures 8.6.6(c), (d) and (e). It is clear that these are of distinct homeochromatic types and so the enumeration is completed. A similar investigation could be carried out for $k \geq 4$.

In the case of 2-colored tilings the color-incidence symbol can be simplified without losing any essential information; we may omit the color of the defining tile from the tile symbol, and in the adjacency symbol we may omit () and replace (12) by $*$. Thus the symbol

$$[a^+b^+c^+d^+(1); a^-(12)b^-(12)d^+()c^+()]$$

of Figure 8.6.6(b) becomes, in simplified form,

$$[a^+b^+c^+d^+; a_*^-b_*^-d_*^+c_*^+].$$

This notation is used in Table 8.2.2 which lists primitively colored isohedral tilings whose symmetry groups are the 46 periodic 2-color groups. Table 8.2.3 contains the corresponding information for three colors. The fact that each color group can be represented in this way as the color symmetry group of a primitively colored isohedral tiling leads to an easy and systematic, if laborious, way of enumerating all the k -color groups for each k .

We can now explain how to determine all the homeochromatic types of k -chromatic colorings of a given isohedral tiling $\mathcal{T}$ (and not, as previously, only the primitively-colored ones). We begin by determining the pattern type of the tiling pattern $\mathcal{P}_h(\mathcal{T})$ (see column (10) of Table 6.2.1), and then all the motif-transitive subgroups of this pattern can be read off from Table 5.2.3. These groups are tile-transitive for $\mathcal{T}$. For each such group H we add marks to the tiles of $\mathcal{T}$ and so obtain a marked tiling $\mathcal{T}_0$ whose symmetry group coincides with H . Using the method of color-incidence symbols explained above, we determine all the primitive k -chromatic colorings of the tilings $\mathcal{T}_0$. Finally we delete the marks on the tiles and so, by Statement 8.6.1, we are left with all possible homeochromatic types of colorings of $\mathcal{T}$; repetitions may occur but these are readily eliminated.

The procedure is illustrated in Figures 8.6.7 to 8.6.10. In this example we consider a tiling of type IH90 (Figure 8.6.7(a)) of which the corresponding pattern type is PP44. There are three motif-transitive subgroups for this pattern type and hence three tile-transitive subgroups of $\mathcal{T}$, namely two of type $p6$ and one of type $p2$. In Figures 8.6.7(b), (c) and (d) some of the centers of rotation for each of these groups are indicated, and we also show corresponding marked tilings $\mathcal{T}_0$ whose symmetry groups are these three tile-transitive subgroups. The incidence symbols are also indicated.

For the tiling of Figure 8.6.7(b) it is easy to see that for $k = 2$ the only possible color-incidence symbol is $[a^+a^+a^+(1); a^+(12)]$, and no primitive k -colorings are possible for $k > 2$ (see Figure 8.6.8).

For the tiling of Figure 8.6.7(c) more possibilities arise. For $k = 2$, potential incidence symbols like

$$[a^+b^+c^+(1); c^+()b^+(12)a^+()]$$

are inconsistent and the only possible primitive 2-coloring corresponds to

$$[a^+b^+c^+(1); c^+(12)b^+(12)a^+(12)].$$

For $k = 3$, we obtain four possibilities and for $k = 4$ there are three possibilities as shown, along with the corresponding color-incidence symbols, in Figure 8.6.9.

For the tiling of Figure 8.6.7(d) we find three tilings with $k = 2$, four with $k = 3$ and eight with $k = 4$. These are shown in Figure 8.6.10.

At every stage we must be careful to avoid repetitions of types. Thus for the tiling of Figure 8.6.7(c) the symbols

$$[a^+b^+c^+(1); c^+(23)b^+(12)a^+(23)]$$

$$[a^+b^+c^+(1); c^+(23)b^+(13)a^+(23)]$$

differ trivially since one can be obtained from the other by interchanging colors 2 and 3, leaving color 1 fixed. Figure 8.6.9(h) shows the corresponding colored tiling. In this particular case, however, we can say more; the two symbols lead to identical colored tilings since one

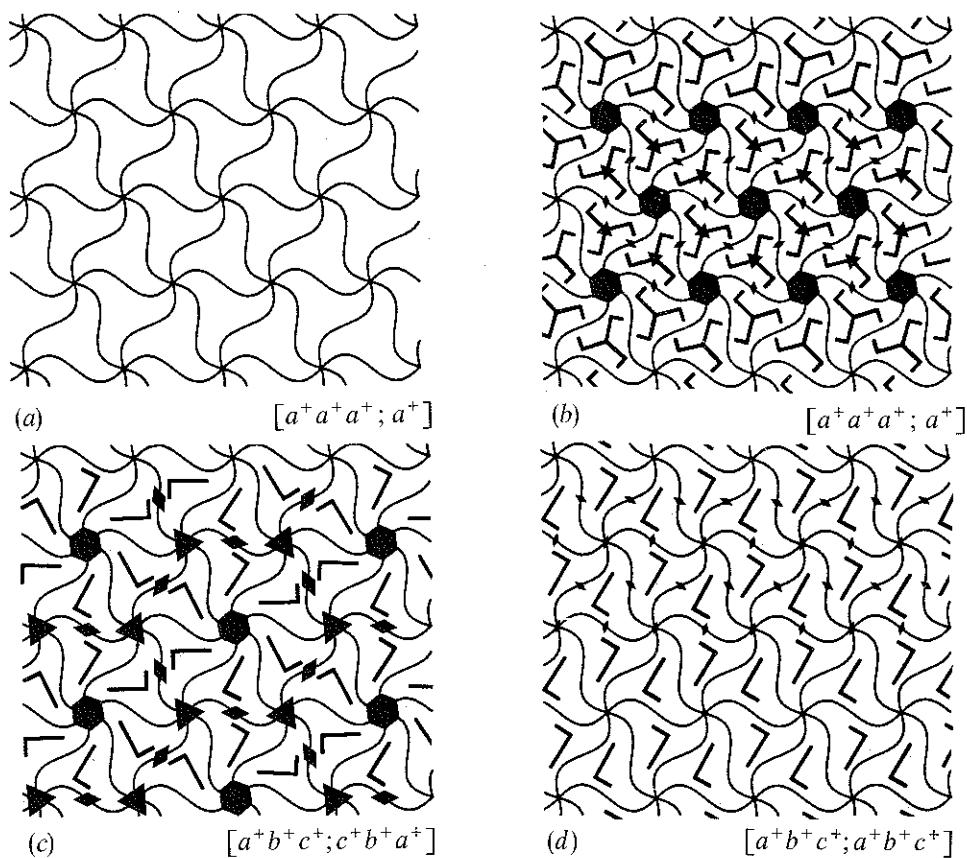


Figure 8.6.7

A tiling of isohedral type IH90 and the corresponding marked tilings that correspond to the tile-transitive subgroups of its symmetry group.

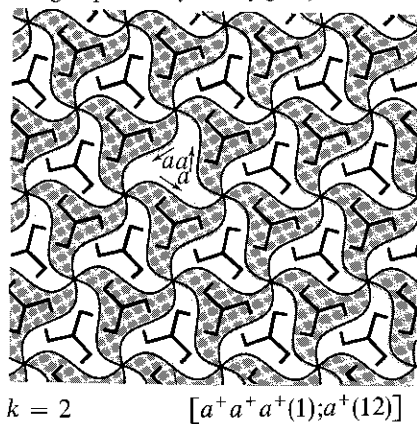
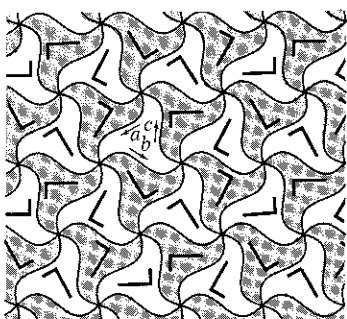


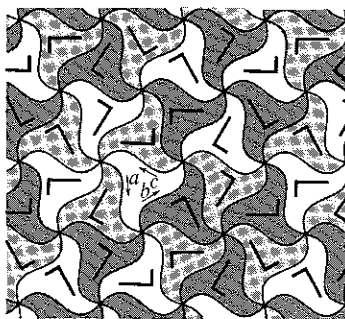
Figure 8.6.8

The only primitive coloring of the marked tiling shown in Figure 8.6.8(b). No other k -colorings are possible with $k \geq 2$.



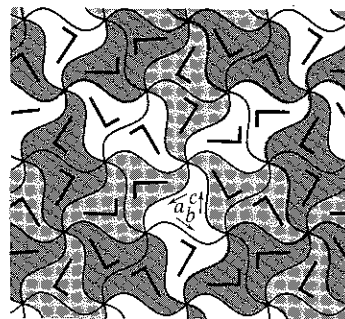
(a) $k = 2$

$$[a^+b^+c^+(1); c^+(12)b^+(12)a^+(12)]$$



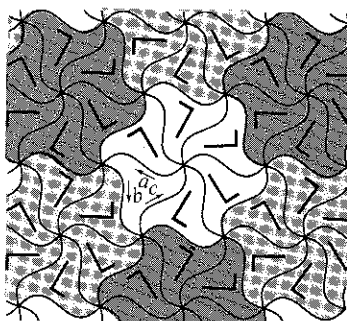
(b) $k = 3$

$$[a^+b^+c^+(1); c^+(12)b^+(13)a^+(12)]$$



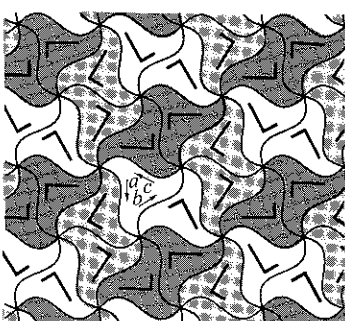
(c) $k = 3$

$$[a^+b^+c^+(1); c^+(12)b^+(23)a^+(12)]$$



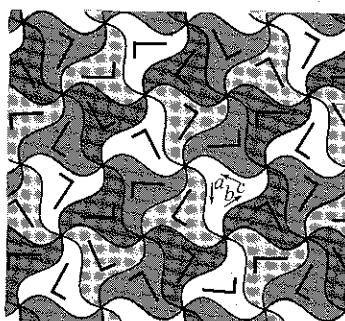
(d) $k = 3$

$$[a^+b^+c^+(1); c^+(23)b^+(12)a^+(23)]$$



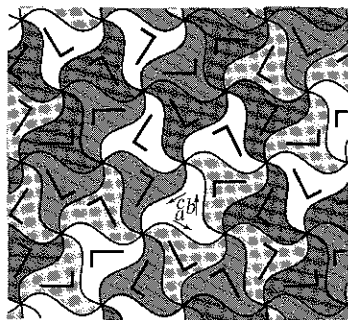
(e) $k = 3$

$$[a^+b^+c^+(1); c^+(123)b^+()a^+(321)]$$



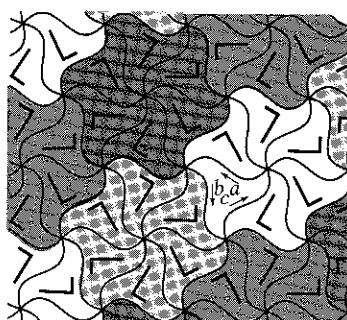
(f) $k = 4$

$$[a^+b^+c^+(1); c^+(123)b^+(14)(23)a^+(132)]$$



(g) $k = 4$

$$[a^+b^+c^+(1); c^+(123)b^+(12)(34)a^+(132)]$$

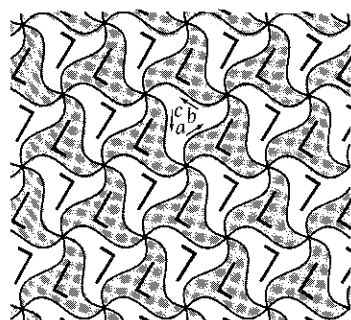


(h) $k = 4$

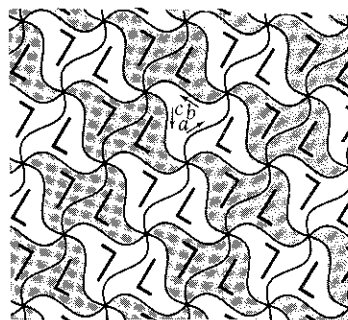
$$[a^+b^+c^+(1); c^+(432)b^+(12)(34)a^+(234)]$$

Figure 8.6.9

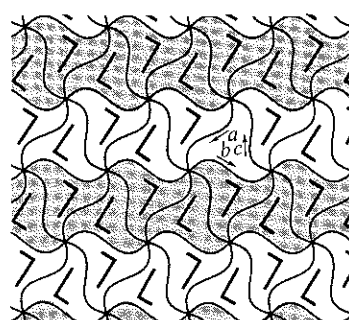
The primitive 2-colorings, 3-colorings and 4-colorings of the marked tiling shown in Figure 8.6.7(c).



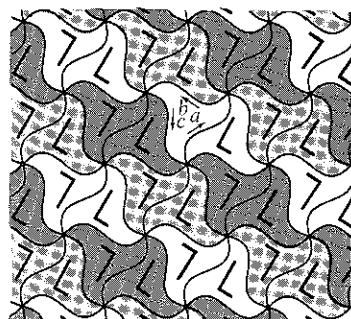
(a) $k = 2$
 $[a^+b^+c^+(1); a^+(12)b^+(12)c^+(12)]$



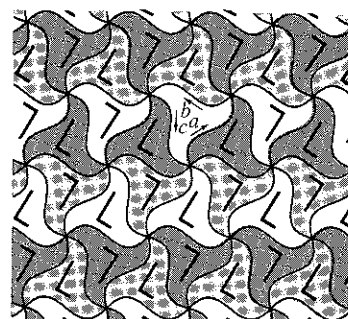
(b) $k = 2$
 $[a^+b^+c^+(1); a^+()b^+(12)c^+(12)]$



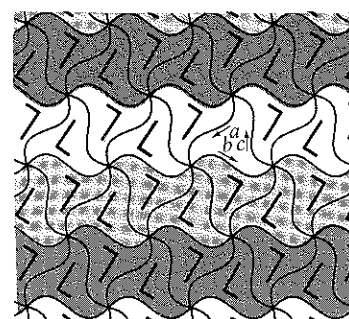
(c) $k = 2$
 $[a^+b^+c^+(1); a^+()b^+(12)c^+()]$



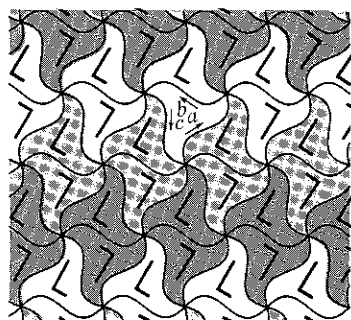
(d) $k = 3$
 $[a^+b^+c^+(1); a^+(23)b^+(12)c^+(13)]$



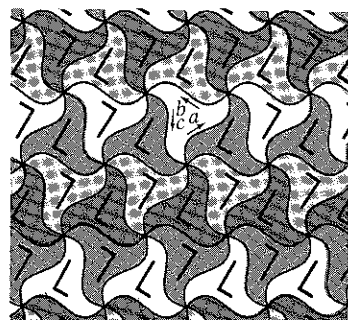
(e) $k = 3$
 $[a^+b^+c^+(1); a^+(13)b^+(12)c^+(13)]$



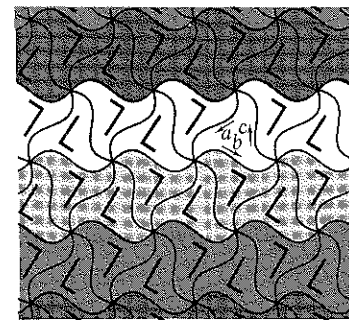
(f) $k = 3$
 $[a^+b^+c^+(1); a^+(23)b^+(12)c^+(23)]$



(g) $k = 3$
 $[a^+b^+c^+(1); a^+(12)b^+(23)c^+(12)]$



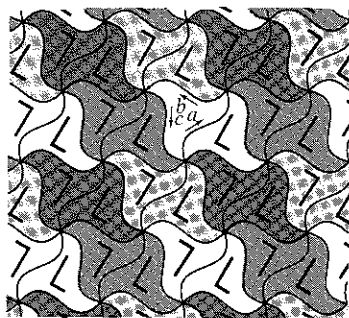
(h) $k = 4$
 $[a^+b^+c^+(1); a^+(13)b^+(12)(34)c^+(13)]$



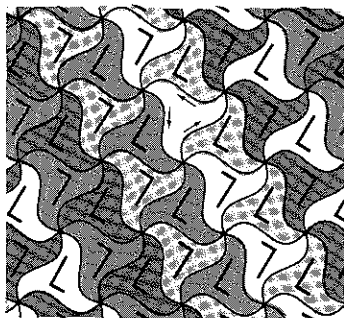
(i) $k = 4$
 $[a^+b^+c^+(1); a^+(24)b^+(12)(34)c^+(24)]$

Figure 8.6.10

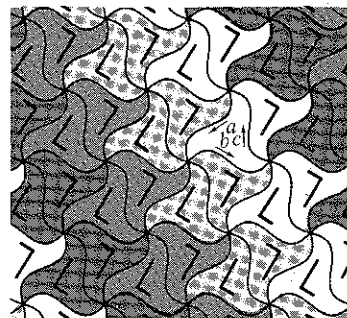
The primitive 2-colorings, 3-colorings and 4-colorings of the marked tiling shown in Figure 8.6.7(d).



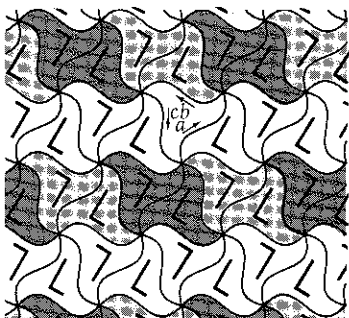
(j) $k = 4$
 $[a^+b^+c^+(1); a^+(24)b^+$
 $(12)(34)c^+(13)]$



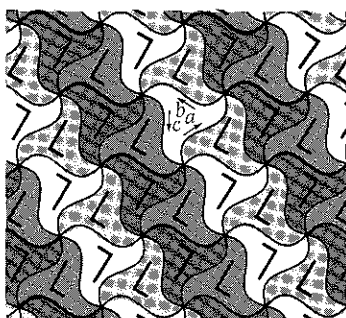
(k) $k = 4$
 $[a^+b^+c^+(1); a^+(12)(34)b^+$
 $(12)(34)c^+(13)]$



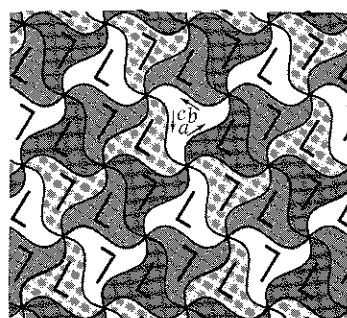
(l) $k = 4$
 $[a^+b^+c^+(1); a^+(12)(34)b^+$
 $(12)(34)c^+(24)]$



(m) $k = 4$
 $[a^+b^+c^+(1); a^+()b^+(12)(34)c^+$
 $(13)(24)]$



(n) $k = 4$
 $[a^+b^+c^+(1); a^+(12)(34)b^+$
 $(12)(34)c^+(13)(24)]$



(o) $k = 4$
 $[a^+b^+c^+(1); a^+(14)(23)b^+$
 $(13)(24)c^+(12)(34)]$

can be obtained from the other by choosing a different defining tile of color 1. In other words, if two copies of the tiling are colored in accordance with the above incidence symbols, then one is the image of the other under a color-preserving isometry.

This is not, however, generally the case. For example, consider the two colorings of the tiling of Figure 8.6.7(c) corresponding to the incidence symbols

$$[a^+b^+c^+(1); c^+(123)b^+(12)(34)a^+(132)]$$

and $[a^+b^+c^+(1); c^+(123)b^+(13)(24)a^+(132)]$.

These are shown in Figures 8.6.9(g) and 8.6.11, respectively. Interchanging colors 2 and 3 leads to incidence symbols that differ only in that the cycles attached to c^+ and a^+ are replaced by their inverses. It is easy to see that the colorings cannot be brought into coincidence by applying any permutation of colors or isometry of

the tiling; nevertheless they are homeochromatic. The color-preserving homeomorphism required in the definition of homeochromatic type is more difficult to describe; it corresponds to a transformation that alters the shape of each tile in one of the tilings into its mirror image followed by reflection of the whole tiling in a suitable line. The interchange of colors 2 and 3 then leads to the required color-preserving homeomorphism.

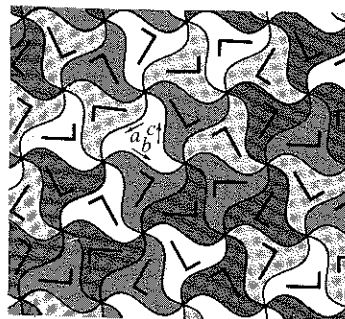
Finally, we remove the markings from the tilings shown in Figures 8.6.8, 8.6.9 and 8.6.10. When this is done several repetitions occur. For example, the colorings of Figures 8.6.8, 8.6.9(a) and 8.6.10(a) are identical, as are the colorings of Figures 8.6.9(f) and 8.6.10(o). In each of these cases the coloring of the unmarked tiling is perfect, though in the latter it is not primitive since the tiling admits, as a color symmetry, a 3-fold rotation about the center of each tile. This is associated

with a cyclic permutation of three of the colors so that $S_c(\mathcal{T}^\wedge | T^\wedge) \neq S(\mathcal{T}^\wedge | T^\wedge)$.

Eliminating repetitions leaves us with a total of three types of 2-colorings, eight types of 3-colorings and ten types of 4-colorings. This is a complete enumeration of the homeochromatic types of k -chromatic colorings of a tiling of type IH90 with two, three or four colors.

The procedure we have just described for determining the homeochromatic types of colorings of isohedral tilings is completely straightforward. It is therefore surprising to report that for no value of k has the enumeration been carried out. We can, however, assert the following.

8.6.3 For any given value of $k \geq 2$ and any isohedral tiling $\mathcal{T}$, the number of homeochromatic types of k -chromatic colorings of $\mathcal{T}$ is finite.



$k = 4$

$$[a^+b^+c^+(1);c^+(123)b^+(13)(24)a^+(132)]$$

Figure 8.6.11

This 4-coloring of the marked tiling shown in Figure 8.6.7(c) is of the same homeochromatic type as that in Figure 8.6.9(g) in spite of the fact that it is not possible to make the colorings coincide by applying a permutation of the colors in either tiling.

EXERCISES 8.6

1. Find examples, other than that shown in Figure 8.6.1, in which the classification by color-homeomeric type and by homeochromatic type differ.
- *2. Find color-incidence symbols for all the 2-chromatic and 3-chromatic colorings of the regular tilings.
3. (a) Using the methods of this section determine all the homeochromatic types of k -chromatic colorings ($2 \leq k \leq 4$) of the Laves tiling [3.6.3.6].
- *(b) Which of the homeochromatic types found in part (a) have the same (i) equicolor type, (ii) color-homeomeric type, (iii) color pattern type, or (iv) color symmetry group?
- **4. List the homeochromatic types of 2-chromatic colorings of all 93 types of marked isohedral tilings.

8.7 PERFECT COLORINGS OF TILINGS

In Section 8.1 we pointed out that in general a given pattern or tiling cannot be perfectly k -colored for every value of k . In Grünbaum & Shephard [1977d] it was shown that every isohedral tiling and every pattern can

be perfectly k -colored for infinitely many values of k ; the permissible values of k were determined explicitly by Senechal [1979a], and the reader is referred to this paper for further details.

Here we consider only perfect colorings of the three regular tilings; we shall prove the following results.

8.7.1 The regular tiling by squares (4^4) admits a perfect k -coloring if and only if $k = n^2$ or $k = 2n^2$ for some positive integer n .

8.7.2 The regular tiling by hexagons (3^6) admits a perfect k -coloring if and only if $k = n^2$ or $k = 3n^2$.

8.7.3 The regular tiling by triangles (6^3) admits a perfect k -coloring if and only if $k = 2n^2$, $k = 6n^2$, $k = (3n - 2)^2$ or $k = (3n - 1)^2$.

In each case we shall construct the perfect colorings explicitly and each coloring will be unique within a permutation of the colors or a symmetry of the tiling.

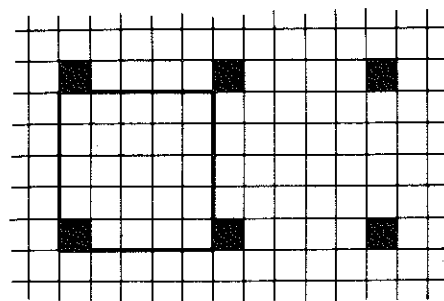
We begin by considering the tiles of color class 1 in a perfect coloring of (4^4). Let m be the smallest positive integer such that a horizontal translation by m tiles brings a tile of color 1 into coincidence with another tile of color 1. The fact that the color classes are congruent enables us to deduce that every tile obtainable from a tile of color j by a horizontal translation of m tiles also has color j . Further, considering rotation about the center of a tile through 90° we see that exactly the same assertion follows for vertical translations also. Thus, for each j , color j is assigned to, at least, all tiles that form a square lattice Λ_j of "mesh" m .

Considering the color class 1, two possibilities arise (see Figure 8.7.1 where color class 1 is indicated by black squares; $m = 5$ in (a) and $m = 6$ in (b)):

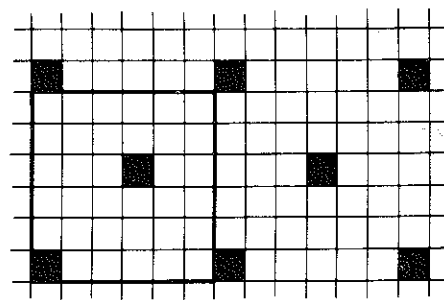
- (i) either all the tiles of color 1 belong to the lattice Λ_1 , or
- (ii) some other tile has color 1.

In case (i) we see that all the m^2 tiles in a mesh of Λ_1 must have different colors, and that these same colors are repeated in the same way in every mesh (see Figure 8.7.2(a) for the case $m = 5$). Hence $k = m^2$ colors occur.

In case (ii), consider reflections in the horizontal and vertical lines that pass through the centers of the tiles of color 1. Due to the minimality of m the only possible positions for the extra tiles of color 1 are at the centers of the meshes, and so this situation can arise only if



(a)



(b)

Figure 8.7.1

The possible arrangements of tiles of color class 1 in perfect colorings of the regular tiling (4^4). The cases $m = 5$ and $m = 6$ are illustrated.

$m = 2n$ is even. Hence each mesh of $(2n)^2 = 4n^2$ tiles contains two tiles of color 1, the number of colors is $k = 2n^2$ and we obtain an arrangement of colors like that shown in Figure 8.7.2(b) for the case $m = 6$.

It is easily verified that each of the colorings described above and illustrated in Figure 8.7.2 is perfect, and hence $k = m^2$ or $k = 2n^2$. The proof of Statement 8.7.1 is thus completed.

For the hexagonal tiling (3^6) the procedure is similar. Orienting the tiling as in Figure 8.7.3, so that one third of the edges are vertical, we again let m be the smallest positive integer such that a horizontal translation by m tiles brings a hexagonal tile of color 1 into coincidence with another tile of color 1. Then we can deduce, as before, that the tiles of color 1 form at least a lattice with triangular mesh of side m , see Figure 8.7.3. A fundamental region R for the corresponding group of translations is rhomb-like (indicated in Figure 8.7.3 by thickened lines) and hence, if there are no other tiles of color 1, we have $k = m^2$, this being the number of tiles in R (see Figures

10	6	7	8	9	10	6	7	8	9	10	6	7
5	1	2	3	4	5	1	2	3	4	5	1	2
25	21	22	23	24	25	21	22	23	24	25	21	22
20	16	17	18	19	20	16	17	18	19	20	16	17
15	11	12	13	14	15	11	12	13	14	15	11	12
10	6	7	8	9	10	6	7	8	9	10	6	7
5	1	2	3	4	5	1	2	3	4	5	1	2
25	21	22	23	24	25	21	22	23	24	25	21	22

(a)

7	8	9	10	11	12	7	8	9	10	11	12	7
1	2	3	4	5	6	1	2	3	4	5	6	1
16	17	18	13	14	15	16	17	18	13	14	15	16
10	11	12	7	8	9	10	11	12	7	8	9	10
4	5	6	1	2	3	4	5	6	1	2	3	4
13	14	15	16	17	18	13	14	15	16	17	18	13
7	8	9	10	11	12	7	8	9	10	11	12	7
1	2	3	4	5	6	1	2	3	4	5	6	1

(b)

Figure 8.7.2

The perfect colorings of (4^4) corresponding to the arrangement of tiles of color class 1 shown in Figure 8.7.1. In (a) $m^2 = k = 25$ colors are used and in (b) $\frac{1}{2}m^2 = k = 18$ colors are used.

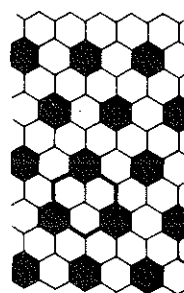
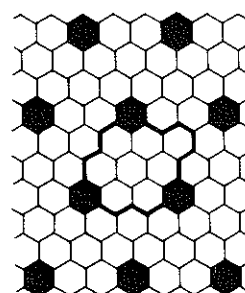
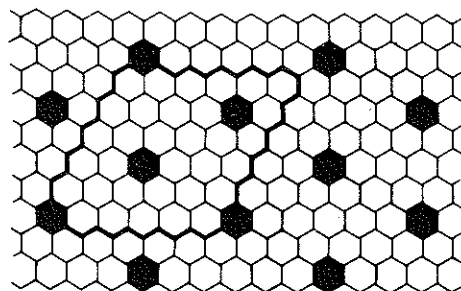
(a) $k = 4$ (b) $k = 9$ (c) $k = 12$

Figure 8.7.3

The possible arrangements of tiles of color class 1 in perfect colorings of the regular tiling (3^6) . Cases $m = 2, 3$ and 6 are shown and the corresponding values of k are indicated.

8.7.3(a) and (b) for the cases $m = 2$ and 3). On the other hand, if another tile has color 1, then considering reflections in the altitudes of the triangles forming the meshes, we see that the only position it can occupy is the center of one (and therefore every) "triangle" in the mesh, see Figure 8.7.3(c). This case can only arise when $m = 3n$ for some integer n —in Figure 8.7.3(c) we have $n = 2$. The fact that each of the other color classes is a translation of this makes it evident how the coloring is to be completed. In all, $3n^2$ colors will be used. As each of the colorings we have described is perfect, the proof of Statement 8.7.2 is completed.

For the triangular tiling (6^3) the situation is slightly more complicated, due to the fact that the tiles occur in two different "orientations" or aspects. Considering the tiles of only one aspect, we again define m' to be the smallest positive integer such that a horizontal translation through a distance mb (where b is the length of the side of a tile) brings a triangular tile of color 1 into coin-

cidence with another tile of color 1. Again we deduce that all the tiles of one aspect of color 1 form, at least, a lattice with triangular mesh of side m . A fundamental region R for the corresponding group of translations is a rhomb (indicated in Figure 8.7.4 by thickened lines) and hence if there are no further tiles of color 1 we have $k = 2m^2$ (see Figures 8.7.4(a) and (b) for the cases $m = 2$ and $m = 3$). If there is another tile of color 1 in R then the only position it can occupy is the center of one of the triangles of the mesh. Three cases arise:

- (i) If $m = 3n$ then the triangle in the left half of the rhomb R has a central tile of the same aspect as the tiles forming the mesh. Hence the number of colors is $k = 6n^2$ (see Figures 8.7.4(c) and (d) for the cases $n = 1$ and $n = 2$).
- (ii) If $m = 3n - 2$ then the triangle in the left half of the rhomb R has a central tile of the opposite aspect to the tiles forming the mesh. Hence the

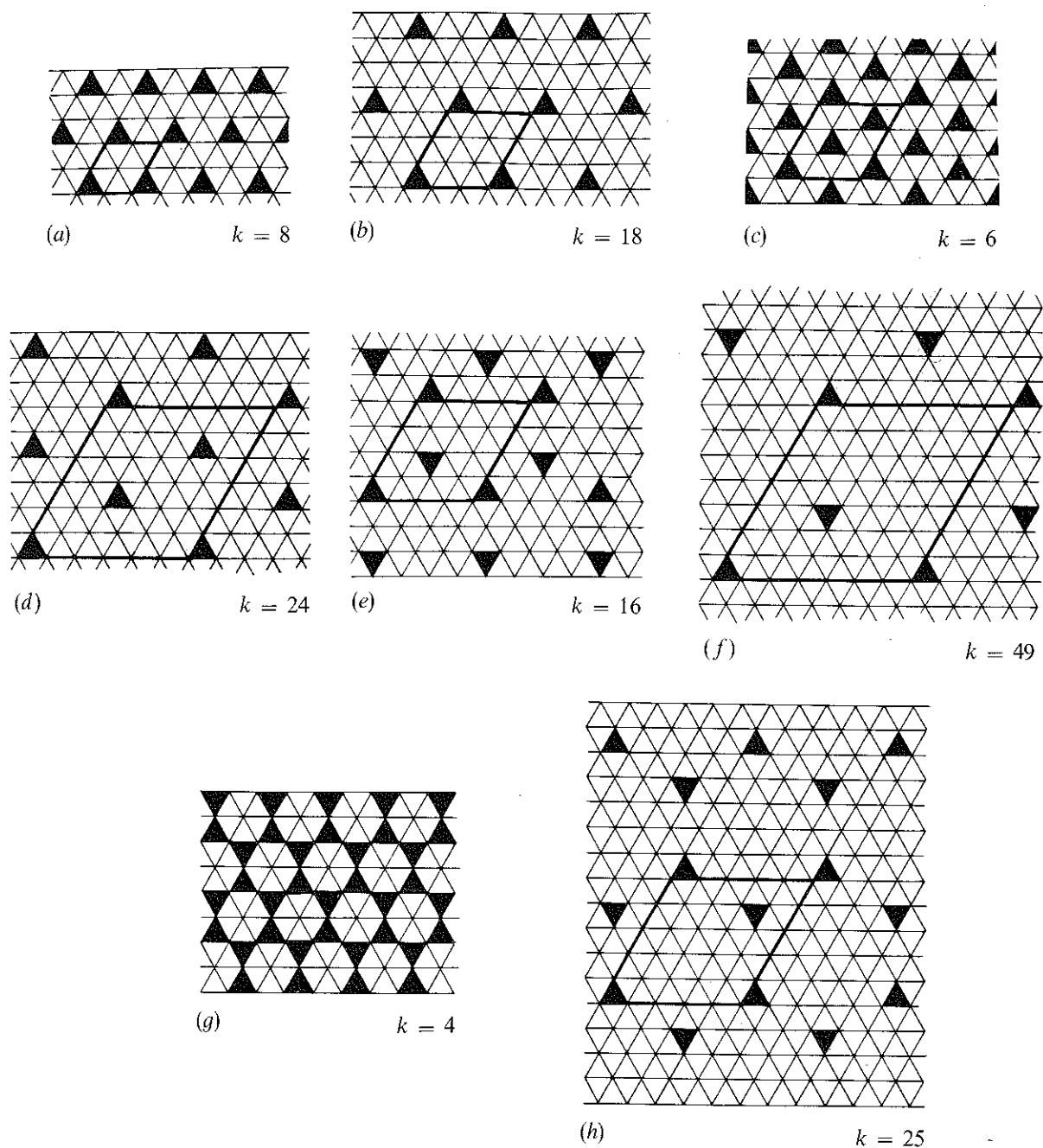


Figure 8.7.4
The possible arrangements of tiles of color class 1 in perfect colorings of the regular tiling (6^3) . The corresponding values of m are: (a) and (g) $m = 2$, (b) and (c) $m = 3$, (e) $m = 4$, (h) $m = 5$, (d) $m = 6$ and (f) $m = 7$.

number of colors is $k = (3n - 2)^2$ (see Figures 8.7.4(e) and (f) for the cases $n = 2$ and $n = 3$).

- (iii) Finally, if $m = 3n - 1$ then each of the triangles in the right half of the rhomb R has a central tile of the opposite aspect to the tiles forming the mesh. Hence the number of colors is $k = (3n - 1)^2$ (see Figure 8.7.4(g) and (h) for the cases $n = 1$ and $n = 2$).

In Figure 8.7.4 we have indicated only the color class 1, but it is evident how, as before, the coloring of the tiling can be completed in a perfect manner by k colors where k is of the stated form. Thus Statement 8.7.3 is proved, and all perfect colorings of the regular tilings have been described.

We conclude with a few remarks about perfect colorings of other isohedral tilings. If $\mathcal{T}_1$ and $\mathcal{T}_2$ are two such tilings of the same topological type, and every symmetry of $\mathcal{T}_1$ is a symmetry of $\mathcal{T}_2$, then it follows from the definitions that to every perfect coloring of $\mathcal{T}_2$ will correspond a perfect coloring of $\mathcal{T}_1$. Roughly speaking,

the smaller the symmetry group, the more perfect colorings will exist. For example, a tiling of type IH62 (see Figure 6.2.4) is of the same topological type as the regular tiling (4^4) , its symmetry group consists of the *direct* isometries (translations and rotations) in the symmetry group of (4^4) . We deduce from Statement 8.7.1 that each tiling of type IH62 admits perfect k -colorings when $k = n^2$ or $2n^2$. However other values of k are possible; it can be shown that there also exist perfect k -colorings when $k = n^2 + p^2$ where n and p are integers such that $n > 0$ and $n \geq p \geq 0$.

This same number $k = n^2 + p^2$ also arises as a permissible value of k for perfect k -colorings of isohedral tilings of type IH28. A perfect 5-coloring of a very decorative tiling of this type is shown in Grünbaum & Shephard [1977d, Figure 10]. This tiling was designed by M. C. Escher who, however, did not seem to be aware of the possibility of using five colors—a perfect 4-coloring of this tiling by Escher is shown in MacGillavry [1965].

EXERCISES 8.7

- *1. Give a detailed proof of the assertion that every isohedral tiling admits perfect k -colorings for infinitely many values of k .
- *2. Determine all the types of isohedral tilings which

admit perfect k -colorings for every $k \geq 2$.

- *3. Determine all the values of k for which there exists a perfect k -coloring of isohedral tilings of type (a) IH37 (b) IH72; (c) IH73.
-

8.8 NOTES AND REFERENCES

In every civilization and culture, colored tilings and patterns appear among the earliest decorations. Splendid examples are shown in Jones [1856], d'Avennes [1877], Bossert [1928] and in other accounts of early art. In par-

ticular, 2-color patterns arose—early and frequently—through a device known as “counterchange” (see, for example, Christie [1929, Chapter 10]). An early paper with remarkable counterchange designs formed by diagonally divided squares—one-half black, one-half white—was published by Truchet [1704]. For a more recent treatment, with many illustrations, see Cullinane

[1976]. However, all these were more or less "accidental" occurrences, independently reinvented many times, and passed from generation to generation by artists and artisans. The only artist who deliberately and consistently tried to investigate colored patterns (more specifically, colored tilings) was M. C. Escher (see Escher [1958], [1960], [1971], [1982], [1983], MacGillavry [1965], Ernst [1976]). The idea of devising isohedral tilings in which the tiles have shapes of animals or various objects, and are colored by two or more colors, has in recent years become a popular tool in mathematical education (see, for example, Ranucci & Teeters [1977], Maletsky [1974], Teeters [1974], Haak [1976], Brisse [1981]). (A different method of drawing families of mutually related tilings of the Escher type is investigated by Bøggild [1980a], [1980b].) In view of this, it is rather surprising to note that, except in Escher's work, k -chromatic patterns with $k \geq 3$ occur very rarely in art and decoration. Moreover, most of those that do occur are of the very simplest kinds (typically, a strip pattern in which the colors are cyclically permuted, or a periodic pattern generated by two translations, one of which leaves the colors unchanged while the other permutes them cyclically). Very few examples have the complexity of the 3-chromatic tilings shown in Figure 5.2.13. (Much of the startling and attractive art of Hans Hinterreiter is based on the idea of homeomorphically transforming colored tilings or patterns with a crystallographic group of symmetries; see Hinterreiter [1977] and Makovicky [1979].)

In spite of the widespread use of colored patterns and tilings in art, theoretical investigations of the various kinds of colorings have been largely ignored by mathematicians and crystallographers; there are, however, some exceptions to this statement. For example, in a series of papers, Woods [1935a], [1935b], [1935c], [1936] gives the first complete, explicit and deliberate enumeration of strip and periodic 2-color groups. (See Crowe [1985] for an account of Woods' work.) Unfortunately, these nicely written papers seem to have had equally little influence on mathematics as on design. Then there is the work of Escher, already mentioned. Its influence on mathematicians and, even more, on crystal-

lographers and designers seems to have been considerable, but there appears to be no account available of the systematic results which Escher obtained in his investigations of colored tilings. (It is to be hoped that the efforts of Doris Schattschneider in this respect will be published soon; see Schattschneider [1985].) In a different direction leads the work of Washburn [1977], who developed a theory of 2-chromatic patterns motivated by designs on the pottery of Southwest American Indian tribes. Washburn was apparently not aware of the existence of other works on 2-color groups and patterns. Although her classification of 2-chromatic patterns may be appropriate for its intended use, it is not quite consistent. For example, Washburn's classification of 2-chromatic strip patterns into 21 types falls between the classifications by the 17 2-color strip groups and by the 28 2-color pattern types.

For an account of an interesting variety of 2-colored strip patterns, see Bérczi [1985].

In a somewhat different sense, the history of 2-color groups started in the late 1920's, even before the work of Woods mentioned above. In order to describe this we shall begin by explaining some of the situations in which 2-color groups are useful. Instead of referring (as we shall do) to black-and-white patterns (or to patterns of some other two colors), it is clear that we may consider patterns in which each copy of the motif has one of two mutually exclusive properties, and we can interpret 2-color groups as expressing (generalized) symmetries of such patterns. In Figure 8.8.1 we illustrate several instances of this kind, which show the wide applicability of the concept of 2-color groups.

One further modification is of considerable geometric and historic interest. It is related to the last example in Figure 8.8.1 and deals with patterns in a plane which is assumed to be situated in 3-dimensional space and to have two sides. Thus a copy of the motif can be situated on either of the two sides, and symmetries of the pattern may involve *reflection in the plane itself*. It is obviously possible to relate 2-chromatic patterns in the usual sense to such (uncolored) patterns on the two-sided plane by placing motifs of one color (say white) on one "side" of the plane, those of the other color (say black) on the

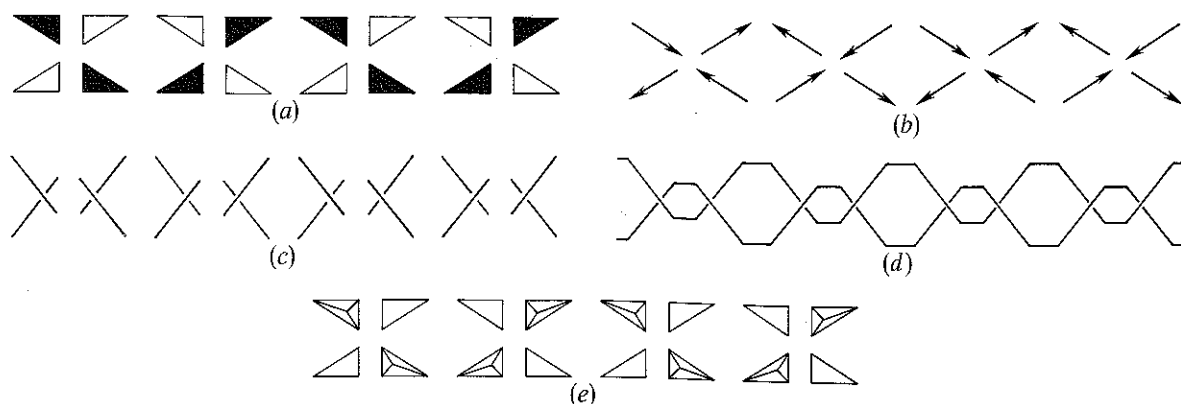


Figure 8.8.1

Strip patterns in which each copy of the motif is endowed with one of two mutually exclusive properties. The pattern in (a) is 2-chromatic. In (b) we show a pattern consisting of magnetic rods (taken as segments) in which the north pole is indicated by an arrowhead; each symmetry either preserves or else reverses all magnetic orientations. Parts (c) and (d) represent a layered pattern (see Section 5.7) of segments and a "layered filamentary pattern"; here a symmetry of the pattern may be accompanied by the reversal of the over-under relationship. The pattern in (e) consists of tetrahedra (three-sided pyramids) in 3-dimensional space, such that their bases form a strip pattern in the plane; some of the symmetries of the pattern of tetrahedra involve a reflection in the plane of the paper, while others do not. It is clear that the symmetry groups of these patterns are isomorphic and can be identified with the 2-color group of the 2-chromatic pattern in (a).

other "side"; but the converse correspondence has two shortcomings. A pattern on the two-sided plane may happen to be situated in its entirety on one side and thus correspond to a monochromatic (that is, uncolored) pattern in the usual sense. On the other hand, each copy of the motif of a pattern on the two-sided plane may be accompanied by a copy obtained by reflection in the plane itself—so that in trying to give a 2-chromatic representation in the ordinary plane we would have to assign to each copy of the motif *both* colors, white and black. Following Šubnikov (see below) this quandary is frequently resolved by saying that such patterns of the two-sided plane correspond to *gray* patterns of the ordinary plane. From this discussion it is clear that the number of strip or periodic groups of patterns on the two-sided plane equals the number of 2-color strip or periodic groups, plus twice the number of (uncolored) strip or periodic groups. Hence there are $31 = 17 + 2 \cdot 7$ strip groups, and $80 = 46 + 2 \cdot 17$ periodic ones. Similar relations hold for the numbers of strip or periodic pattern types. Moreover, it is not hard to verify that these 31 or 80 groups can be obtained from 3-dimensional crystallographic groups by restricting attention to those

groups of the 3-dimensional space which map a given strip or plane onto itself. (For additional comments on this topic, see Grünbaum & Shephard [1983b].)

The historic interest of these considerations lies in the fact that 2-color groups first appeared in the literature in the guise of representations of symmetry groups of strip and periodic patterns on the two-sided plane, with no concern for or connection with 2-chromatic patterns. Speiser [1927] was the first to propose the investigation of groups of symmetries of the two-sided plane, and he showed that there are 31 strip groups. Partly as a consequence of Speiser's suggestion, several crystallographers determined the 80 periodic groups of the two-sided plane independently and simultaneously in 1929 (see Alexander & Herrmann [1929], Heesch [1929], Hermann [1929], Weber [1929]). It is interesting to note that Weber used 46 2-chromatic patterns (composed of black and white triangles) to represent by diagrams those groups of the two-sided plane which correspond neither to monochromatic nor to gray patterns. But Weber, as well as the other authors, continued stressing the 80 periodic symmetry groups of the 2-sided plane even when discussing black-and-white

patterns. They clearly were not considering colored patterns and their groups as objects of independent interest. Similar attitudes were exhibited by many later authors (see, for example, Šubnikov [1930], [1940], [1946], [1951], Müller [1944], Burckhardt [1945], Dornberger-Schiff [1956], [1959], Holser [1958], [1961], Neronova & Belov [1961a], Wood [1964], Šubnikov & Kopcik [1972], and many of the references mentioned below). Brief surveys of the symmetry groups of finite, strip and periodic 2-color patterns, together with numerous references to earlier papers, can be found in Le Corre [1958] and Nowacki [1960]. Šubnikov introduced the term "antisymmetry" to denote any operation which involves interchanging the sides of the plane (in other words, which corresponds to a color-reversing symmetry in the 2-chromatic representation), and the expression soon found wide currency. To us this seems unfortunate, since this terminology may well have contributed to the delay in understanding (and even in considering) the symmetries of colored patterns. (The treatment of color groups in Jaswon & Rose [1983] is confused and misleading.)

Belov [1956b] noted the symmetry and color-symmetry groups of some Moorish patterns. He and his collaborators also used tilings colored with three or more colors (see Belov & Tarhova [1956a], [1956b], Belov [1956a], [1959], Belov & Belova [1957]), but their understanding of the concept of color symmetry seems to have been incomplete. For example, although the last of these papers utilizes 2-colored tilings to represent the 46 periodic 2-color groups, there is no mention of isohedral 2-colored tilings; moreover, in several cases in which such tilings could have been used, more complicated 2-colored tilings are shown without any explanation.

Another discovery made, apparently, by Belov and his co-workers led to some k -chromatic tilings with $k > 2$ but also caused an unfortunate narrowing of the scope of investigation and so contributed to delays in the development of color-symmetry groups. The discovery in question amounts to the observation that each 2-chromatic pattern in the plane can be interpreted as an uncolored periodic pattern in 3-dimensional space.

To do this it is only necessary to place the white copies of the motif, for example, in the plane $z = 0$ and also, by translation along the z -axis, in the planes $z = n$ for each integer n ; the black copies of the motif are translated to the planes $z = n + \frac{1}{2}$ in a similar manner. Conversely, to any given periodic pattern in 3-space, in which all copies of the motif are contained in planes of the form $z = n$ and $z = n + \frac{1}{2}$ (copies in planes differing by an integral value of z being translates of each other parallel to the z -axis) there corresponds a 2-chromatic planar pattern (or a monochromatic one, or a gray one). This led Belov and co-workers to the idea of representing certain 3-dimensional patterns by k -colored tilings of the plane. The 3-dimensional patterns they considered are those in which all copies of the motif are situated in k different equidistant planes, and translation along the z -axis through k times the distance between neighboring planes is a symmetry of the pattern. In such a situation the motifs in each of k consecutive planes are assigned one of k colors, and everything is collapsed (by orthogonal projection) to a k -colored pattern in the (x,y) -plane. It was found that there are fifteen such representations of crystallographic groups in 3-space by multicolor patterns in the plane. These are shown in several publications (see Belov & Tarhova [1956a], [1956b], Belov & Belova [1957], Belov, Belova & Tarhova [1958], [1964], Šubnikov & Kopcik [1972]; there are slight differences between the various publications, and the last two should be considered as definitive). Observing that among these fifteen multicolor patterns there are four pairs which involve "mere reversal of the colour sequence", Lockwood & Macmillan [1978, Chapter 11] assert that "in fact there are only eleven polychromatic types that allow rotations". In any case, even a very hasty glance at the colored tilings in these papers makes it clear that the groups of color symmetries of chromatic planar patterns were not in their authors' minds. Indeed, the specific colors were used only as a code for the sequence of parallel planes, and so, for example, the two 4-chromatic tiling patterns in Figure 8.8.2 are shown as representations of the same 3-dimensional crystallographic group, although these 4-chromatic patterns clearly have different 4-color

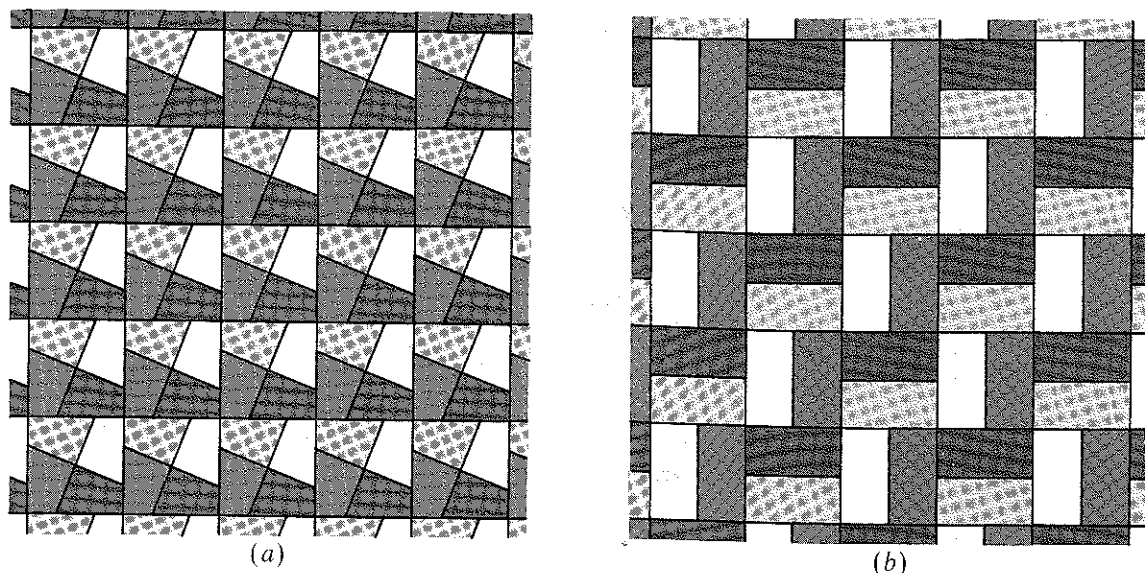


Figure 8.8.2

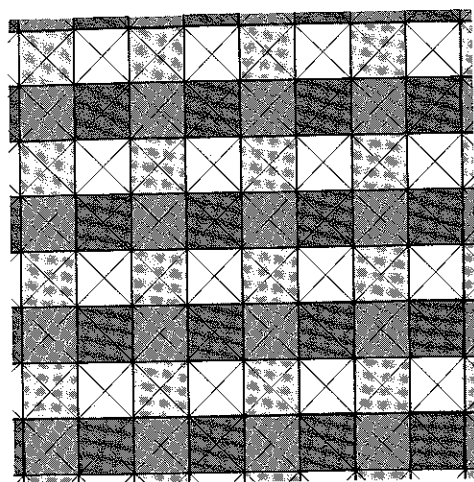
Two 4-chromatic patterns, used by Belov and co-workers to represent the same 3-dimensional crystallographic group (see Belov, Belova & Tarhova [1964, Plate I], Šubnikov & Kopcik [1972, Figure 230]). The 4-color groups of these patterns are clearly different, since the pattern in (a) has $p4$ as its underlying group $S_C(\emptyset)$, and the one in (b) has $p4g$.

groups. Extensions of the investigations of Belov and co-workers can be found in Zamorzaev [1969], Palistrant & Zamorzaev [1971], Kopcik & Kužukeev [1972], Palistrant [1972], Zamorzaev, Galyarskii & Palistrant [1978]. It seems that the unwillingness of many crystallographers to consider any symmetries except those arising from isometries of the 3-dimensional space, hampered their understanding of color symmetries in the plane. This, in turn, led to delays in understanding the 3-dimensional analogues.

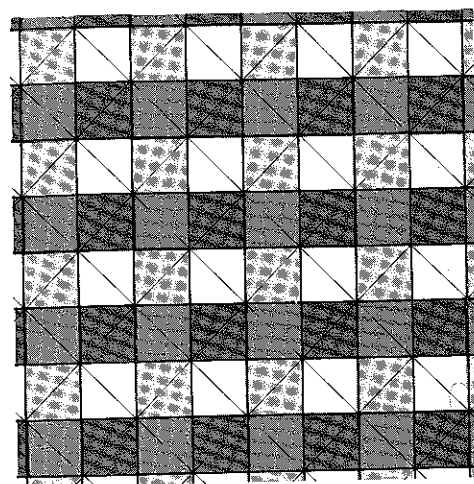
The change in attitude among crystallographers—to an acceptance of color groups as “symmetries” in which each geometric isometry is paired with a systematic change in the labels attached to the objects under consideration—seems to have occurred mainly in the 1960’s, although some explicit indications occur earlier, for example in Cochran [1952], Mackay [1957], Niggli [1959]. This attitude, which is essentially the one adopted in this chapter, is fully present, for example, in van der Waerden & Burckhardt [1961], Bohm [1963], MacGillavry [1965] and Burckhardt [1966]. However, other attitudes continue to be advocated and since the same words are often used in various meanings, some confusion is inevitable.

We shall now give an account of those works which are directly concerned with colored patterns in the plane. There is an abundance of literature on various aspects of color symmetries in the 3-dimensional space, but no references are given as this is outside the scope of the book.

Loeb [1971a] gives an original, interesting and satisfactory account of the 2-color groups, using the word “configuration” instead of “group”. Unfortunately, when discussing multicolor patterns, Loeb restricts the admissible color changes so severely that he obtains a total of only 54 periodic k -color “configurations” with $k \geq 3$, and many of the 2-chromatic patterns classified by Loeb earlier in the book do not qualify as 2-color “configurations”. Moreover, his classification does not apply to a colored pattern (or tiling) unless a set of symmetry elements (centers of rotation, lines of reflection, etc.) compatible with the restricted types of color symmetries admitted, has been designated. As shown by the example in Figure 8.8.3, the same 4-chromatic coloring of the square tiling belongs to different “color configurations” depending on the choice of symmetry elements. Although the impression is given that color groups are being con-



244', 013, $N = 4$, $s = 0$



244', 111, $N = 4$, $s = 3$

Figure 8.8.3

The same 4-chromatic coloring of the square tiling can be interpreted as representing two of Loeb's "color configurations"; Loeb's symbols are indicated near each. His "color configurations" depend not only on the 4-chromatic tiling but also on the choice of symmetry elements (centers of rotation, lines of reflection, etc); hence the classification does not lead to k -color groups.

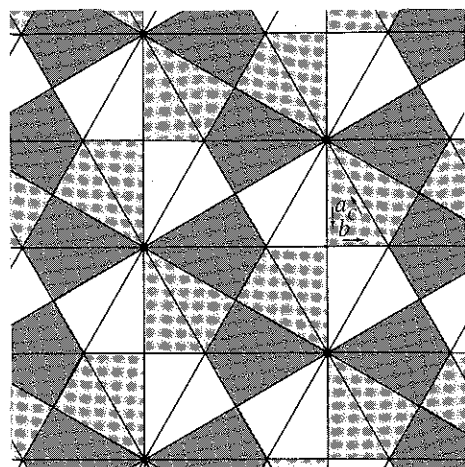
sidered, this is not what actually happens. (This impression is reinforced by the remarks of W. T. Holser in the Foreword to Loeb [1971a, page x]. He says: "The enumeration of plane color groups [in this book] concludes with a detailed analysis of some of M. C. Escher's drawings".) In particular, colored tilings which have not only the same color groups but are even of the same color pattern type are assigned by Loeb to different "color configurations" (see Figure 8.8.4).

Macdonald & Street [1977], [1978a], [1978b] expand Loeb's investigations by removing his restrictions on the color symmetries and discuss colorings of "friezes" (strip patterns) and "crystals" (periodic patterns).

In complete analogy to the way in which isohedral or homeohedral tilings can be generalized to 2-isohedral or 2-homeohedral tilings, or monomotif patterns to dimotif patterns, it is possible to generalize k -chromatic patterns to (k_1, k_2) -chromatic patterns. By this we mean a monomotif or dimotif pattern $\mathcal{P}$ and a coloring $\mathcal{P}^c$ of $\mathcal{P}$ such that the colored copies of the motif form two transitivity classes under the color symmetries of $\mathcal{P}^c$. As examples we show in Figure 8.8.5 several (k_1, k_2) -colorings of the uniform tilings (3⁶) and (3.6.3.6). In Section 4.8 we mentioned (1,1)-chromatic colorings of the Laves

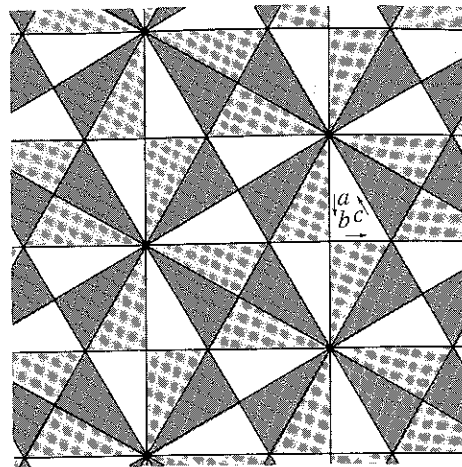
tilings in connection with the enumeration of 2-homeohedral tilings. Other examples of (k_1, k_2) -chromatic patterns are given in Macdonald [1979] and Roth [1982], [1983], [1984], where also more general considerations are presented. In view of its wide applicability and the challenging open problems it presents, this topic may attract considerable attention in the future.

As generalizations of 2-color symmetries (understood as "antisymmetries" in the sense of Šubnikov), several authors have investigated the "symmetry" groups of "patterns" in which the copies of the motif have not just one, but $n \geq 2$ distinct properties each of which can be present or absent ("antisymmetries of several kinds"). In principle, these are clearly special cases of the general k -colorings and could be found from them by selection. In practice, since $k = 2^n$, in most instances it is not practical to proceed in this manner; also, most of the work on "multiple antisymmetries" occurred before the general concept of color groups was well understood. Mackay [1957] dealt with periodic groups for $n = 2$ antisymmetry operations which are interchangeable to some extent. Roman [1959], Pabst [1962], [1963], Šubnikov [1962a], [1962b], [1963a], Belov, Kuncević & Neronova [1962] determined the number of strip groups for $n = 2$ (that



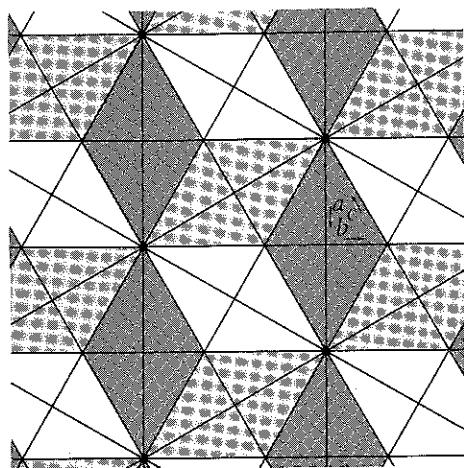
PP46[3]₁ $[a^+b^+c^+(2);a^-(12)b^-(12)c^-(13)]$

Loeb: $\underline{2} \ \underline{3} \ \underline{6}$, 014, $N = 3$, $s = 0$



PP46[3]₁ $[a^+b^+c^+(1);a^-(12)b^-(12)c^-(13)]$

Loeb: $\underline{2} \ \underline{3} \ \underline{6}$, 014, $N = 3$, $s = 1$



PP46[3]₁ $[a^+b^+c^+(3);a^-(12)b^-(12)c^-(13)]$

Loeb: $\underline{2} \ \underline{3} \ \underline{6}$, 014, $N = 3$, $s = 2$

Figure 8.8.4

Three 3-color “configurations” of Loeb [1971a, Figure 88], redrawn to use the same three colors. In Loeb’s interpretation they represent distinct “3-color configurations”, although they clearly have the same 3-color groups ($p6m[3]_1$ in our notation) and are of the same 3-color pattern types PP46[3]₁. Near each tiling we noted Loeb’s symbols, as well as its color-incidence symbol in the notation of Section 8.6. The tilings belong to different homeochromatic types.

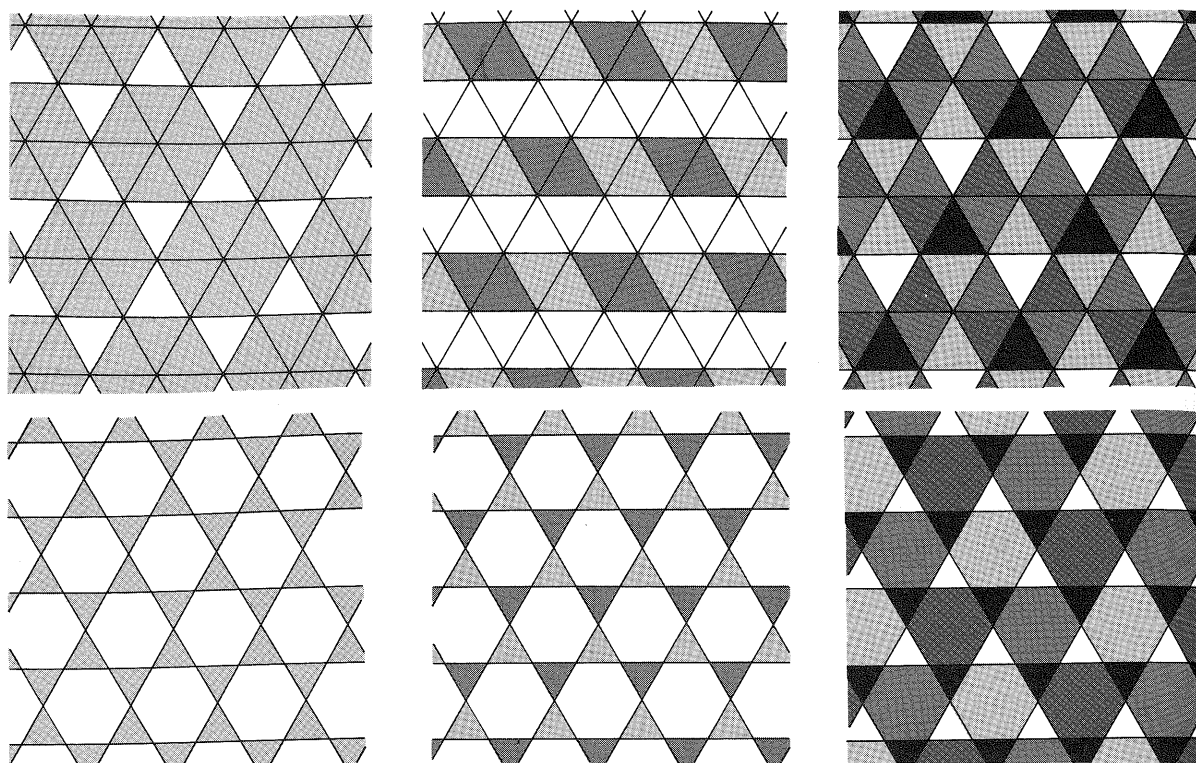


Figure 8.8.5
Examples of (k_1, k_2) -chromatic tilings with $(k_1, k_2) = (1, 1), (1, 2)$ and $(2, 3)$. In the illustrations we use white and black for the first set of colors, shades of gray for the second. The underlying tilings are the uniform tilings (3^6) and $(3.6.3.6)$.

is, the number of black-and-white groups of the 2-sided strip); Palistrant & Zamorzaev [1964] extended this to all $n \geq 2$. Zamorzaev & Palistrant [1960] determined the periodic groups for all $n \geq 2$, and illustrated those corresponding to $n = 2$ by 4-colored tilings (in Zamorzaev & Palistrant [1961]); other determinations of the periodic groups with $n = 2$ kinds of "antisymmetry" were carried out by Palistrant & Zamorzaev [1963] and Palistrant [1963]. Related material is presented in Pawley [1961], Neronova & Belov [1961a], [1961b], Šubnikov [1963a], Roman [1963], [1970], [1971], Palistrant [1965], [1966], [1967], [1968], [1972], Neronova [1966], [1967], Zamorzaev [1967]. Detailed surveys of these and related investigations are given by Donnay [1967], Kopcik [1967], [1975], Zamorzaev & Palistrant [1980], and Zamorzaev [1976]; Zamorzaev's book presents many of the enumerations in extensive tables.

The 23 periodic 3-color groups were first presented

in Grünbaum [1976] and in Grünbaum & Shephard [1977e]. An independent enumeration of these groups, together with the enumeration of all periodic k -color groups for $k \leq 60$, was performed by Wieting [1981]. Using a different approach, Senechal [1979a] enumerates the periodic k -color groups for many infinite families of integers k (but not for all integers). In the same very interesting paper, for each periodic pattern type all values of k are determined for which the pattern admits a perfect k -coloring.

The color-incidence symbols first appeared in Grünbaum & Shephard [1979a] and were used in listing the 2-color and 3-color groups. The description of the color-incidence symbols given in this paper is valid only for primitively colored tilings. The 3-color pattern types were first illustrated in Grünbaum & Shephard [1983b]. A recent exposition of color symmetries and color patterns, and their history, can be found in Schwarzenberger [1984].